

Bayesian Hierarchical Generalized Poisson Model

Load the data

```
library("haven")
library("dplyr")
wm <- read_sav("wm.sav")
wm <- wm %>% arrange(HH1, HH2, WM3)
hh <- read_sav("hh.sav")
df_hh <- hh %>% select(HH1, HH2, HHSEX, HHAGE)
wm <- merge(x=wm, y=df_hh, by = c("HH1", "HH2"))
```

Clean the data

(one line of code)

```
wm <- wm %>% filter(MA1 == 1 & MT1!=9 & MT2!=9 & MT3!=9 & MA2!=98 & MA2!=99 & MA3!=9)
```

68709 women before cleaning the data

MA1: "Currently married"

```
table(wm$MA1,useNA="ifany")
```

```
##
##      1
## 51331
```

```
wm <- wm %>% filter(MA1==1)
```

```
1      3  <NA>
51426 12951 4332
```

We exclude 12951 women who have never married and 4332 women with unknown status. In total, 17283 samples were excluded.

Technology usage (22): Missing data: media_newspaper (5), media_radio (3), media_tv(14)

```
table(wm$MT1,useNA="ifany")
```

```
##
##      0      1      2      3
## 48411 1364   884   672
```

```
table(wm$MT2,useNA="ifany")
```

```
##
##      0      1      2      3
## 50608   316   271   136
```

```
table(wm$MT3,useNA="ifany")
```

```
##
##      0      1      2      3
## 18295 2321 4162 26553

attr(wm$MT1, "labels")

##          NOT AT ALL LESS THAN ONCE A WEEK  AT LEAST ONCE A WEEK          ALMOST EVERY DAY
##                  0                      1                      2                      3
##          NO RESPONSE
##                  9

wm <- wm %>% filter(MT1!=9 & MT2!=9 & MT3!=9)

other wives: (30)

table(wm$MA3,useNA="ifany")

##
##      1      2
## 1618 49713

wm <- wm %>% filter(MA3!=9)

Age of husband (44) (98 for "DK"; 99 for "NO RESPONSE")

table(wm$MA2,useNA="ifany")

##
##   15   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31
##    1    3   14   73   95  237  302  479  527  742 1038 1042 1055 1448 1201 2178 1495
##   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47   48
## 1909 1456 1556 2558 1828 1567 2035 1588 2212 1485 1559 1162 1148 1822 1375 1368 1620
##   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63   64   65
## 1137 1429 1029 978  819  715 1051  801  553  532  343  544  246  199  150  105  164
##   66   67   68   69   70   71   72   73   74   75   76   77   78   79   80   82   84
##   72   41   51   19   59   22   16    8    9   21   10    5    6    7    2    3    1
##   85   88   90
##    2    1    3

wm <- wm %>% filter(MA2!=98 & MA2!=99)

HHAGE

table(wm$HHAGE,useNA="ifany")

##
##   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31   32
##    3    7   28   40   89  154  231  258  383  590  624  674  934  846 1470 1142 1327
##   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47   48   49
## 1158 1311 1863 1588 1427 1732 1491 1799 1445 1399 1146 1113 1640 1376 1435 1633 1212
##   50   51   52   53   54   55   56   57   58   59   60   61   62   63   64   65   66
## 1450 1198 1078 984  902 1320 1108  810  835  705 1095  611  606  511  363  649  379
##   67   68   69   70   71   72   73   74   75   76   77   78   79   80   81   82   83
##  308  364  218  499  257  212  145   94  211  128   95  102   61  137   55   42   33
##   84   85   86   87   88   89   90   92   93   94   95
##   21   56   26   27   10    9   19    9    1    2   18
```

No missing values for the following predictors:

```
table(wm$HH6,useNA="ifany")
```

```
##
##      1      2
## 10123 41208
```

```
table(wm$HH7,useNA="ifany")
```

```
##
##    10    20    30    40    45    50    55    60
## 4539  9135 10364  8410  2670  6363  6430  3420
```

```
table(wm$HH7A,useNA="ifany")
```

```
##
##      1      3      4      6      9     10     12     13     15     18     19     22     26     27     29     30     32
## 804  600  770  726  749  790  765  677 1675  876 1094  686 1963  804  720  816  707
##  33   35   36   38   39   41   42   44   46   47   48   49   50   51   52   54   55
## 644  642  878  779  588  882  707  861  642  755  629  808  899  695  819  618  817
##  56   57   58   59   61   64   65   67   68   69   70   72   73   75   76   77   78
## 696  885  874  782  844  769  775  811  756  754  835  657  846  839  816  806  847
##  79   81   82   84   85   86   87   88   89   90   91   93   94
## 740  802  746  646  739  657  856  818  581  831  837  700  901
```

```
#table(wm$sumweight,useNA="ifany")
```

```
table(wm$WAGEM,useNA="ifany")
```

```
##
##      5      6      7      8      9     10     11     12     13     14     15     16     17     18     19     20     21
##      1      2      1      5     16     40    614   1730   3813   5751   7125   7463   6783   5599   3721   2647   1764
##     22     23     24     25     26     27     28     29     30     31     32     33     34     35     36     37     38
## 1252  889  559  473  332  221  169  105   77   56   29   35   15    8    9    8    3
##     39     41     42     43     44     45     48
##      4      4      3      1      2      1      1
```

```
table(wm$WB4,useNA="ifany")
```

```
##
##     15     16     17     18     19     20     21     22     23     24     25     26     27     28     29     30     31
## 183  356  450 1359 1483 1580 1509 1669 1707 1745 1838 1955 1753 1858 1762 2205 2015
##    32    33    34    35    36    37    38    39    40    41    42    43    44    45    46    47    48
## 1900 1848 1803 1799 2045 1630 1777 1480 1433 1405 1293 1172 1084 1021 1190 1135 1153
##    49
##   736
```

```
table(wm$windex5,useNA="ifany")
```

```
##
##      1      2      3      4      5
## 10718 10732 10702 10336  8843
```

```
table(wm$welevel,useNA="ifany")
```

```
##
##      0      1      2      3
## 9186 13235 22289  6621
```

```
table(wm$ethnicity,useNA="ifany")
```

```
##
##      1      2
## 50184 1147
```

```
table(wm$CM11,useNA="ifany")
```

```
##
##      0      1      2      3      4      5      6      7      8      9     10     11     12
## 4663 10933 16565 10670 5029 2029 885 337 145 52 18 3 2
```

Summary

```
women_age <- wm$WB4
women_age2 <- women_age^2
husband_age <- wm$MA2
#husband_age2 <- husband_age^2
HH_age <- wm$HHAGE
#HH_age2 <- wm$HHAGE^2
women_agem <- wm$WAGEM
division <- wm$HH7
ethnicity <- wm$ethnicity # Fixed
area <- wm$HH6 # Fixed
women_edu <- wm$welevel
HH_sex <- wm$HHSEX
other_wives <- wm$MA3
windex <- wm$windex5
media_newspaper <- ifelse(wm$MT1>0, 1, 0)
media_radio <- ifelse(wm$MT2>0, 1, 0)
media_tv <- ifelse(wm$MT3>0, 1, 0)
media <- media_newspaper|media_radio|media_tv
media <- ifelse(media,1,0)
# Response variable
CEB <- wm$CM11
# District
district <- wm$HH7A
# Sampling weight
wmweight <- wm$wmweight
```

```
mean(women_age)
```

```
## [1] 32.02186
```

```
sd(women_age)
```

```
## [1] 8.701892
```

```
mean(husband_age)
```

```
## [1] 39.94711
```

```
sd(husband_age)
```

```
## [1] 10.29503
```

```
mean(HH_age)
```

```
## [1] 45.12719
```

```

sd(HH_age)

## [1] 12.78071
mean(women_agem)

## [1] 16.85521
sd(women_agem)

## [1] 3.300367
total <- dim(wm)[1]
table(division)

## division
##      10      20      30      40      45      50      55      60
## 4539  9135 10364  8410  2670  6363  6430  3420
round(table(division)/total,4)

## division
##      10      20      30      40      45      50      55      60
## 0.0884 0.1780 0.2019 0.1638 0.0520 0.1240 0.1253 0.0666
table(area)

## area
##      1      2
## 10123 41208
round(table(area)/total,4)

## area
##      1      2
## 0.1972 0.8028
attr(wm$welevel, "labels")

## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9
table(women_edu)

## women_edu
##      0      1      2      3
## 9186 13235 22289 6621
round(table(women_edu)/total,4)

## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
attr(wm$welevel, "labels")

## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9

```

```

table(women_edu)

## women_edu
##      0      1      2      3
## 9186 13235 22289 6621
round(table(women_edu)/total,4)

## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
attr(wm$HHSEX, "labels")

##      Male  Female Missing
##          1      2      9
table(wm$HHSEX)

##
##      1      2
## 47345 3986
round(table(wm$HHSEX)/total,4)

##
##      1      2
## 0.9223 0.0777
attr(other_wives, "labels")

##      YES      NO NO RESPONSE
##          1      2      9
table(other_wives)

## other_wives
##      1      2
## 1618 49713
round(table(other_wives)/total,4)

## other_wives
##      1      2
## 0.0315 0.9685
attr(windex, "labels")

## Poorest  Second  Middle  Fourth  Richest
##          1      2      3      4      5
table(windex)

## windex
##      1      2      3      4      5
## 10718 10732 10702 10336 8843
round(table(windex)/total,4)

## windex
##      1      2      3      4      5
## 0.2088 0.2091 0.2085 0.2014 0.1723

```

```

table(media)

## media
##      0      1
## 17834 33497

round(table(media)/total,4)

## media
##      0      1
## 0.3474 0.6526

attr(wm$ethnicity, "labels")

##      Bengali      Other Missing/DK
##           1           2           9

table(ethnicity)

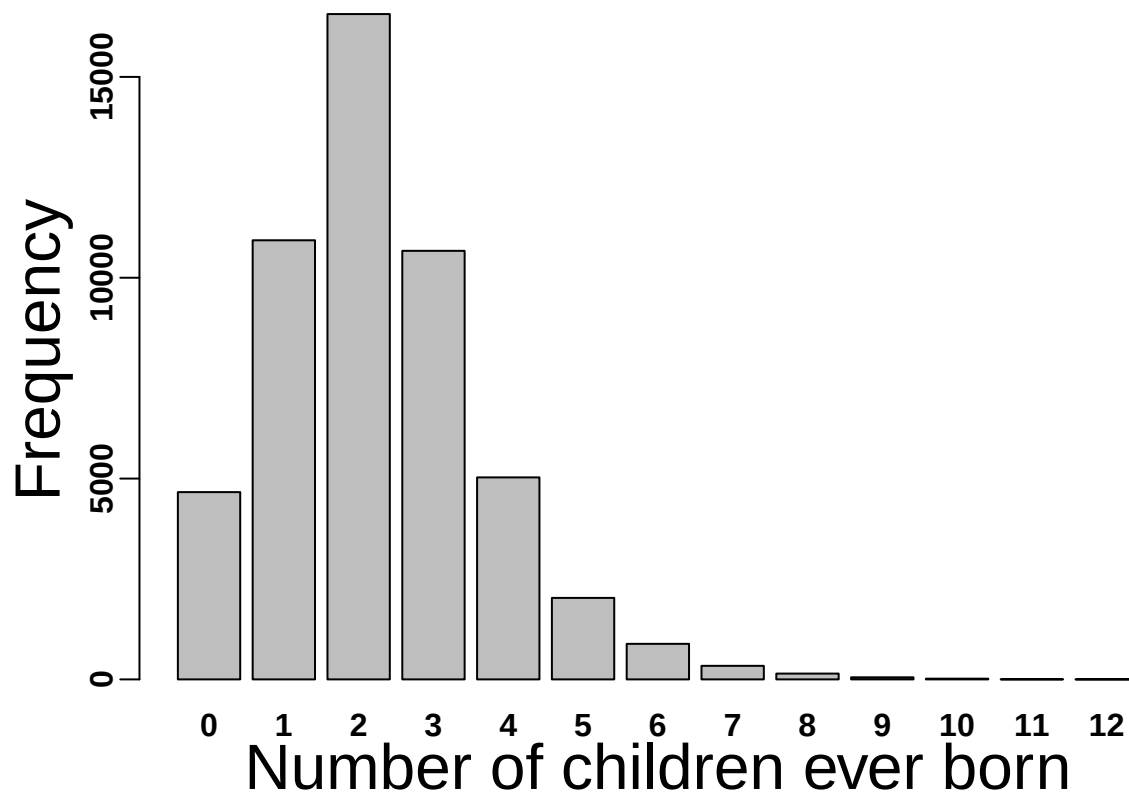
## ethnicity
##      1      2
## 50184 1147

round(table(ethnicity)/total,4)

## ethnicity
##      1      2
## 0.9777 0.0223

IMAGEOUT <- "./pics/BHGP/"
par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
barplot(table(CEB), ylab="Frequency",
        xlab="Number of children ever born", cex.lab = 2, names.arg=c(0:12), font.axis = 2.5)

```



```
dev.print(pdf,
  paste(IMAGEOUT,"CEB_histo.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

```
boxplot(CEB ~ area, xlab = NULL, #xlab="Place of residence",
  ylab="Number of ever born children", xaxt = "n", yaxt = "n", cex.lab = 1.2)
axis(side = 1, labels = FALSE)
axis(side = 2, las = 2, mgp = c(3, 0.75, 0))
```

```
anames <- c("Urban", "Rural")
text(x = 1:length(unique(area)),
  y = par("usr")[3] - 0.45,
  labels = unique(anames),
  xpd = NA,
  srt = 45,
  cex = 1.2,
  adj = 1)
```

```
dev.print(device = postscript, paste(IMAGEOUT, "CEB_vs_area.pdf",sep=""),width=8,height=8, horizontal =
```

```
boxplot(CEB ~ media, xlab = NULL, #xlab="Media exposure",
  ylab="Number of ever born children",
  xaxt = "n", yaxt = "n", cex.lab = 1.2)
axis(side = 1, labels = FALSE)
axis(side = 2, las = 2, mgp = c(3, 0.75, 0))
```

```
mnames <- c("No", "Yes")
text(x = 1:length(unique(media)),
```



```

y = par("usr")[3] - 0.45,
# labels = sort(unique(dataBD$division)),
labels = sort(unique(mnames)),
xpd = NA,
## Rotate the labels by 35 degrees.
srt = 45,
cex = 1.2,
adj = 1)

y <- CEB
division <- factor(division, levels = c(10,20,30,40,45,50,55,60), labels = c("Barishal", "Chattogram", "
division <- relevel(division, ref=3)

division <- model.matrix(~division, data = division)
colnames(division) <- c(
  "Dhaka.ref",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymenshing",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)

district_names <- c("Bagerhat", "Bandarban", "Barguna", "Barishal", "Bhola", "Bogura", "Brahmanbaria", "
district_codes <- c(1, 3, 4, 6, 9, 10, 12, 13, 15, 18, 19, 22, 26, 27, 29, 30, 32, 33, 35, 36, 38, 39, 4

district <- factor(district, levels = district_codes, labels = district_names)

women_edu <- factor(women_edu)
women_edu <- model.matrix(~women_edu, data = women_edu)

windex <- factor(windex)
windex <- relevel(windex, ref=3)
windex <- model.matrix(~windex, data = windex)
colnames(windex) <- c(
  "windex.Middle.ref",
  "windex.Poorest",
  "windex.Poor",
  "windex.Rich",
  "windex.Richest"
)

area <- ifelse(area==2, 0, 1)
area <- factor(area)
area <- relevel(area, ref = 1)

other_wives <- ifelse(other_wives==2, 0, 1)
other_wives <- factor(other_wives)

media <- factor(media)
media <- relevel(media, ref = 1)

```

Helper functions

```
# Helper functions
# bisection to find root (copied from VGAM package)
bisection.basic <-
  function(f, a, b, tol = 1e-9, nmax = NULL, ...) {

    if (any(is.infinite(b))) {
      warning("replacing 'b' values of Inf by a large value")
      b[is.infinite(b)] <- .Machine$double.xmax / 4
    }

    if (is.null(nmax)) {
      nmax <- round(log2(max(b - a)) - log2(min(tol))) + 4
      if (!is.finite(nmax))
        nmax <- log2(.Machine$double.xmax) - 5
    }
    signtest <- (sign(f(a, ...)) * sign(f(b, ...)) <= 0)

    allsign <- all(signtest, na.rm = TRUE)
    if (!allsign || any(is.na(signtest))) {
      warning("roots do not exist between 'a' and 'b'. ",
        "Some answers may be misleading.")
    }

    N <- 1
    while (N <= nmax) {
      mid <- (a + b) / 2
      save.f <- f(mid, ...)
      if (all(save.f == 0 | (b - a)/2 < tol)) {
        return(mid)
      }
      N <- N + 1
      vecTF <- sign(save.f) == sign(f(a, ...))
      a[ vecTF] <- mid[ vecTF]
      b[!vecTF] <- mid[!vecTF]
    }

    warning("did not coverge. Returning final root")
    mid
  } # bisection.basic

is.Numeric <-
  function(x, length.arg = Inf,
    integer.valued = FALSE, positive = FALSE)
  if (all(is.numeric(x)) && all(is.finite(x)) &&
    (if (is.finite(length.arg))
      length(x) == length.arg else TRUE) &&
    (if (integer.valued) all(x == round(x)) else TRUE) &&
    (if (positive) all(x > 0) else TRUE)) TRUE else FALSE
```

Define our genpois2 functions

```
dgenpois0 <- function(x, theta, lambda = 0, log = FALSE) {
  if (!is.logical(log.arg <- log) || length(log) != 1)
    stop("bad input for argument 'log'")
  rm(log)

  LLL <- max(length(x), length(theta), length(lambda))
  if (length(x) != LLL) x <- rep_len(x, LLL)
  if (length(theta) != LLL) theta <- rep_len(theta, LLL)
  if (length(lambda) != LLL) lambda <- rep_len(lambda, LLL)

  bad0 <- !is.finite(theta) | !is.finite(lambda) |
    theta < 0 | 1 <= lambda
  bad <- bad0 | !is.finite(x) | !is.finite(lfactorial(x))

  logpdf <- x + lambda + theta

  if (any(!bad)) {
    negind <- (theta + x * lambda) < 0
    logpdf[!bad&!negind] <- -x[!bad&!negind] * lambda[!bad&!negind] - theta[!bad&!negind] +
      (x[!bad&!negind] - 1) * log(theta[!bad&!negind] + x[!bad&!negind] * lambda[!bad&!negind]) +
      log(theta[!bad&!negind]) - lfactorial(x[!bad&!negind])
  }

  if (any(ind1 <- (lambda < 0))) {
    epsilon <- 1.0e-9 # Needed to handle a "<" rather than a "<=".
    mmm <- pmax(4, floor(theta/abs(lambda) - epsilon))
    logpdf[ind1 & mmm < pmax(-1, -theta/mmm)] <- NaN
    logpdf[ind1 & mmm < x] <- log(0) # probability 0, not NaN
  }

  logpdf[!bad0 & is.infinite(x)] <- log(0)
  logpdf[!bad0 & is.infinite(lfactorial(x))] <- log(0)
  logpdf[!bad0 & x < 0] <- log(0)
  logpdf[!bad0 & x != round(x)] <- log(0)
  logpdf[bad0] <- NaN

  if (log.arg) logpdf else exp(logpdf)
} # dgenpois0

pgenpois0 <-
function(q, theta, lambda = 0, lower.tail = TRUE) {
  q <- floor(q)
  LLL <- max(length(q), length(theta), length(lambda))
  if (length(q) != LLL) q <- rep_len(q, LLL)
  if (length(theta) != LLL) theta <- rep_len(theta, LLL)
  if (length(lambda) != LLL) lambda <- rep_len(lambda, LLL)

  bad0 <- !is.finite(theta) | !is.finite(lambda) |
    theta < 0 | 1 <= lambda
  bad <- bad0 | !is.finite(q)

  if (all(is.finite(lambda)) && all(lambda == 0))
```

```

    return(ppois(q, theta, lower.tail = lower.tail))

ans <- q + lambda + theta
ok3 <- !bad & 0 <= q
if (any(ok3)) {
  ans[ok3] <- mapply(function(q, theta, lambda) {
    xx <- 0:q
    if (any(theta + xx * lambda <= 0)){
      return(1)
    }else{
      return(sum(exp(-xx * lambda - theta + log(theta) +
        (xx - 1) * log(theta + xx * lambda) -
        lfactorial(xx))))
    }
  },
  q = q[ok3],
  theta = theta[ok3],
  lambda = lambda[ok3])
  ans <- unlist(ans)
}
ans[!bad0 & is.infinite(q)] <- 1
ans[!bad0 & q < 0] <- 0
ans[bad0] <- NaN
if (!lower.tail)
  ans <- 1 - ans
ans
} # pgenpois0

qgenpois0 <- function(p, theta, lambda = 0) {
  LLL <- max(length(p), length(theta), length(lambda))
  if (length(p) != LLL) p <- rep_len(p, LLL)
  if (length(theta) != LLL) theta <- rep_len(theta, LLL)
  if (length(lambda) != LLL) lambda <- rep_len(lambda, LLL)

  ans <- p + lambda + theta

  bad0 <- !is.finite(theta) | !is.finite(lambda) |
    theta < 0 | 1 <= lambda |
    is.na(p) | is.na(lambda) |
    is.na(theta)
  bad <- bad0 | !is.finite(p) | p <= 0 | 1 <= p

  lo <- rep_len(0, LLL) - 0.5
  approx.ans <- lo # True at lhs
  hi <- 2 * lo + 10.5
  dont.iterate <- bad
  done <- dont.iterate | p <= pgenpois0(hi, theta, lambda = lambda)
  iter <- 0
  max.iter <- round(log2(.Machine$double.xmax)) - 2
  max.iter <- round(log2(1e300)) - 2
  while (!all(done) && iter < max.iter) {
    lo[!done] <- hi[!done]
    hi[!done] <- 2 * hi[!done] + 10.5 # Bug fixed
  }
}

```

```

    done[!done] <- (p[!done] <= pgenpois0(hi[!done],
      theta[!done], lambda = lambda[!done]))
    iter <- iter + 1
  }

foo <- function(q, theta, lambda, p)
  pgenpois0(q, theta, lambda = lambda) - p

lhs <- dont.iterate |
  p <= dgenpois0(0, theta, lambda = lambda)

if (any(!lhs)) {
  approx.ans[!lhs] <-
    bisection.basic(foo, lo[!lhs], hi[!lhs],
      tol = 1/16, p = p[!lhs],
      theta = theta[!lhs],
      lambda = lambda[!lhs])
  faa <- floor(approx.ans[!lhs])
  tmp <-
    ifelse(
      pgenpois0(faa, theta[!lhs],
        lambda = lambda[!lhs]) < p[!lhs] &
      p[!lhs] <= pgenpois0(faa+1, theta[!lhs],
        lambda = lambda[!lhs]),
      faa+1,
      faa)
  ans[!lhs] <- tmp
} # any(!lhs)
vecTF <- !bad0 & !is.na(p) &
  p <= dgenpois0(0, theta, lambda = lambda)
ans[vecTF] <- 0

ans[!bad0 & !is.na(p) & p == 0] <- 0
ans[!bad0 & !is.na(p) & p == 1] <- Inf
ans[!bad0 & !is.na(p) & p < 0] <- NaN
ans[!bad0 & !is.na(p) & p > 1] <- NaN
ans[bad0] <- NaN
ans
} # qgenpois0

rgenpois0 <-
function(n, theta, lambda = 0) {
  use.n <- if ((length.n <- length(n)) > 1) length.n else
    if (!is.Numeric(n, integer.valued = TRUE,
      length.arg = 1, positive = TRUE))
      stop("bad input for argument 'n'") else n
  if (length(theta) > use.n)
    warning("length of 'theta' exceeds 'n'. Truncating it.")
  if (length(lambda) > use.n)
    warning("length of 'lambda' exceeds 'n'. Truncating it.")

```

```

theta <- rep_len(theta, use.n)
lambda <- rep_len(lambda, use.n)

bad0.a <- !is.finite(lambda) | !is.finite(theta)
if (any(bad0.a))
  stop("cannot have NAs or NaNs in 'theta' or 'lambda'.")
bad0.b <- theta <= 0 | 1 <= lambda # zz theta < 0
if (any(bad0.b))
  stop("some values of 'theta' or 'lambda' are out of range.")

if (all(lambda == 0))
  return(rpois(use.n, theta))

return(qgenpois0(runif(n), theta, lambda = lambda))
} # rgenpois0

dgenpois2 <- function(x, meanpar, disppar = 0, log = FALSE) {
  dgenpois0(x, theta = meanpar / (1 + disppar * meanpar),
    lambda = disppar * meanpar / (1 + disppar * meanpar),
    log = log)
} # dgenpois2

pgenpois2 <- function(q, meanpar, disppar = 0, lower.tail = TRUE) {
  pgenpois0(q, theta = meanpar / (1 + disppar * meanpar),
    lambda = disppar * meanpar / (1 + disppar * meanpar),
    lower.tail = lower.tail)
} # pgenpois2

qgenpois2 <- function(p, meanpar, disppar = 0) {
  qgenpois0(p, theta = meanpar / (1 + disppar * meanpar),
    lambda = disppar * meanpar / (1 + disppar * meanpar))
} # qgenpois2

rgenpois2 <- function(n, meanpar, disppar = 0) {
  rgenpois0(n, theta = meanpar / (1 + disppar * meanpar),
    lambda = disppar * meanpar / (1 + disppar * meanpar))
} # rgenpois2

```

Hierarchical Poisson Model

Utility functions and data

```

#####utility functions#####
## mvnormal simulation
rmvnorm<-function(n,mu,Sigma)
{
  E<-matrix(rnorm(n*length(mu)),n,length(mu))
  t( t(E%*%chol(Sigma)) +c(mu))
}

```

```

## Wishart simulation
rwish<-function(n,nu0,S0)
{
  sS0 <- chol(S0)
  S<-array( dim=c( dim(S0),n ) )
  for(i in 1:n)
  {
    Z <- matrix(rnorm(nu0 * dim(S0)[1]), nu0, dim(S0)[1]) %%% sS0
    S[, ,i]<- t(Z)%%Z
  }
  S[, ,1:n]
}

## mvnrm log density
ldmvnorm<-function(X,mu,Sigma,iSigma=solve(Sigma),dSigma=det(Sigma))
{
  Y<-t( t(X)-mu)
  sum(diag(-.5*t(Y)%%Y%%iSigma)) -
  .5*( prod(dim(X))*log(2*pi) + dim(X)[1]*log(dSigma) )
}

## Calculate matrix multiplication: (I,Z) %%% Beta
matmul <- function(Z, Beta){
  nZ <- nrow(Z)
  as.matrix(cbind(rep(1,nZ),Z))%%as.matrix(Beta)
}

#####utility functions#####

#####design matrix#####
# Design matrix for all variables (random x14 | fixed x21)
X <- as.matrix(cbind(
  ### random predictors
  women_age,
  women_age2,
  husband_age,
  #husband_age2,
  HH_age,
  #HH_age2,
  women_agem,
  women_edu[, -1],
  windex[, -1],
  media,
  other_wives,
  area,
  HH_sex,
  ### fixed predictor
  division[, -1], # 7
  ethnicity
))

# Design matrix for random variabe
# (the intercept will be add later)
Z <- as.matrix(cbind(
  women_age,
  women_age2,
  husband_age,

```

```

#husband_age2,
HH_age,
#HH_age2,
women_agem,
women_edu[,-1],
windex[,-1],
media,
other_wives,
area,
HH_sex
))

# Concat response variable y with design matrix of X
yX <- cbind(y,X)
#####design matrix#####

```

Init

```

#####initial values#####
q<-dim(Z)[2] + 1# number of columns in Z (13) + 1 (intercept)
p<-dim(X)[2] + 1# number of columns in X (20) + 1 (intercept)
m<-length(unique(district)) # number of district

GAMMA <- matrix(0, nrow = m, ncol = q)

# eta0: inverse-Wishart prior for Sigma
eta0 <- q+2
# The initial mean of the MVN prior of theta
model <- glm(y~., data = data.frame(yX), family=poisson)
THETA <- mu0 <- model$coefficients

# The initial variance of the MVN prior for the variance of theta
# The prior covariance matrix  $\Sigma_0$  is specified as  $m$  times the variance-covariance matrix of t
PLAMBDA <- m*vcov(model)
LAMBDA <- vcov(model)

# THETA1 stores m rows of ad hoc estimate of regression coefficients theta1
THETA1 <- NULL

# traverse each districts (total of 64)
for (dis in district_codes){
dis_data = yX[which(wm$HH7A==dis),]

captured <- tryCatch({
  model <- glm(y~., data = data.frame(dis_data), family=poisson)
  THETA1<-rbind(THETA1, model$coeff[c(1:q)])
}, warning = function(w) {
})
#print(captured)
}

S0 <- cov(THETA1)

```



```

# Prepare the list to store data in each stratum separately.
y.list <- list()
Z.list <- list()
X.list <- list()
wgt.list <- list()
N <- NULL
j <- 1
# traverse all 64 districts
for (dis in district_codes){
y.list[[j]] = y[which(wm$HH7A == dis)]
N[[j]]<-length(which(wm$HH7A == dis))
Z.list[[j]] = Z[which(wm$HH7A == dis),]
X.list[[j]] = X[which(wm$HH7A == dis),]
wgt.list[[j]] = wmweight[which(wm$HH7A == dis)]
j<-j+1
}

#summary(unlist(N))
#barplot(unlist(N))

# inverse Lambda_0, inverse S_0
iL0<-iSigma<-solve(S0)

# Alpha stores the current Alpha vector.
Alpha <- rep(0,m)
mu.alpha.0 <- 0
kappa.0 <- 1
kappa.m <- 1+m
nu.0 <- 1
nu.m <- 1+m

# s <- seq(0.00001, 6, by = 0.00001)
# for(i in 1:length(s)){
#   quant <- qnorm(0.975, mean=0, sd=s[i])
#   if(quant > 0.2){
#     cat("q = ", quant, "\n")
#     cat("sd = ", s[i], "\n")
#     break
#   }
# }

sigma2.alpha.0 <- 0.0104142

## MCMC
# The accepted count of proposed gamma_j for each stratum (j from 1 to 66)
ACCEPT.count <- rep(0,m)
# The accepted count of proposed theta
ACCEPT.Theta.count <- 0
ACCEPT.Alpha.count <- rep(0,m)
SIGMA.post<-NULL
GAMMA.post<- list()
THETA.post <- list()
ALPHA.post <- NULL

```

```

MU.ALPHA.post <- NULL
SIGMA2.ALPHA.post <- NULL
S=55000
B=5000
thin=50
#####initial values#####

#####simulation#####
set.seed(123)
start.time <- Sys.time()
for(s in 1:S)
{
  ##update Sigma
  iSigma<-rwish(1,eta0+m, solve( S0+t(GAMMA)%*(GAMMA)) )

  ##update sigma_alpha^2
  s2 <- var(Alpha)
  sigma2.alpha.m <- 1/nu.m*(nu.0 * sigma2.alpha.0 + (m-1)*s2 + kappa.0*m*(mean(Alpha)-mu.alpha.0)^2/kappa.m)
  sigma2.alpha <- 1/rgamma(1,shape=nu.m/2, rate=nu.m/2*sigma2.alpha.m)
  SIGMA2.ALPHA.post <- c(SIGMA2.ALPHA.post,sigma2.alpha)

  ##update mu_alpha
  mu.alpha <- rnorm(1,sum(Alpha)/kappa.m, sqrt(sigma2.alpha/kappa.m))
  MU.ALPHA.post <- c(MU.ALPHA.post, mu.alpha)

  ## update Alpha
  K_alpha = rep(0.5,m)
  K_alpha[2] <- 1
  K_alpha[6] <- 0.1
  K_alpha[9] <- 0.1
  K_alpha[11] <- 0.1
  K_alpha[13] <- 0.05
  K_alpha[14] <- 0.3
  K_alpha[22] <- 0.1
  K_alpha[23] <- 0.1
  K_alpha[40] <- 0.1
  K_alpha[42] <- 0.1
  K_alpha[48] <- 0.1
  K_alpha[49] <- 0.1
  K_alpha[56] <- 0.1
  for(j in 1:m)
  {
    # beta proposed (beta^(*)) from multivariate normal (beta_j^(s), V_J^(s))
    alpha.p <-rnorm(1,Alpha[j],sqrt(K_alpha[j]*sigma2.alpha))
    lr<-sum(
      wgt.list[[j]] * dgenpois2(y.list[[j]],
        meanpar=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],THETA)), disppar=alpha.p),
      -wgt.list[[j]] * dgenpois2(y.list[[j]],
        meanpar=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],THETA)), disppar=Alpha[j]),
      dnorm(alpha.p, sqrt(sigma2.alpha), log=TRUE),
      -dnorm( Alpha[j], sqrt(sigma2.alpha), log=TRUE)
    )

    if( is.Numeric(lr) && log(runif(1))<lr ) {
      ACCEPT.Alpha.count[j] <- ACCEPT.Alpha.count[j]+1
    }
  }
}

```

```

        Alpha[j]<-alpha.p
    }
}

##update theta
# theta proposed
K_theta <- 0.05
theta.p<-t(rmvnorm(1,THETA,K_theta*LAMBDA))
lr<-0
for(j in 1:m)
{
    lr <- lr+sum(
        wgt.list[[j]] * dgenpois2(y.list[[j]],
            meanpar=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],theta.p)), disppar=Alpha[j],
        -wgt.list[[j]] * dgenpois2(y.list[[j]],
            meanpar=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],THETA)), disppar=Alpha[j],
    )
}
lr <- lr + ldmvnorm(t(theta.p),mu0,LAMBDA) - ldmvnorm(t(THETA),mu0,LAMBDA)
#print("lr is")
#print(lr)
if( is.Numeric(lr) && log(runif(1))<lr ) {# update theta
    ACCEPT.Theta.count<- ACCEPT.Theta.count+1
    THETA<-theta.p
}

##update gamma
Sigma<-solve(iSigma) ; dSigma<-det(Sigma)
K_gamma = rep(0.1,m)
K_gamma[9] = 0.05
K_gamma[11] = 0.05
K_gamma[13] = 0.05

for(j in 1:m)
{
    # gamma proposed
    gamma.p<-t(rmvnorm(1,GAMMA[j,],K_gamma[j]*Sigma))
    lr<-sum(
        wgt.list[[j]] * dgenpois2(y.list[[j]],
            meanpar=exp(matmul(Z.list[[j]],gamma.p) + matmul(X.list[[j]],THETA)), disppar=Alpha[j],
        -wgt.list[[j]] * dgenpois2(y.list[[j]],
            meanpar=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],THETA)), disppar=Alpha[j],
        ldmvnorm( t(gamma.p),rep(0,q),Sigma,iSigma=iSigma,dSigma=dSigma ),
        -ldmvnorm( t(GAMMA[j,]),rep(0,q),Sigma,iSigma=iSigma,dSigma=dSigma )
    )

    if( is.Numeric(lr) && log(runif(1))<lr ) { # update gamma
        #print(sprintf("Gamma %d Accepted", j ))
        ACCEPT.count[j] <- ACCEPT.count[j]+1
        GAMMA[j,]<-gamma.p
    }
}
}

```

```

##store some output
if((s %% thin == 0) & (s > B)) # saving every thinth value
{
  cat("Current iteration index = ", s, "\n")
  # Print current acceptance rate
  cat("Current acceptance rate of Theta = ", round((ACCEPT.Theta.count/s)*100, digits = 2), "\n")
  print("Current acceptance rate of Alpha:")
  print(round(ACCEPT.Alpha.count/s*100,digits=2))
  print("Current acceptance rate of Gamma:")
  print(round(ACCEPT.count/s*100,digits=2))
  #print("Alpha is:")
  #print(round(Alpha,digits=4))
  #print("Sigam2 is:")
  #print(sigma2.alpha)
  # save thinned coefficients
  THETA.post<-rbind(THETA.post,t(THETA))
  SIGMA.post<-rbind(SIGMA.post,c(Sigma))
  ALPHA.post<-rbind(ALPHA.post,c(Alpha))
  GAMMA.post[[length(GAMMA.post)+1]]<-GAMMA
}
}
end.time <- Sys.time()
#####simulation#####

```

Save and load results

```

RESULTPATH = "./data_output/123/"
save(THETA.post, file=paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
save(SIGMA.post, file=paste(RESULTPATH,"SIGMA.post.S=",S,".RData",sep=""))
save(ALPHA.post, file=paste(RESULTPATH,"ALPHA.post.S=",S,".RData",sep=""))
save(ACCEPT.count, file=paste(RESULTPATH,"ACCEPT.count.S=",S,".RData",sep=""))
save(ACCEPT.Alpha.count, file=paste(RESULTPATH,"ACCEPT.Alpha.count.S=",S,".RData",sep=""))
save(ACCEPT.Theta.count, file=paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))
save(GAMMA, file=paste(RESULTPATH,"GAMMA.S=",S,".RData",sep=""))
save(GAMMA.post, file=paste(RESULTPATH,"GAMMA.post.S=",S,".RData",sep=""))
save(MU.ALPHA.post, file=paste(RESULTPATH,"MU.ALPHA.post.S=",S,".RData",sep=""))
save(SIGMA2.ALPHA.post, file=paste(RESULTPATH,"SIGMA2.ALPHA.post.S=",S,".RData",sep=""))

```

```

RESULTPATH = "./data_output/123/"
load(paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"SIGMA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ALPHA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.count.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.Alpha.count.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"GAMMA.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"GAMMA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"MU.ALPHA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"SIGMA2.ALPHA.post.S=",S,".RData",sep=""))

```

Plot results

Assess the convergence of the Markov chain

```
## stationarity plot - boxplot
stationarity.plot<-function(x,...){
  S<-length(x)
  scan<-1:S
  ng<-min( round(S/100),10)
  group<-S*ceiling( ng*scan/S) /ng
  boxplot(x~group,...)
}

# convert the result (from list) into a 47x(S-B) matrix
thin2 = 1
THETA.post.mat <- NULL
for (i in c(1:dim(THETA.post)[1])){
  if(i %% thin2 == 0){
    # Thin again
    THETA.post.mat <- cbind(THETA.post.mat, as.matrix(unlist(t(THETA.post)[,i])) )
  }
}
GAMMA.post.mat <- list()
j <- 1
for (dis in district_codes){
  gamma_j <- NULL
  for (s in c(1:length(GAMMA.post))){
    if(i %% thin2 == 0){
      # Thin again
      gamma_j <- rbind(gamma_j, GAMMA.post[[s]][j, ])
    }
  }
  GAMMA.post.mat[[j]] <- gamma_j
  j<-j+1
}
# GAMMA.post.mat Usage: matmul(Z, apply(GAMMA.post.mat[[9]],2,mean))
SIGMA.post.mat <- NULL
for (i in c(1:dim(SIGMA.post)[1])){
  if(i %% thin2 == 0){
    # Thin again
    SIGMA.post.mat <- cbind(SIGMA.post.mat, SIGMA.post[i,] )
  }
}

ACCEPT.count/S

## [1] 0.3772000 0.4498545 0.3993091 0.3021455 0.3256000 0.2565273 0.2826182 0.3019818
## [9] 0.3129273 0.3790909 0.3129455 0.2902545 0.3181273 0.2416182 0.3155273 0.3361455
## [17] 0.3000182 0.3142545 0.3857455 0.3351455 0.4102000 0.3251455 0.2759636 0.4341818
## [25] 0.3065091 0.4214545 0.3333636 0.2776909 0.3080727 0.3084727 0.3208000 0.3647091
## [33] 0.3662182 0.3911818 0.3425091 0.4246727 0.3399455 0.3630000 0.2363091 0.2903091
## [41] 0.4230727 0.3081455 0.3071455 0.3428727 0.3256545 0.3120364 0.3138545 0.2382182
## [49] 0.3076182 0.3790182 0.3403818 0.4015636 0.3089091 0.3661818 0.4199273 0.2502364
## [57] 0.3849273 0.3063636 0.2753818 0.3694000 0.2727091 0.2712545 0.2500545 0.3566000
```

```

#stationarity.plot(THETA.post[, 1],xlab="iteration",ylab=expression(theta[1]))
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
for (i in 1:p) {
stationarity.plot(THETA.post.mat[i, ],xlab="iteration",ylab=expression(theta))
}
dev.print(pdf,
  paste(IMAGEOUT,"THETA.post.boxplot.pdf",sep=""),
  width=16,height=12)

```

```

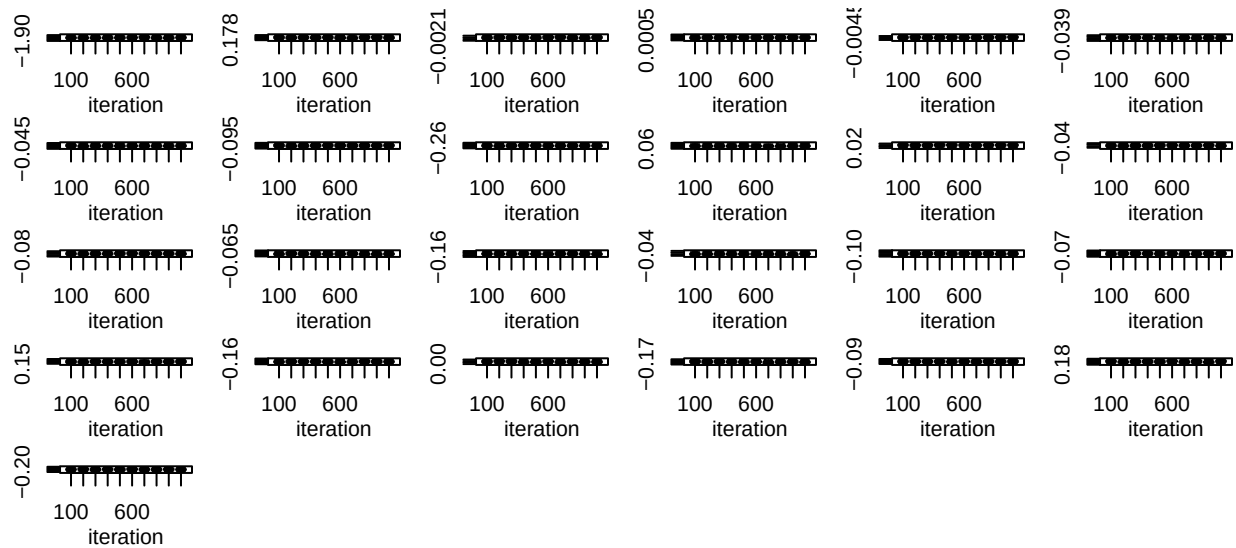
## pdf
## 2

```

```

# Traceplot
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))

```



```

for (i in 1:p) {
plot(seq(1,dim(THETA.post.mat)[2],by=1), THETA.post.mat[i,], type = "l", xlab = "Iterations", ylab = expression(theta[i]))
}
dev.print(pdf,
  paste(IMAGEOUT,"THETA.post.pdf",sep=""),
  width=16,height=12)

```

```

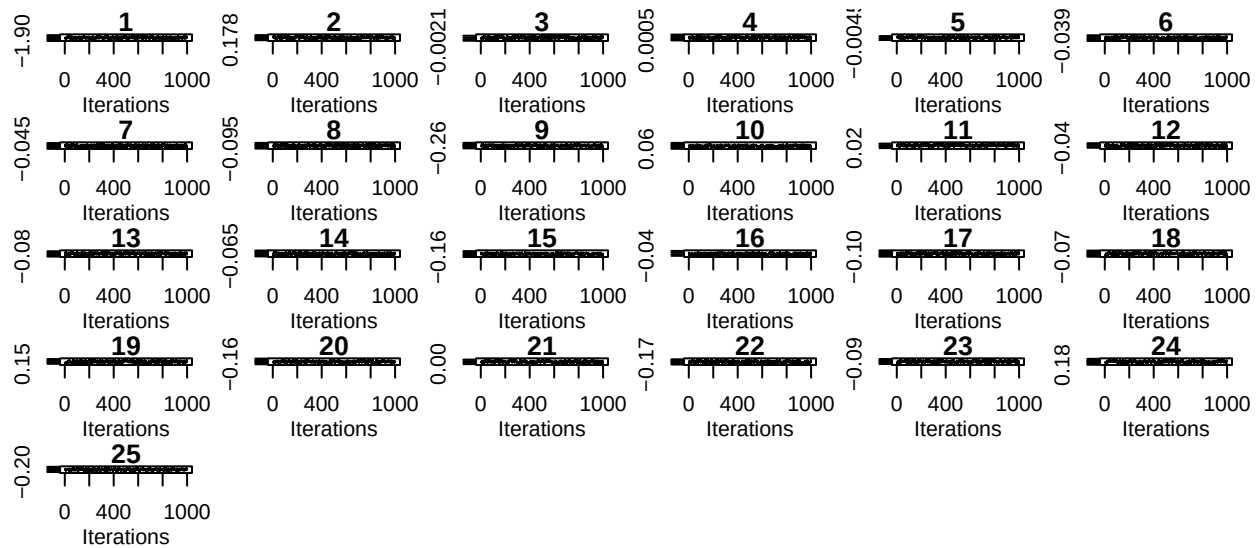
## pdf
## 2

```

```

# ACF plot
library(coda)
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))

```



```
for (i in 1:p){
  esize <- effectiveSize(THETA.post.mat[i,])
  acf_plot <- acf(THETA.post.mat[i,], plot = FALSE)
  t <- paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=t)
}
dev.print(pdf,
  paste(IMAGEOUT,"ACF.THETA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

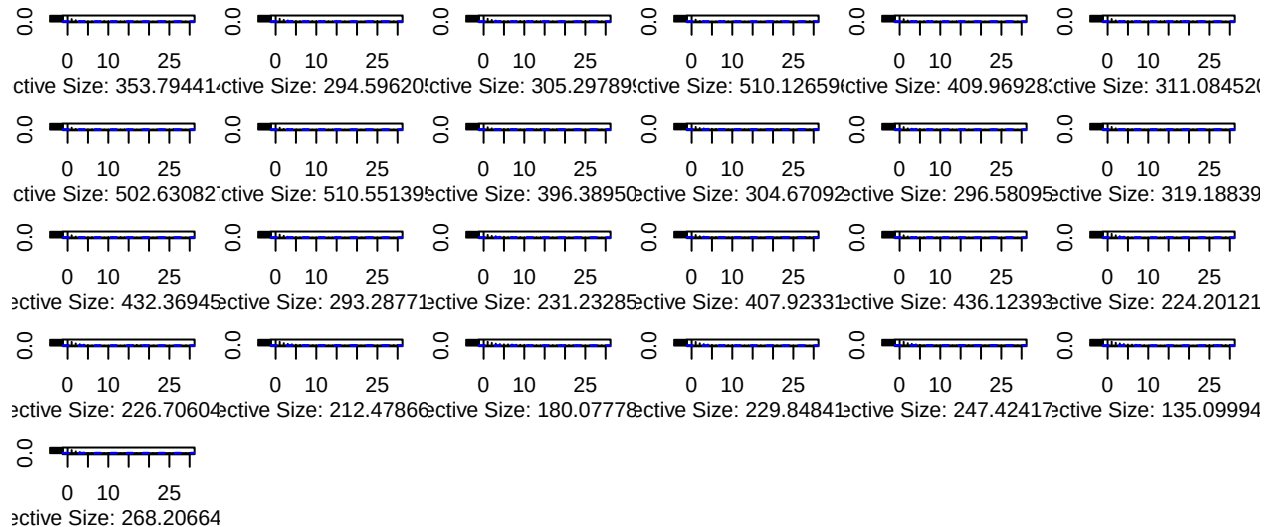
```
apply(SIGMA.post, 2, effectiveSize)
```

```
##      [1] 161.88134 131.22928 132.89374 105.43305 112.83719 105.55875 102.93819 144.04993
##      [9] 105.55604 120.89435 84.07905 69.31581 53.56640 132.62627 109.72564 153.26134
##     [17] 109.24318 131.22928 98.74670 96.87677 117.52612 93.85785 94.97207 90.90317
##     [25] 111.25732 90.54638 153.46257 73.96498 70.65136 51.59101 117.03919 49.62164
##     [33] 161.85210 107.08495 132.89374 96.87677 0.00000 127.32635 109.08673 84.02464
##     [41] 93.26726 99.71990 70.42940 176.38938 83.53452 76.12503 60.16901 104.50671
##     [49] 54.41528 142.54516 121.99699 105.43305 117.52612 127.32635 152.58078 140.11887
##     [57] 119.53943 127.85749 108.11623 134.13968 209.27126 107.56555 175.29635 169.46969
##     [65] 100.63342 106.85621 97.31421 108.91423 112.83719 93.85785 109.08673 140.11887
##     [73] 91.90259 155.06051 116.74911 76.22228 111.90933 114.86891 93.31217 187.01796
##     [81] 145.01532 113.31763 60.77587 75.68827 219.15550 105.55875 94.97207 84.02464
##     [89] 119.53943 155.06051 162.07620 104.68362 146.39530 175.82956 206.44935 138.54388
##     [97] 130.76923 144.12342 174.33901 110.91127 201.70973 119.06934 102.93819 90.90317
##    [105] 93.26726 127.85749 116.74911 104.68362 84.34521 110.24405 150.49575 192.65506
##    [113] 126.67076 130.79642 196.24685 127.12543 116.09306 100.74431 164.67942 144.04993
##    [121] 111.25732 99.71990 108.11623 76.22228 146.39530 110.24405 170.60012 191.68683
##    [129] 182.10215 81.89665 110.40883 119.93185 198.52691 86.94890 128.82580 125.13170
##    [137] 105.55604 90.54638 70.42940 134.13968 111.90933 175.82956 150.49575 191.68683
##    [145] 163.23088 124.26431 115.69371 143.94420 150.30260 148.19147 71.20301 174.01205
##    [153] 158.75560 120.89435 153.46257 176.38938 209.27126 114.86891 206.44935 192.65506
##    [161] 182.10215 124.26431 212.04394 88.48510 108.93058 181.02119 280.51240 96.64556
##    [169] 134.38718 245.29305 84.07905 73.96498 83.53452 107.56555 93.31217 138.54388
```

```
## [177] 126.67076 81.89665 115.69371 88.48510 121.65445 134.67040 115.16378 129.85651
## [185] 76.71629 86.30990 170.43167 69.31581 70.65136 76.12503 175.29635 187.01796
## [193] 130.76923 130.79642 110.40883 143.94420 108.93058 134.67040 106.87024 67.18105
## [201] 96.86014 197.37753 62.58986 126.15585 53.56640 51.59101 60.16901 169.46969
## [209] 145.01532 144.12342 196.24685 119.93185 150.30260 181.02119 115.16378 67.18105
## [217] 98.37090 100.87899 134.54223 121.60413 151.36638 132.62627 117.03919 104.50671
## [225] 100.63342 113.31763 174.33901 127.12543 198.52691 148.19147 280.51240 129.85651
## [233] 96.86014 100.87899 192.10588 78.46552 159.58557 144.92578 109.72564 49.62164
## [241] 54.41528 106.85621 60.77587 110.91127 116.09306 86.94890 71.20301 96.64556
## [249] 76.71629 197.37753 134.54223 78.46552 94.49480 99.13066 146.56539 153.26134
## [257] 161.85210 142.54516 97.31421 75.68827 201.70973 100.74431 128.82580 174.01205
## [265] 134.38718 86.30990 62.58986 121.60413 159.58557 99.13066 122.46444 186.07063
## [273] 109.24318 107.08495 121.99699 108.91423 219.15550 119.06934 164.67942 125.13170
## [281] 158.75560 245.29305 170.43167 126.15585 151.36638 144.92578 146.56539 186.07063
## [289] 161.03128
```

```
# index of sigma
q<-dim(Z)[2] + 1# number of columns in Z + 1 (intercept)
ind_sigma <- NULL
for (i in 1:q){
  ind_sigma <- c(ind_sigma,(i-1)*q+i)
}
```

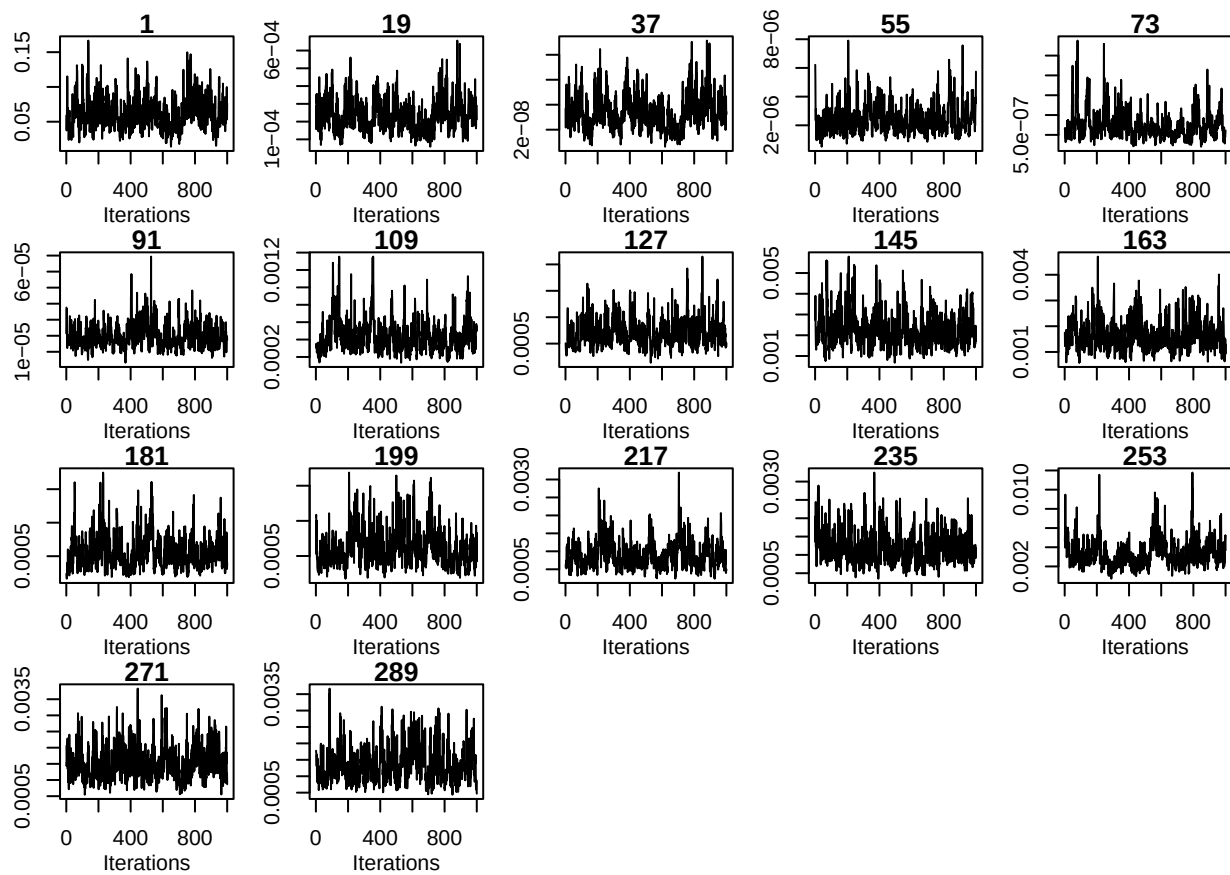
```
par(mfrow=c(4,5))
```



```
for(i in ind_sigma){
  plot(seq(1,dim(SIGMA.post.mat)[2],by=1), SIGMA.post.mat[i,], type = "l", xlab = "Iterations", ylab = exp)
}
dev.print(pdf,
  paste(IMAGEOUT,"SIGMA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

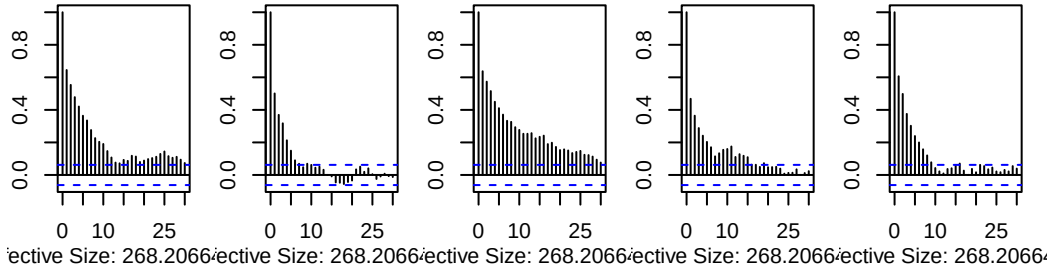
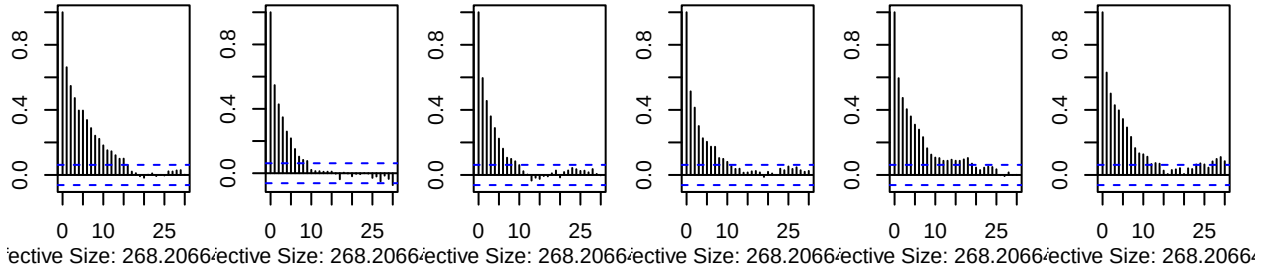
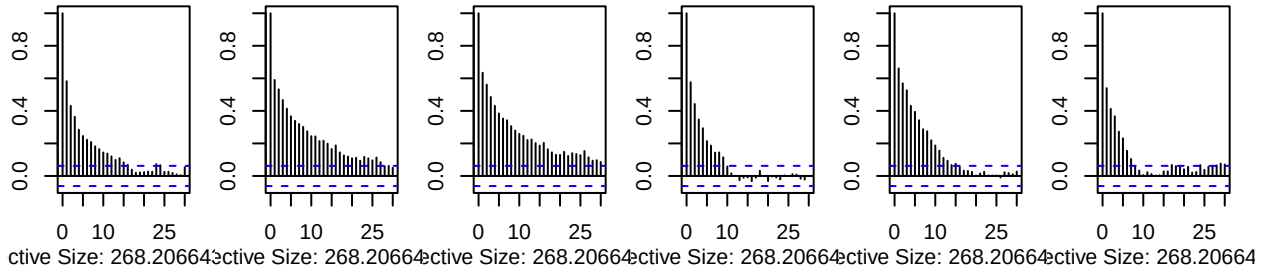
```
par(mfrow=c(3,6))
```

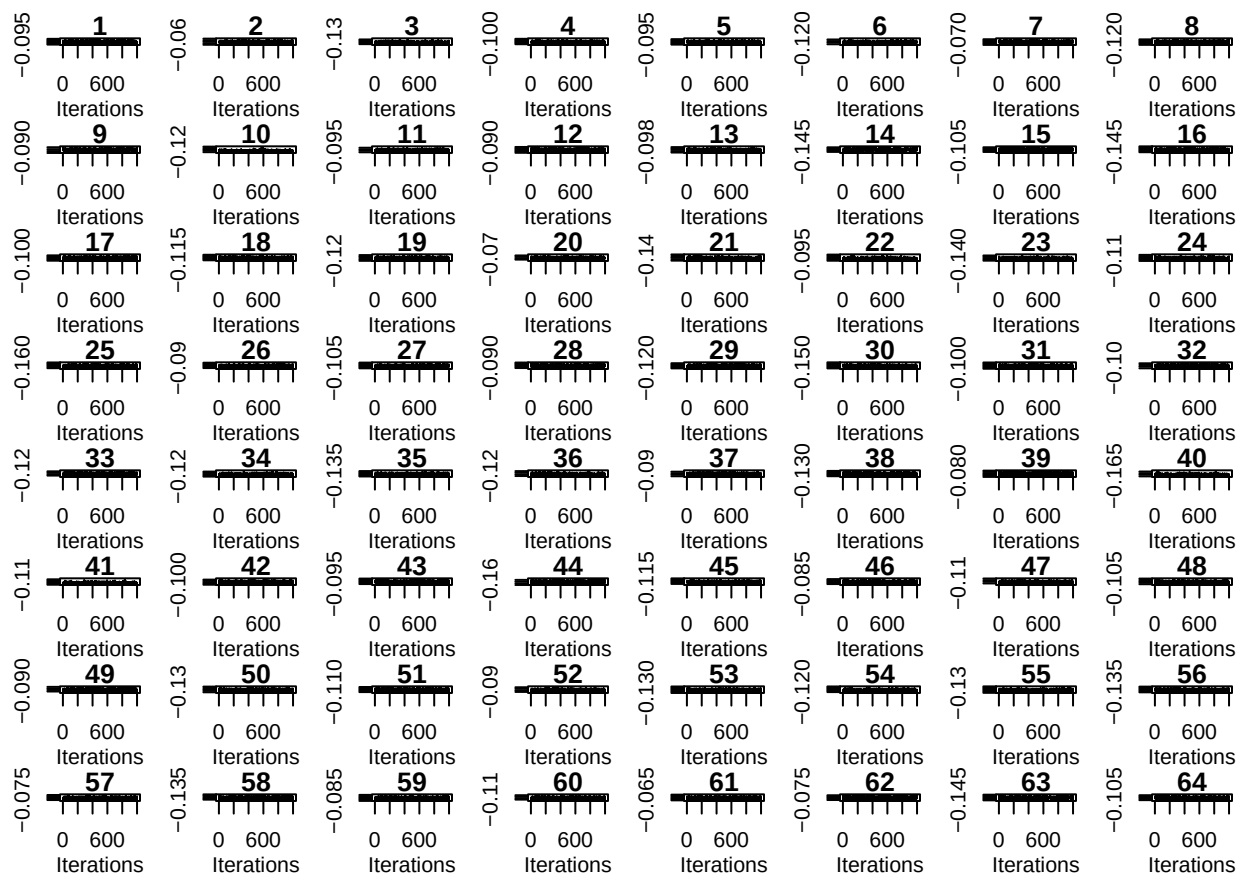
```
for(i in ind_sigma){
  acf_plot <- acf(SIGMA.post.mat[i,], plot=FALSE)
  t=paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=expression(paste("acf(",sigma^2,")",sep="")))
}
dev.print(pdf,
  paste(IMAGEOUT,"ACF.SIGMA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

```
# Traceplot of alpha
par(mfrow = c(8, 8), mar=c(3,2,1,1),mgp=c(2,1,0))
```



```
for (i in 1:m) {
plot(seq(1,dim(ALPHA.post)[1],by=1), ALPHA.post[,i], type = "l", xlab = "Iterations", ylab = expression
}
```



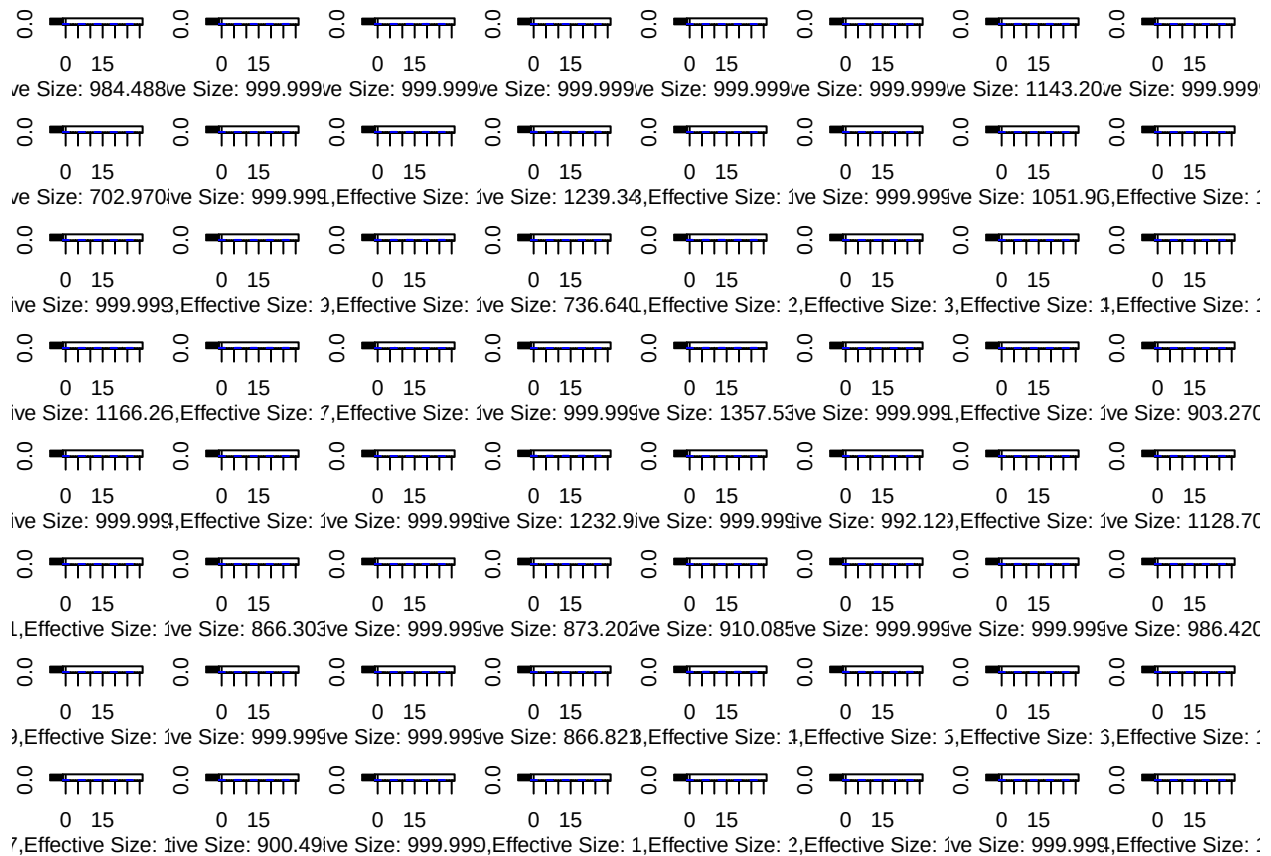
```
dev.print(pdf,
  paste(IMAGEOUT, "ALPHA.post.pdf", sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

```
apply(ALPHA.post, 2, effectiveSize)
```

```
## [1] 984.4881 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000 1143.2039 1000.0000
## [9] 702.9705 1000.0000 1000.0000 1239.3425 1000.0000 1000.0000 1000.0000 1051.9047 1000.0000
## [17] 1000.0000 1000.0000 1000.0000 736.6403 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000
## [25] 1166.2676 1000.0000 1000.0000 1000.0000 1000.0000 1357.5383 1000.0000 1000.0000 903.2702
## [33] 1000.0000 1000.0000 1000.0000 1232.9227 1000.0000 992.1227 1000.0000 1128.7070
## [41] 1000.0000 866.3032 1000.0000 873.2021 910.0859 1000.0000 1000.0000 1000.0000 986.4202
## [49] 1000.0000 1000.0000 1000.0000 866.8215 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000
## [57] 1000.0000 900.4957 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000 1000.0000
```

```
# ACF plot
library(coda)
par(mfrow = c(8, 8), mar=c(3,2,1,1),mgp=c(2,1,0))
for (i in 1:m){
  esize <- effectiveSize(ALPHA.post[,i])
  acf_plot <- acf(ALPHA.post[,i], plot = FALSE)
  t <- paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=t)
}
```

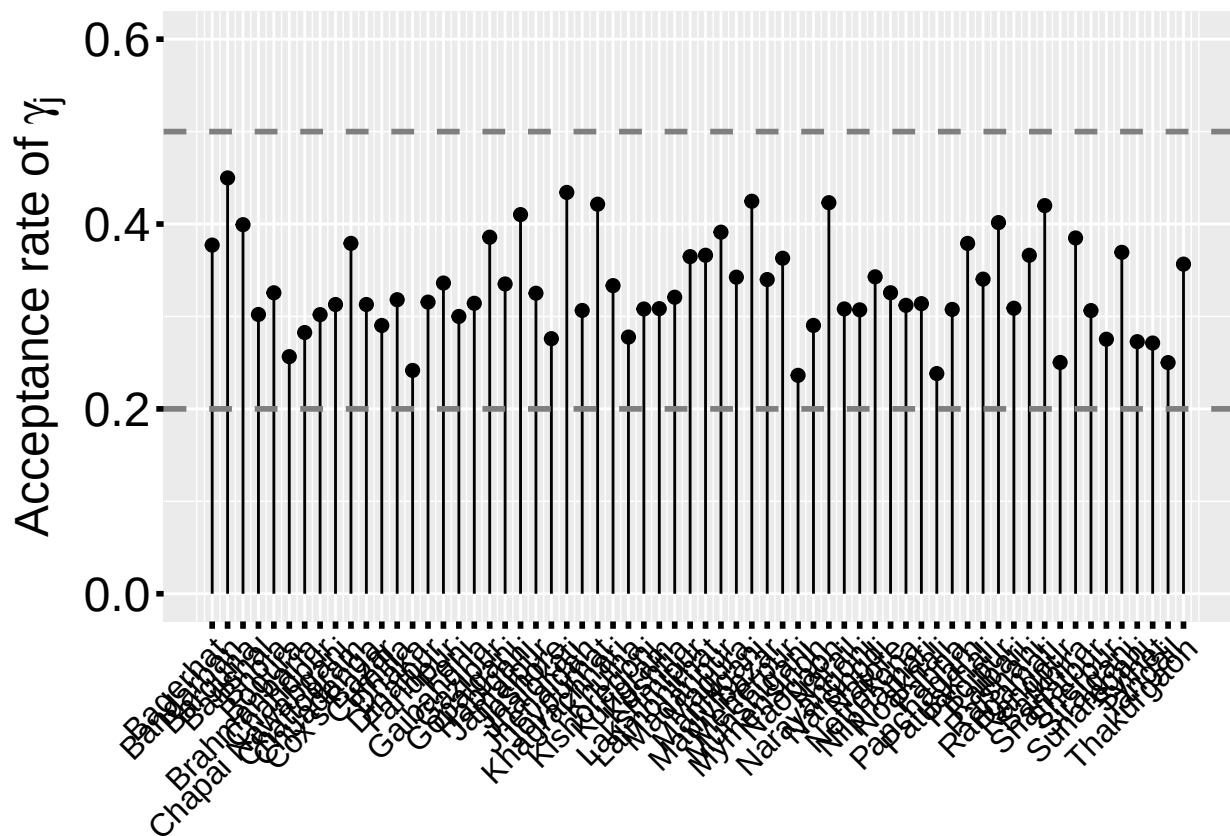


```
dev.print(pdf,
  paste(IMAGEOUT, "ACF.ALPHA.post.pdf", sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

The acceptance rate of each gamma

```
library(ggplot2)
index <- 1:m
df <- data.frame(x = 1:m,
  L = rep(0,m),
  U = ACCEPT.count/S)
abc <- ggplot(df, aes(x = x, y = U)) +
  labs(y = expression(paste("Acceptance rate of ", gamma[j], sep="")), x = NULL) +
  geom_point(size = 2) +
  geom_linerange(aes(ymin = L)) +
  geom_hline(yintercept = 0.2, color = "gray50", linetype = "dashed", size = 1) +
  geom_hline(yintercept = 0.5, color = "gray50", linetype = "dashed", size = 1) +
  scale_y_continuous(limits = c(0, 0.6)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
    axis.ticks.x = element_line(color = "black", linewidth = 1),
    axis.text.y = element_text(size = 18, color = "black"),
    axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```

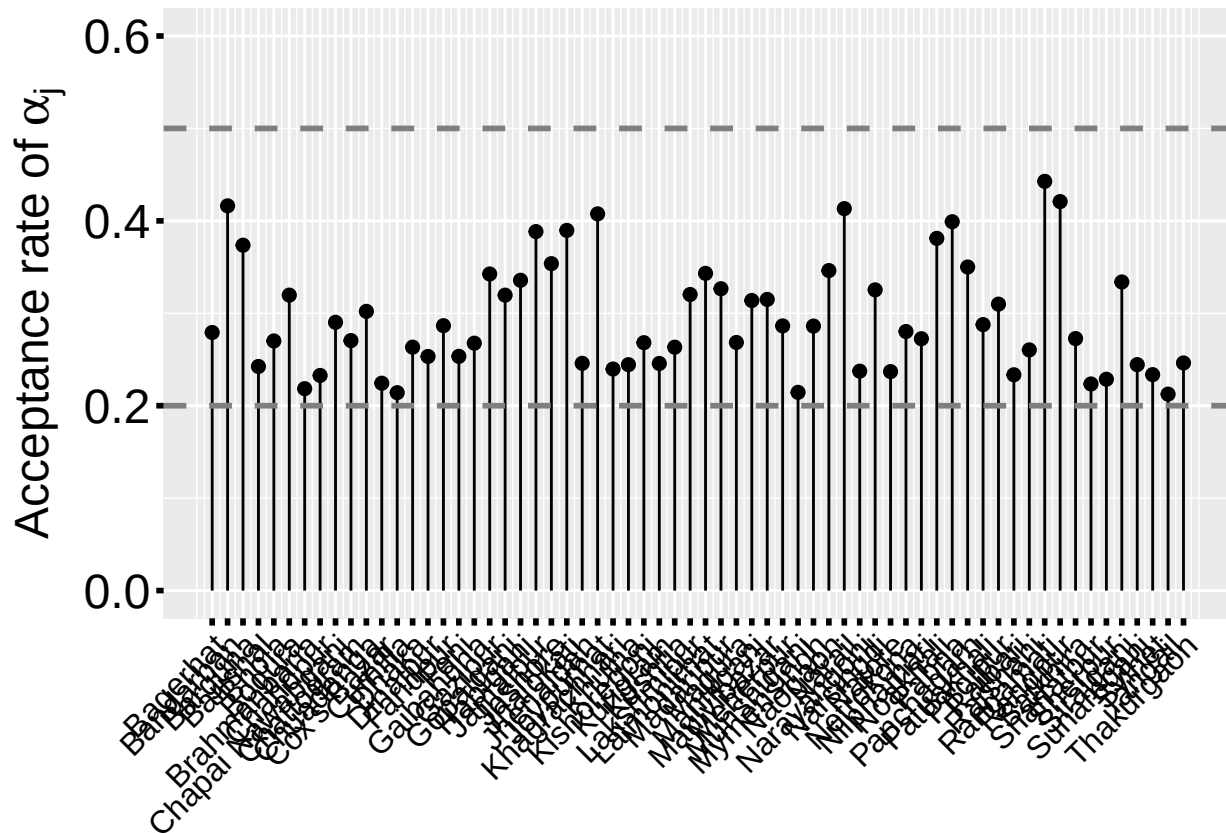


```
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal =
dev.print(pdf, paste(IMAGEOUT, "ACCEPT.rate.GAMMA.pdf",sep=""),width=16,height=8)
```

```
## pdf
## 2
```

The acceptance rate of each alpha

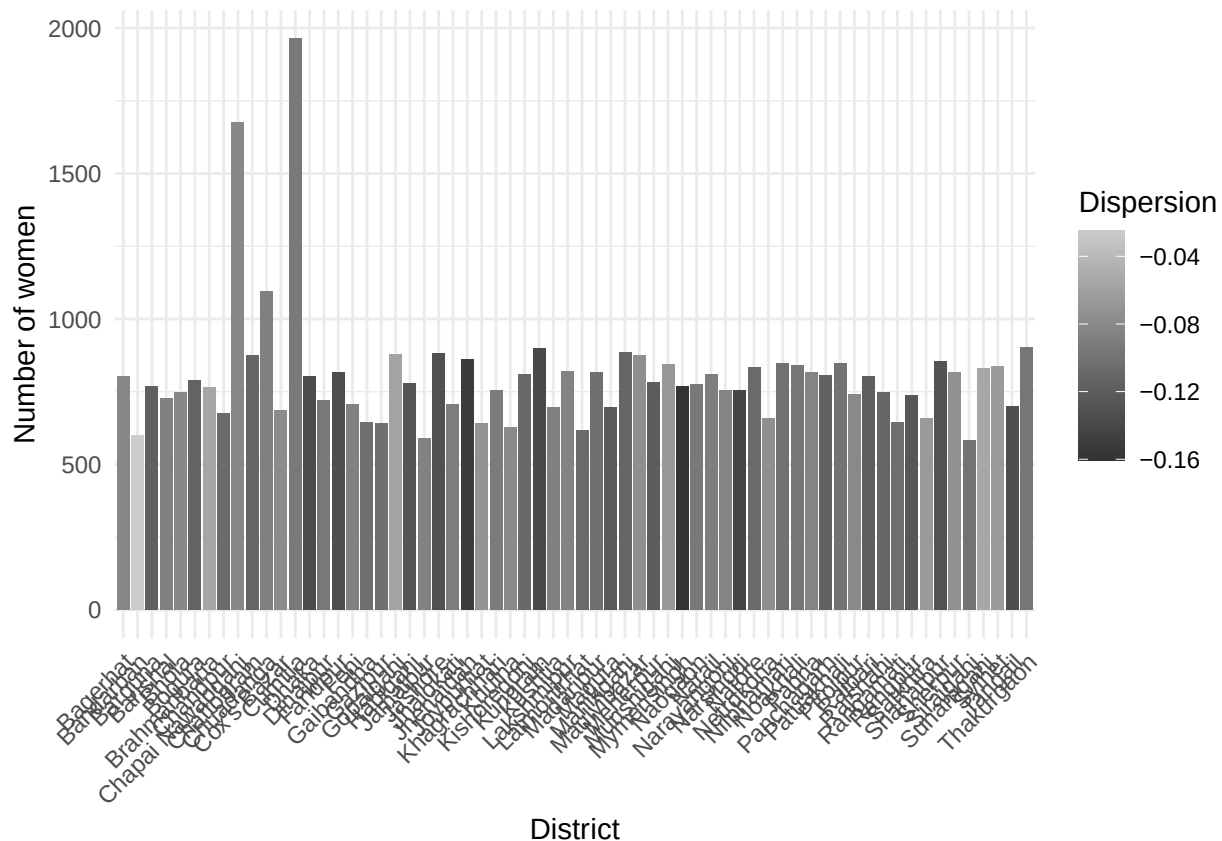
```
index <- 1:m
df <- data.frame(x = 1:m,
                 L = rep(0,m),
                 U = ACCEPT.Alpha.count/S)
abc <- ggplot(df, aes(x = x, y = U)) +
  labs(y = expression(paste("Acceptance rate of ",alpha[j],sep="")), x = NULL) +
  geom_point(size = 2) +
  geom_linerange(aes(ymin = L)) +
  geom_hline(yintercept = 0.2, color = "gray50", linetype = "dashed", size = 1) +
  geom_hline(yintercept = 0.5, color = "gray50", linetype = "dashed", size = 1) +
  scale_y_continuous(limits = c(0, 0.6)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal =
dev.print(pdf, paste(IMAGEOUT, "ACCEPT.rate.ALPHA.pdf",sep=""),width=16,height=8)
```

```
## pdf
## 2
```

```
library(ggplot2)
alpha.post <- apply(ALPHA.post, 2, mean)
district_demographic <- data.frame("name" = district_names, "count"=unlist(N), "Dispersion"=alpha.post)
ggplot(district_demographic, aes(x = name, y = count, fill = Dispersion))+
  scale_fill_gradient(low="gray20", high="gray80") +
  geom_bar(stat="identity", position="dodge")+
  labs(
    x = "District",
    y = "Number of women") +
  scale_x_discrete(
    labels = district_names
  ) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
dev.print(pdf,
  paste(IMAGEOUT,"NumofRecords.pdf",sep=""),
  width=16,height=10)
```

```
## pdf
## 2
```

Dispersion Test

```
fit.mle <- glm(y~., data = data.frame(yX), family=poisson)
library(AER)
sink(paste(IMAGEOUT,"dispersion.txt",sep=""))
dispersiontest(fit.mle, alternative = "less")
sink()
```

```
# Mean and variance for y in each district:
nu_est <- rep(0,m)
for (dis in seq(1,m,1)){
  cat("District: ",district_names[dis])
  y.perdistrict<- y.list[[dis]]
  cat(", length(y): ", length(y.perdistrict))
  X.perdistrict <- X.list[[dis]]
  yX.perdistrict <- cbind(y.perdistrict,X.perdistrict)
  cat(" ,mean: ")
  cat(mean(y.perdistrict))
  cat(" ,var: ")
  print(var(y.perdistrict))
}
```

}

```
## District: Bagerhat, length(y): 804 ,mean: 2.079602 ,var: [1] 1.639982
## District: Bandarban, length(y): 600 ,mean: 2.796667 ,var: [1] 3.467769
## District: Barguna, length(y): 770 ,mean: 2.027273 ,var: [1] 1.558423
## District: Barishal, length(y): 726 ,mean: 2.374656 ,var: [1] 2.325646
## District: Bhola, length(y): 749 ,mean: 2.567423 ,var: [1] 2.719044
## District: Bogura, length(y): 790 ,mean: 2.065823 ,var: [1] 1.303647
## District: Brahmanbaria, length(y): 765 ,mean: 2.901961 ,var: [1] 3.669695
## District: Chandpur, length(y): 677 ,mean: 2.33678 ,var: [1] 1.987003
## District: Chapai Nawabganj, length(y): 1675 ,mean: 2.403582 ,var: [1] 2.405722
## District: Chattogram, length(y): 876 ,mean: 2.031963 ,var: [1] 1.386406
## District: Chuadanga, length(y): 1094 ,mean: 2.501828 ,var: [1] 2.718661
## District: Cox's Bazar, length(y): 686 ,mean: 2.845481 ,var: [1] 3.228644
## District: Cumilla, length(y): 1963 ,mean: 1.868569 ,var: [1] 1.496478
## District: Dhaka, length(y): 804 ,mean: 2.125622 ,var: [1] 1.377724
## District: Dinajpur, length(y): 720 ,mean: 2.356944 ,var: [1] 2.032358
## District: Faridpur, length(y): 816 ,mean: 2.210784 ,var: [1] 1.643859
## District: Feni, length(y): 707 ,mean: 2.356436 ,var: [1] 2.082405
## District: Gaibandha, length(y): 644 ,mean: 1.807453 ,var: [1] 1.580287
## District: Gazipur, length(y): 642 ,mean: 2.252336 ,var: [1] 1.836383
## District: Gopalganj, length(y): 878 ,mean: 2.725513 ,var: [1] 3.458208
## District: Habiganj, length(y): 779 ,mean: 1.765083 ,var: [1] 1.179962
## District: Jamalpur, length(y): 588 ,mean: 2.241497 ,var: [1] 1.904101
## District: Jashore, length(y): 882 ,mean: 1.955782 ,var: [1] 1.311323
## District: Jhalokati, length(y): 707 ,mean: 2.247525 ,var: [1] 1.999551
## District: Jhenaidah, length(y): 861 ,mean: 2.006969 ,var: [1] 1.2953
## District: Joypurhat, length(y): 642 ,mean: 2.390966 ,var: [1] 2.341447
## District: Khagrachhari, length(y): 755 ,mean: 1.97351 ,var: [1] 1.447573
## District: Khulna, length(y): 629 ,mean: 2.561208 ,var: [1] 2.558748
## District: Kishoregonj, length(y): 808 ,mean: 2.079208 ,var: [1] 1.71119
## District: Kurigram, length(y): 899 ,mean: 2.008899 ,var: [1] 1.371858
## District: Kushtia, length(y): 695 ,mean: 2.696403 ,var: [1] 2.825563
## District: Lakshmipur, length(y): 819 ,mean: 2.310134 ,var: [1] 2.013724
## District: Lalmonirhat, length(y): 618 ,mean: 2.433657 ,var: [1] 2.236272
## District: Madaripur, length(y): 817 ,mean: 2.226438 ,var: [1] 1.584692
## District: Magura, length(y): 696 ,mean: 2.014368 ,var: [1] 1.412743
## District: Manikganj, length(y): 885 ,mean: 1.945763 ,var: [1] 1.304747
## District: Maulvibazar, length(y): 874 ,mean: 2.498856 ,var: [1] 2.64891
## District: Meherpur, length(y): 782 ,mean: 2.132992 ,var: [1] 1.536708
## District: Munshiganj, length(y): 844 ,mean: 2.430095 ,var: [1] 2.648725
## District: Mymensingh, length(y): 769 ,mean: 1.778934 ,var: [1] 1.010962
## District: Naogaon, length(y): 775 ,mean: 2.291613 ,var: [1] 1.739141
## District: Narail, length(y): 811 ,mean: 2.02836 ,var: [1] 1.901664
## District: Narayanganj, length(y): 756 ,mean: 2.477513 ,var: [1] 2.490884
## District: Narsingdi, length(y): 754 ,mean: 1.923077 ,var: [1] 1.202574
## District: Natore, length(y): 835 ,mean: 2.261078 ,var: [1] 1.780677
## District: Netrokona, length(y): 657 ,mean: 2.636225 ,var: [1] 2.838503
## District: Nilphamari, length(y): 846 ,mean: 2.347518 ,var: [1] 2.101574
## District: Noakhali, length(y): 839 ,mean: 2.529201 ,var: [1] 2.59312
## District: Pabna, length(y): 816 ,mean: 2.234069 ,var: [1] 1.867844
## District: Panchagarh, length(y): 806 ,mean: 2.183623 ,var: [1] 1.511582
## District: Patuakhali, length(y): 847 ,mean: 2.14876 ,var: [1] 1.845456
## District: Pirojpur, length(y): 740 ,mean: 2.216216 ,var: [1] 1.934243
```



```
## District: Rajbari, length(y): 802 ,mean: 1.849127 ,var: [1] 1.244375
## District: Rajshahi, length(y): 746 ,mean: 2.241287 ,var: [1] 1.700092
## District: Rangamati, length(y): 646 ,mean: 2.126935 ,var: [1] 1.670684
## District: Rangpur, length(y): 739 ,mean: 2.198917 ,var: [1] 1.693441
## District: Satkhira, length(y): 657 ,mean: 2.541857 ,var: [1] 2.898017
## District: Shariatpur, length(y): 856 ,mean: 1.997664 ,var: [1] 1.396486
## District: Sherpur, length(y): 818 ,mean: 2.385086 ,var: [1] 2.486779
## District: Sirajganj, length(y): 581 ,mean: 2.364888 ,var: [1] 2.049386
## District: Sunamganj, length(y): 831 ,mean: 3.022864 ,var: [1] 3.925983
## District: Sylhet, length(y): 837 ,mean: 2.679809 ,var: [1] 3.373432
## District: Tangail, length(y): 700 ,mean: 1.897143 ,var: [1] 1.422882
## District: Thakurgaon, length(y): 901 ,mean: 2.259711 ,var: [1] 1.610253
```

```
# Dispersion test for each district
library(AER)
nu_est <- rep(0,m)
district_alpha <- rep(0,m)
district_sizes <- rep(0,m)
district_dispersion <- rep(0,m)
district_p <- rep(0,m)

for (dis in seq(1,m,1)){
  cat("District: ",district_names[dis])
  y.perdistrict<- y.list[[dis]]
  district_sizes[dis]<-length(y.perdistrict)
  cat(", length(y): ", district_sizes[dis])
  X.perdistrict<- X.list[[dis]]
  yX.perdistrict<- cbind(y.perdistrict,X.perdistrict)
  fit.mle.perdistrict<- glm(y.perdistrict~., data = data.frame(yX.perdistrict), family=poisson)
  test.perdistrict <- dispersiontest(fit.mle.perdistrict, alternative = "less")
  cat(", dispersion: ")
  district_dispersion[dis] <- test.perdistrict$estimate
  cat(district_dispersion[dis], ", p-value: ")
  district_p[dis] <- test.perdistrict$p.value
  print(district_p[dis])
  nu_est[dis] <- 1/test.perdistrict$estimate
  district_alpha[dis] <- test.perdistrict$estimate - 1
  #fit.glm.cmp.perstratum <- glm.cmp(formula = y.perstratum~., data = data.frame(yX.perstratum))
}
```

```
## District: Bagerhat, length(y): 804, dispersion: 0.4217749 , p-value: [1] 1.82494e-68
## District: Bandarban, length(y): 600, dispersion: 0.4781363 , p-value: [1] 4.9286e-46
## District: Barguna, length(y): 770, dispersion: 0.3780662 , p-value: [1] 2.411995e-111
## District: Barishal, length(y): 726, dispersion: 0.3774674 , p-value: [1] 1.476472e-131
## District: Bhola, length(y): 749, dispersion: 0.3833522 , p-value: [1] 4.261355e-132
## District: Bogura, length(y): 790, dispersion: 0.3540433 , p-value: [1] 7.803281e-175
## District: Brahmanbaria, length(y): 765, dispersion: 0.4638326 , p-value: [1] 3.413983e-77
## District: Chandpur, length(y): 677, dispersion: 0.4120742 , p-value: [1] 3.51463e-83
## District: Chapai Nawabganj, length(y): 1675, dispersion: 0.4417889 , p-value: [1] 3.885154e-192
## District: Chattogram, length(y): 876, dispersion: 0.3881319 , p-value: [1] 1.397441e-129
## District: Chuadanga, length(y): 1094, dispersion: 0.4173558 , p-value: [1] 8.730409e-137
## District: Cox's Bazar, length(y): 686, dispersion: 0.4038392 , p-value: [1] 9.931234e-108
## District: Cumilla, length(y): 1963, dispersion: 0.4244183 , p-value: [1] 4.529385e-211
## District: Dhaka, length(y): 804, dispersion: 0.3286623 , p-value: [1] 2.5425e-187
## District: Dinajpur, length(y): 720, dispersion: 0.3619559 , p-value: [1] 1.174323e-134
```

```

## District: Faridpur, length(y): 816, dispersion: 0.334319 , p-value: [1] 2.286972e-181
## District: Feni, length(y): 707, dispersion: 0.412622 , p-value: [1] 2.644688e-93
## District: Gaibandha, length(y): 644, dispersion: 0.4400188 , p-value: [1] 2.842291e-73
## District: Gazipur, length(y): 642, dispersion: 0.3905488 , p-value: [1] 2.854236e-102
## District: Gopalganj, length(y): 878, dispersion: 0.5677132 , p-value: [1] 1.032781e-43
## District: Habiganj, length(y): 779, dispersion: 0.3640835 , p-value: [1] 1.567277e-136
## District: Jamalpur, length(y): 588, dispersion: 0.3705351 , p-value: [1] 3.193756e-108
## District: Jashore, length(y): 882, dispersion: 0.3660053 , p-value: [1] 5.992225e-168
## District: Jhalokati, length(y): 707, dispersion: 0.3845198 , p-value: [1] 2.258918e-126
## District: Jhenaidah, length(y): 861, dispersion: 0.3375504 , p-value: [1] 6.32819e-201
## District: Joypurhat, length(y): 642, dispersion: 0.3970937 , p-value: [1] 2.501032e-116
## District: Khagrachhari, length(y): 755, dispersion: 0.3838446 , p-value: [1] 4.428415e-121
## District: Khulna, length(y): 629, dispersion: 0.447706 , p-value: [1] 8.258444e-67
## District: Kishoregonj, length(y): 808, dispersion: 0.3805787 , p-value: [1] 2.004794e-134
## District: Kurigram, length(y): 899, dispersion: 0.3633405 , p-value: [1] 3.345936e-191
## District: Kushtia, length(y): 695, dispersion: 0.3798917 , p-value: [1] 1.468937e-119
## District: Lakshmipur, length(y): 819, dispersion: 0.4064385 , p-value: [1] 1.781655e-87
## District: Lalmonirhat, length(y): 618, dispersion: 0.3775735 , p-value: [1] 1.192294e-107
## District: Madaripur, length(y): 817, dispersion: 0.3805426 , p-value: [1] 5.556962e-154
## District: Magura, length(y): 696, dispersion: 0.3500967 , p-value: [1] 2.09875e-136
## District: Manikganj, length(y): 885, dispersion: 0.3591526 , p-value: [1] 1.962115e-180
## District: Maulvibazar, length(y): 874, dispersion: 0.4607933 , p-value: [1] 6.681018e-88
## District: Meherpur, length(y): 782, dispersion: 0.3494916 , p-value: [1] 8.129163e-171
## District: Munshiganj, length(y): 844, dispersion: 0.5105164 , p-value: [1] 2.127214e-57
## District: Mymensingh, length(y): 769, dispersion: 0.3344305 , p-value: [1] 2.41314e-184
## District: Naogaon, length(y): 775, dispersion: 0.3688217 , p-value: [1] 5.31998e-142
## District: Narail, length(y): 811, dispersion: 0.4386499 , p-value: [1] 2.746241e-106
## District: Narayanganj, length(y): 756, dispersion: 0.438573 , p-value: [1] 7.977244e-80
## District: Narsingdi, length(y): 754, dispersion: 0.3777582 , p-value: [1] 1.050483e-127
## District: Natore, length(y): 835, dispersion: 0.3709035 , p-value: [1] 2.456347e-149
## District: Netrokona, length(y): 657, dispersion: 0.4521959 , p-value: [1] 8.56763e-72
## District: Nilphamari, length(y): 846, dispersion: 0.3883823 , p-value: [1] 3.543419e-128
## District: Noakhali, length(y): 839, dispersion: 0.378229 , p-value: [1] 4.003237e-156
## District: Pabna, length(y): 816, dispersion: 0.4364161 , p-value: [1] 1.135479e-96
## District: Panchagarh, length(y): 806, dispersion: 0.3446946 , p-value: [1] 2.311047e-176
## District: Patuakhali, length(y): 847, dispersion: 0.3854013 , p-value: [1] 4.367389e-138
## District: Pirojpur, length(y): 740, dispersion: 0.4025212 , p-value: [1] 7.101864e-107
## District: Rajbari, length(y): 802, dispersion: 0.3821182 , p-value: [1] 1.941809e-116
## District: Rajshahi, length(y): 746, dispersion: 0.3432802 , p-value: [1] 7.951682e-166
## District: Rangamati, length(y): 646, dispersion: 0.3675874 , p-value: [1] 1.148077e-124
## District: Rangpur, length(y): 739, dispersion: 0.3711617 , p-value: [1] 1.471603e-137
## District: Satkhira, length(y): 657, dispersion: 0.3941765 , p-value: [1] 5.225774e-103
## District: Shariatpur, length(y): 856, dispersion: 0.3317888 , p-value: [1] 2.058157e-189
## District: Sherpur, length(y): 818, dispersion: 0.513952 , p-value: [1] 2.687049e-47
## District: Sirajganj, length(y): 581, dispersion: 0.41096 , p-value: [1] 3.309462e-91
## District: Sunamganj, length(y): 831, dispersion: 0.5490187 , p-value: [1] 4.356253e-44
## District: Sylhet, length(y): 837, dispersion: 0.5073908 , p-value: [1] 2.74018e-62
## District: Tangail, length(y): 700, dispersion: 0.3855482 , p-value: [1] 1.529989e-121
## District: Thakurgaon, length(y): 901, dispersion: 0.3537309 , p-value: [1] 8.868065e-157

df.district <- data.frame(
  District = district_names,
  Alpha = district_alpha,
  Dispersion = district_dispersion,

```

```

P_value = district_p,
Size = district_sizes
)
write.csv(df.district, file=paste(IMAGEOUT,"df.district.csv",sep=""))

```

Posterior estimates with 95% credible intervals (CI) of the incidence rate ratio (IRR) for the explanatory variables

```

library(xtable)
CI_THETA <- round(t(apply(exp(THETA.post.mat), 1, quantile ,probs=c(0.50, 0.025, 0.975)))), digits = 5)
CI_THETA

```

##	50%	2.5%	97.5%
## (Intercept)	0.17699	0.16228	0.19399
## women_age	1.20221	1.19707	1.20754
## women_age2	0.99800	0.99793	0.99806
## husband_age	1.00174	1.00094	1.00250
## HH_age	0.99614	0.99570	0.99659
## women_agem	0.96383	0.96251	0.96516
## women_edu1	0.97062	0.96070	0.98041
## women_edu2	0.92364	0.91420	0.93367
## women_edu3	0.79364	0.77824	0.80820
## windex.Poorest	1.08618	1.07205	1.10103
## windex.Poor	1.03982	1.02829	1.05271
## windex.Rich	0.97446	0.96223	0.98677
## windex.Richest	0.94351	0.93043	0.95639
## media	0.94888	0.93933	0.95918
## other_wives	0.88841	0.86753	0.90708
## area	0.97102	0.96057	0.98226
## HH_sex	0.92630	0.91257	0.94009
## Barishal	0.96528	0.94306	0.98868
## Chattogram	1.18214	1.15989	1.20266
## Khulna	0.87573	0.85770	0.89188
## Mymensingh	1.03246	1.00719	1.06266
## Rajshahi	0.86526	0.84794	0.88423
## Rangpur	0.93707	0.91709	0.95529
## Sylhet	1.24180	1.21312	1.27167
## ethnicity	0.86045	0.83027	0.89078

```

sink(paste(IMAGEOUT,"CI_THETA.tex",sep=""))
xtable(CI_THETA, digits = 3)
sink()

```

```

# BIC_h for the Hierarchical Model

```

```

n<-length(y)
MLE <- 0
for(j in 1:m)
{
MLE <- MLE + sum(wgt.list[[j]] * dgenpois2(y.list[[j]], meanpar=exp(matmul(Z.list[[j]], apply(GAMMA.pos, 1, FUN=function(x){sum(x)})), digits=5))), digits=5)
}
q<-dim(Z)[2] + 1# number of columns in Z (13) + 1 (intercept)
p<-dim(X)[2] + 1# number of columns in X (20) + 1 (intercept)
BIC_h <- (q+1) * log(m) + (p-q+1) * log(n) - 2 * MLE
print(BIC_h)

```

```
## [1] 138224.8
```

Calculate the expected number of CEB

Expected Children Ever Born - Woman's age

```
Xs <- apply(X, 2, mean)
Xs <- matrix(rep(Xs, 35), nrow = 35, byrow = T)
Xs[,1] <- seq(15, 49, by = 1)
Xs[,2] <- (seq(15, 49, by = 1)^2)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,35),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs%*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 15:49){
  sample_size <- c(sample_size, sum(X[,1]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:49)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 49),range(c(0,qE)),type="n",xlab="Woman's age",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 49, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 49, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 49, by = 1), qE[3,],col="black",lwd=0.5)
```



```
dev.print(pdf,
  paste(IMAGEOUT, "E_women_age.pdf", sep=""),
  width=16, height=16)
```

```
## pdf
## 2
```

Expected Children Ever Born - Husband's age

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 76), nrow = 76, byrow = T)
Xs[,3] <- seq(15, 90, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,76),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

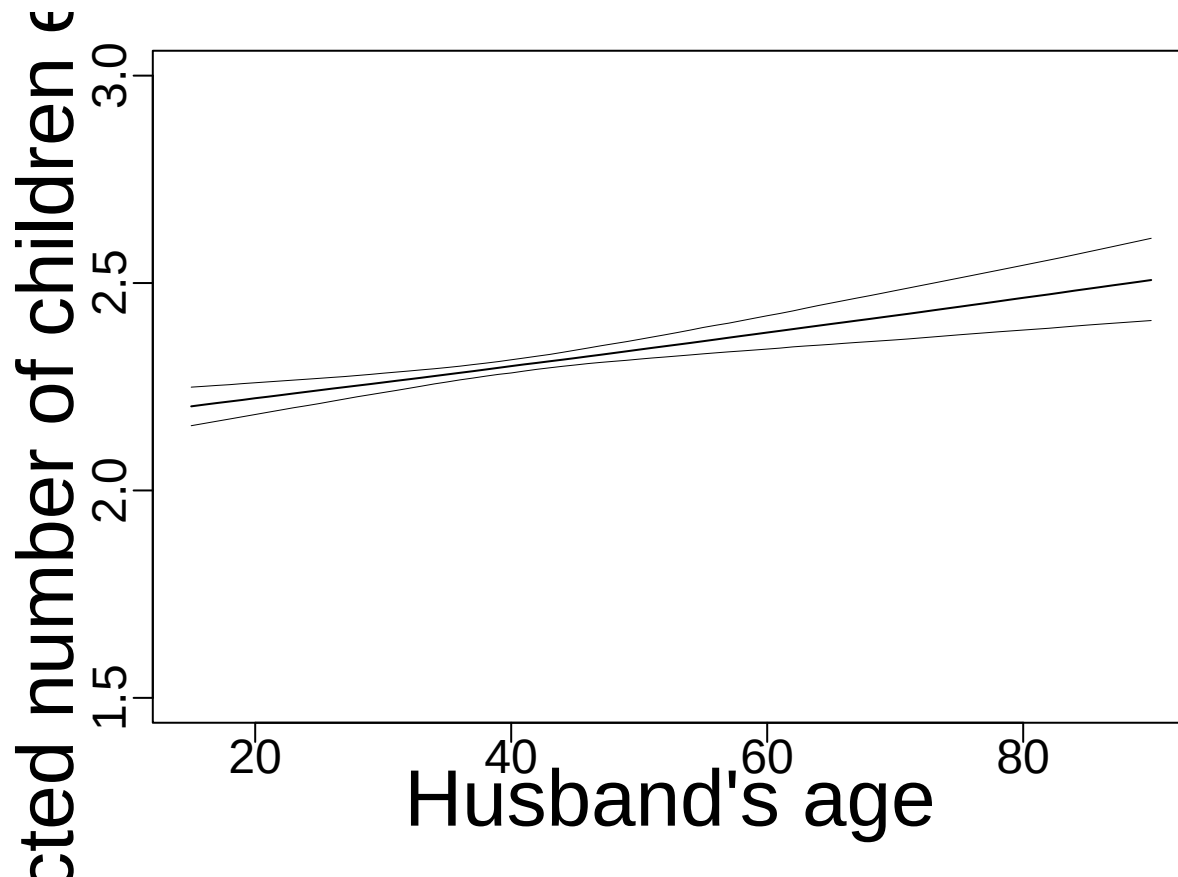
sample_size <- NULL
for (v in 15:90){
  sample_size <- c(sample_size, sum(X[,3]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:90)
write.csv(t(qE), paste(IMAGEOUT, "qE.husband_age.csv", sep=""))
```

```

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 90),range(c(0,qE)),type="n",xlab="Husband's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 90, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 90, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 90, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT, "E_husband_age.pdf", sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

Expected Children Ever Born - Household head's age

```

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 80), nrow = 80, byrow = T)
Xs[,4] <- seq(16, 95, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,80),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL

```

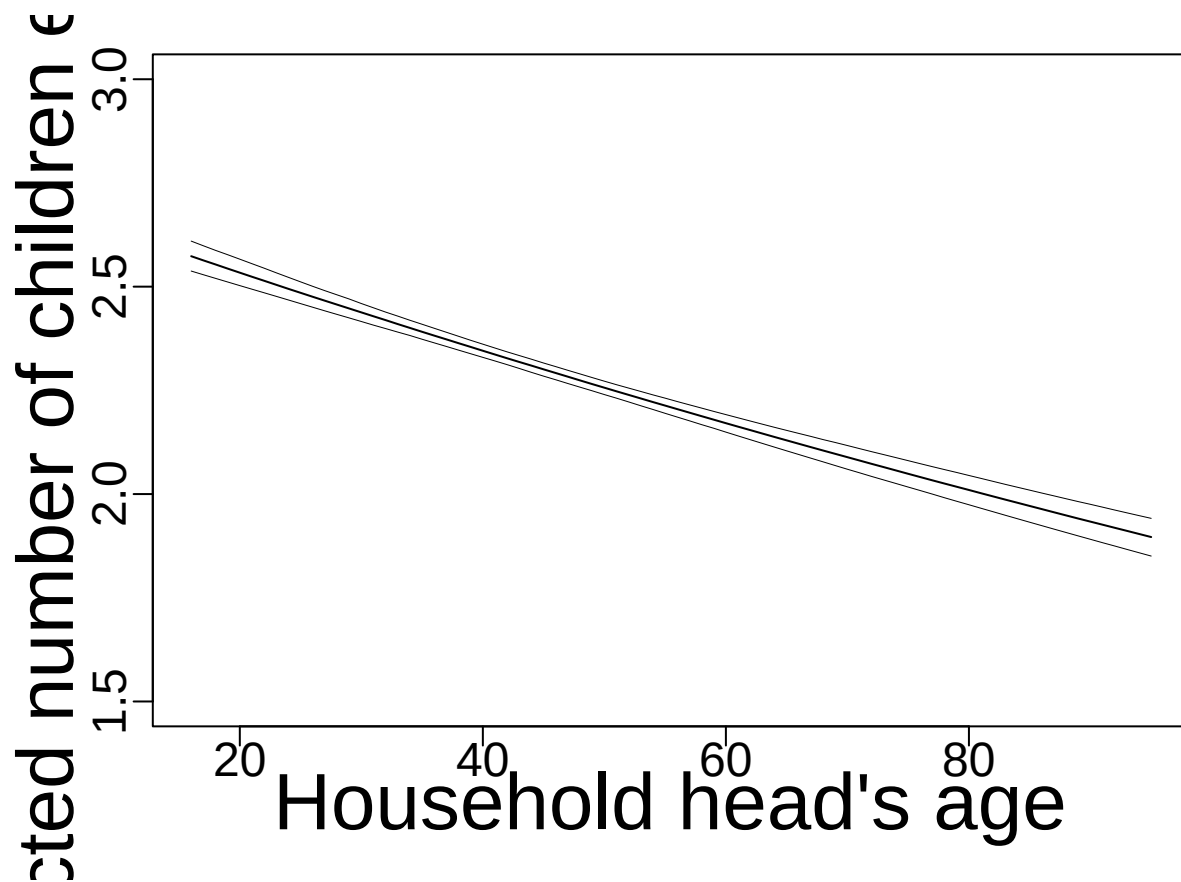
```

for (v in 16:95){
  sample_size <- c(sample_size, sum(X[,4]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(16:95)
write.csv(t(qE), paste(IMAGEOUT,"qE.HH_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(16, 95),range(c(0,qE)),type="n",xlab="Household head's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(16, 95, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(16, 95, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(16, 95, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT,"E_HH_age.pdf",sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

Expected Children Ever Born - Woman's age at marriage

```

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2

```

```

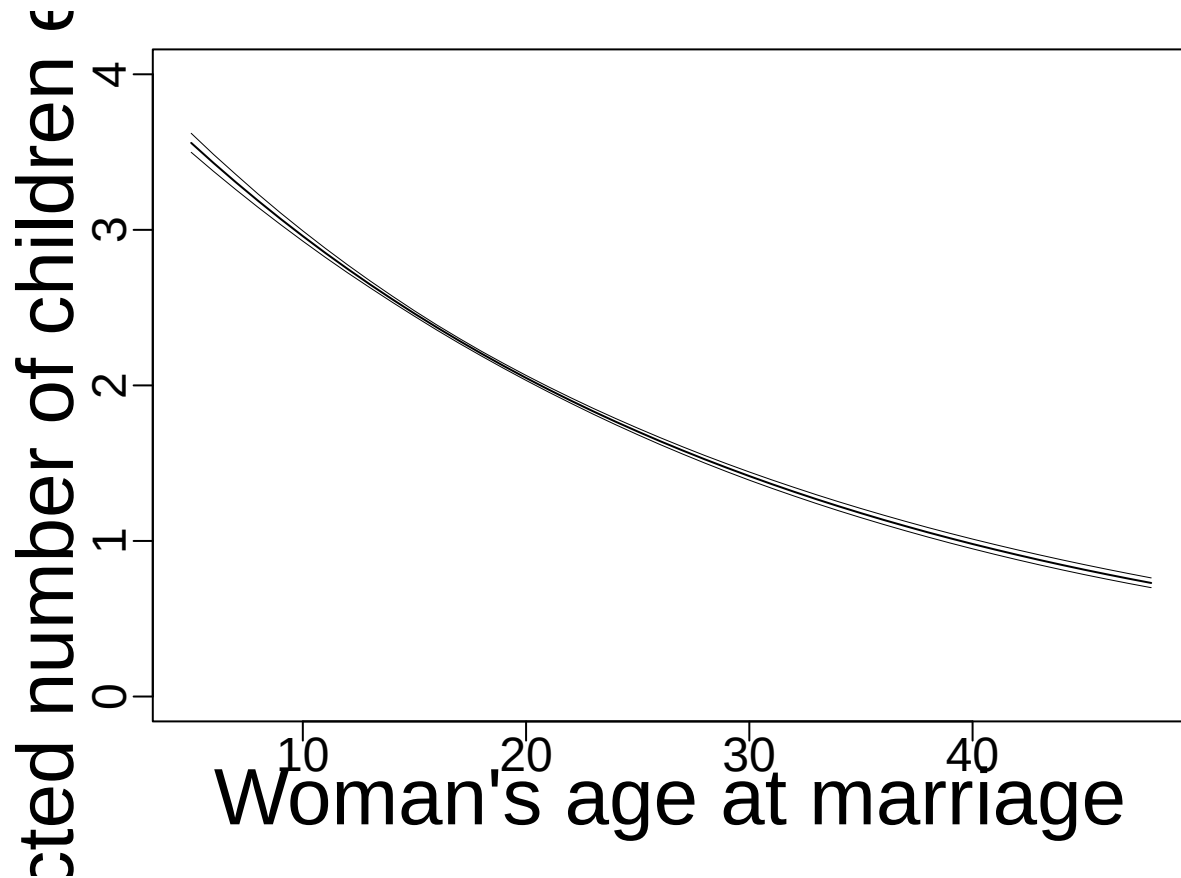
Xs <- matrix(rep(Xs, 44), nrow = 44, byrow = T)
Xs[,5] <- seq(5, 48, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,44),Xs)
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 5:48){
  sample_size <- c(sample_size, sum(X[,5]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(5:48)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_agem.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(5, 48),range(c(0,qE)),type="n",xlab="Woman's age at marriage",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(5, 48, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(5, 48, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(5, 48, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT,"E_women_agem.pdf",sep=""),
  width=16,height=16)

```



```
## pdf
## 2
```

Expected Children Ever Born - Division

```
# Expected Number of Children Ever Born in Bangladesh
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
eXB.post<- exp(t(t(Xs)%*%THETA.post.mat) )
qEA<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
round(qEA, digits = 3)

##          [,1]
## 2.5%    2.283
## 50%     2.300
## 97.5%   2.315

qEA <- rbind(qEA,dim(X)[1])
# Division

dnames <- c(
  "Bangladesh",
  "Dhaka",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymenshing",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 8), nrow = 8, byrow = T)
for (div_index in c(2:8)){
  x <- rep(0,8)
  x[div_index] <- 1
  Xs[, (15+div_index)] <- x
}
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,8),Xs)

eXB.post<- exp(t(Xs)%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))

sample_size <- sum(wm$HH7==30) # count number of women in Herat(reference levels)
for (HH7 in c(10,20,40,45,50,55,60)){
  sample_size <- c(sample_size, sum(wm$HH7==HH7))
}
qE <- rbind(qE,sample_size)

qE.out <- cbind(qEA,qE)
```

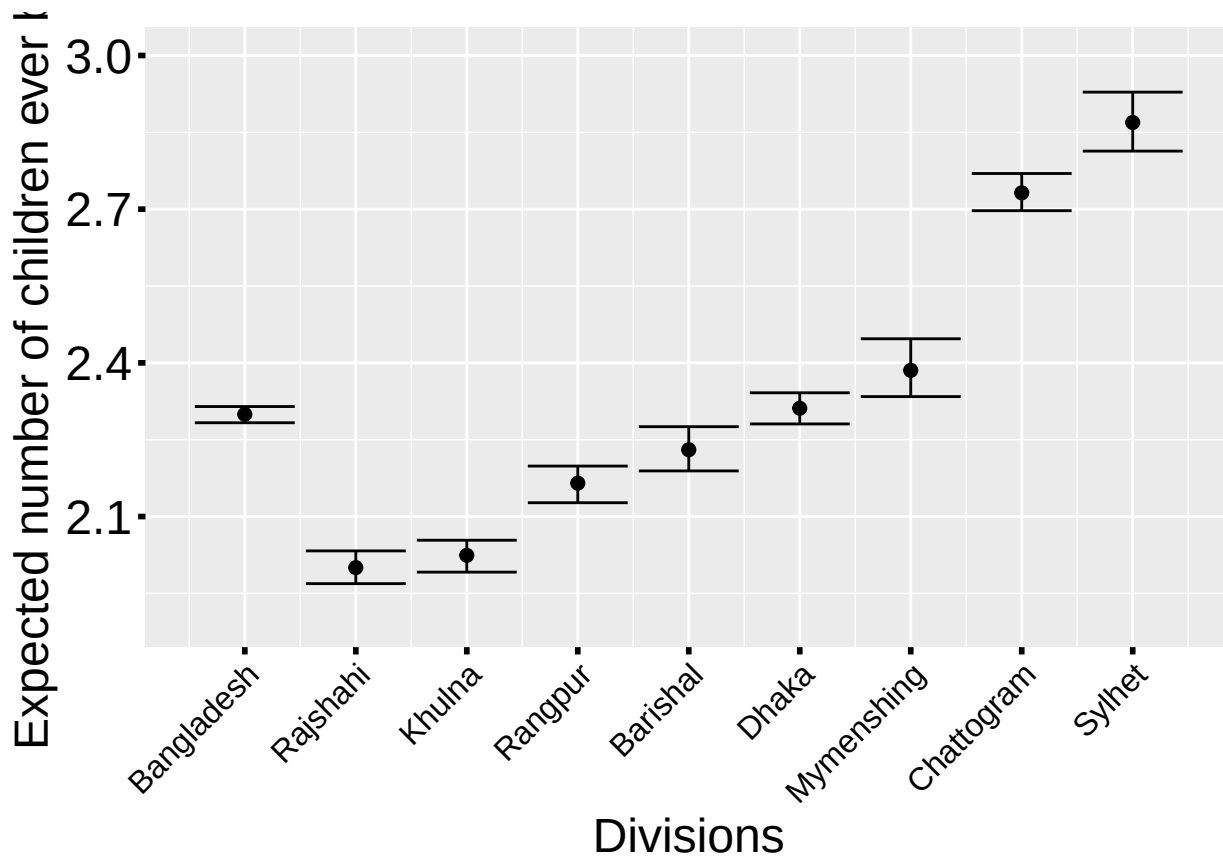
```

colnames(qE.out)<-dnames
write.csv(t(qE.out), paste(IMAGEOUT,"qE.district.csv",sep=""))

index <- order(qE[2,])
index <- c(1, index + 1)

library(ggplot2)
BHGP.df_division <- data.frame(x =1:9,
                               F = qE.out[2,][index],
                               L = qE.out[1,][index],
                               U = qE.out[3,][index],
                               name = dnames[index])
abc <- ggplot(BHGP.df_division, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Divisions") +
  ylim(1.9,3.0) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:9), labels = dnames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHGP.df_division, file=paste(IMAGEOUT,"BHGP.df_division.RData",sep=""))
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal = FALSE)
dev.print(pdf, paste(IMAGEOUT, "E_division.pdf",sep=""),width=8,height=8)

```

```
## pdf
## 2
```

Expected Children Ever Born - Ethnicity

```
anames<- c("Bengali", "Other")
index <- c(1, 2)
anames[index]
```

```
## [1] "Bengali" "Other"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,25] <- c(1, 2) # 2 for Other, 1 for Bengali
```

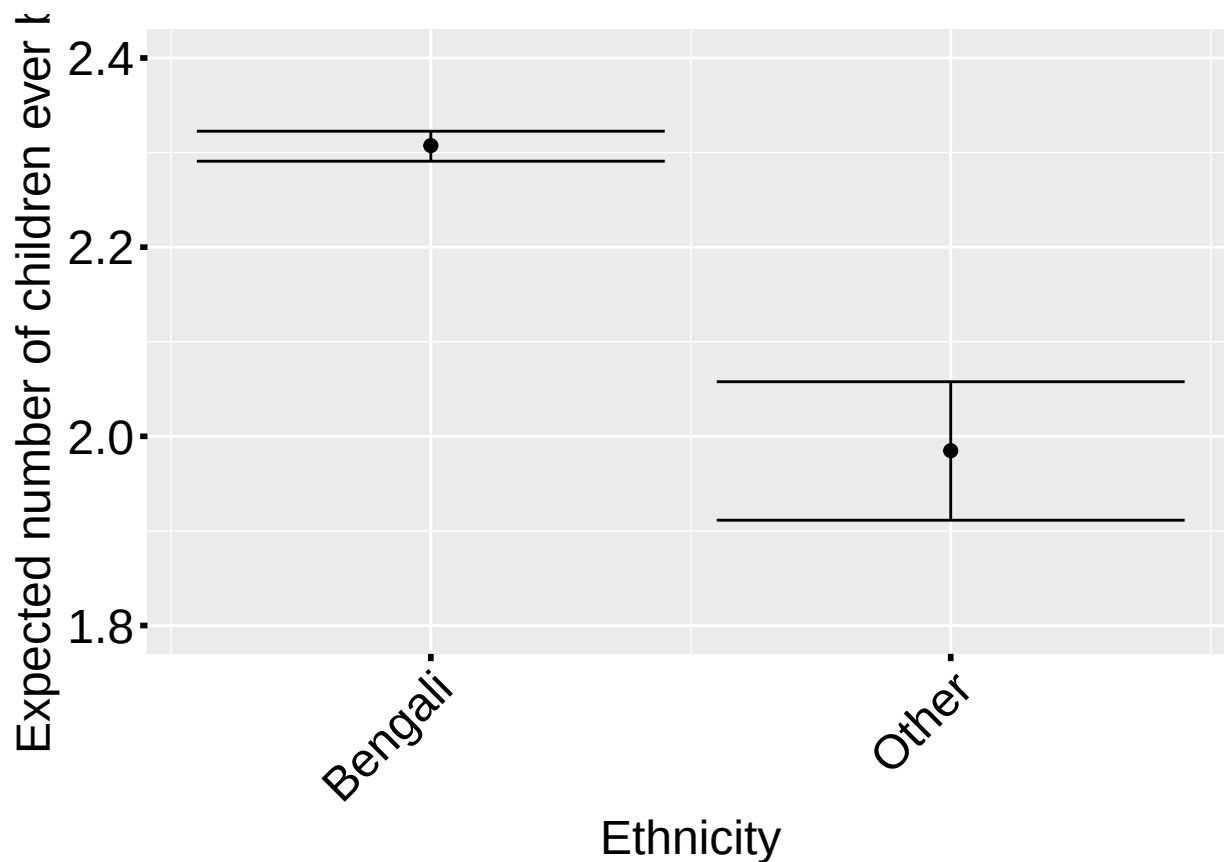
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,24]==1), sum(X[,24]==2)) # Bengali , Other
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.ethnicity.csv",sep=""))
```

```
BHGP.df_ethnicity <- data.frame(x =1:2,
                                F = c(qE[2,])[index],
                                L = c(qE[1,])[index],
                                U = c(qE[3,])[index])
```

```
abc <- ggplot(BHGP.df_ethnicity, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Ethnicity") +
  geom_point(size = 2) +
  ylim(1.8,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BHGP.df_ethnicity, file=paste(IMAGEOUT,"BHGP.df_ethnicity.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_ethnicity.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## Bengali
## 0.1625167
```

Expected Children Ever Born - Area

```
anames<- c("Rural", "Urban")
index <- c(1, 2)
anames[index]
```

```
## [1] "Rural" "Urban"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,16] <- c(1, 2) # 2 for Urban, 1 for Rural
```

```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

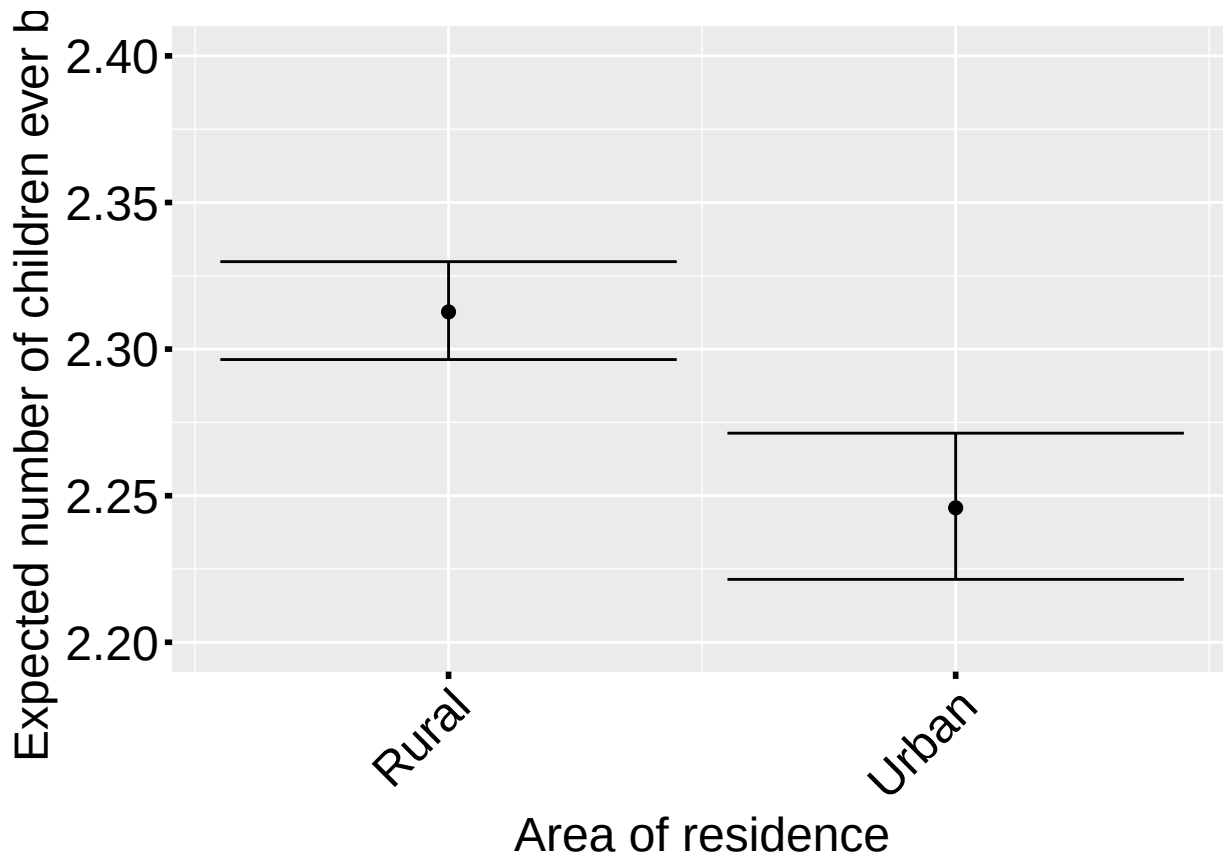
```

sample_size <- c(sum(X[,15]==1), sum(X[,15]==2)) # Rural, Urban
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.area.csv",sep=""))

BHGP.df_area <- data.frame(x =1:2,
                           F = c(qE[2,])[index],
                           L = c(qE[1,])[index],
                           U = c(qE[3,])[index])
abc <- ggplot(BHGP.df_area, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Area of residence") +
  geom_point(size = 2) +
  ylim(2.2,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHGP.df_area, file=paste(IMAGEOUT,"BHGP.df_area.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_area.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##      Rural
## 0.02975342
```

Expected Children Ever Born - Women's education

```
winames <- c("Pre-primary or none", "Primary", "Secondary", "Higher secondary+")
index <- c(1, 2, 3, 4)
winames[index]
```

```
## [1] "Pre-primary or none" "Primary"          "Secondary"
## [4] "Higher secondary+"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 4), nrow = 4, byrow = T)
Xs[,7] <- c(0, 1, 0, 0)
Xs[,8] <- c(0, 0, 1, 0)
Xs[,9] <- c(0, 0, 0, 1)
```

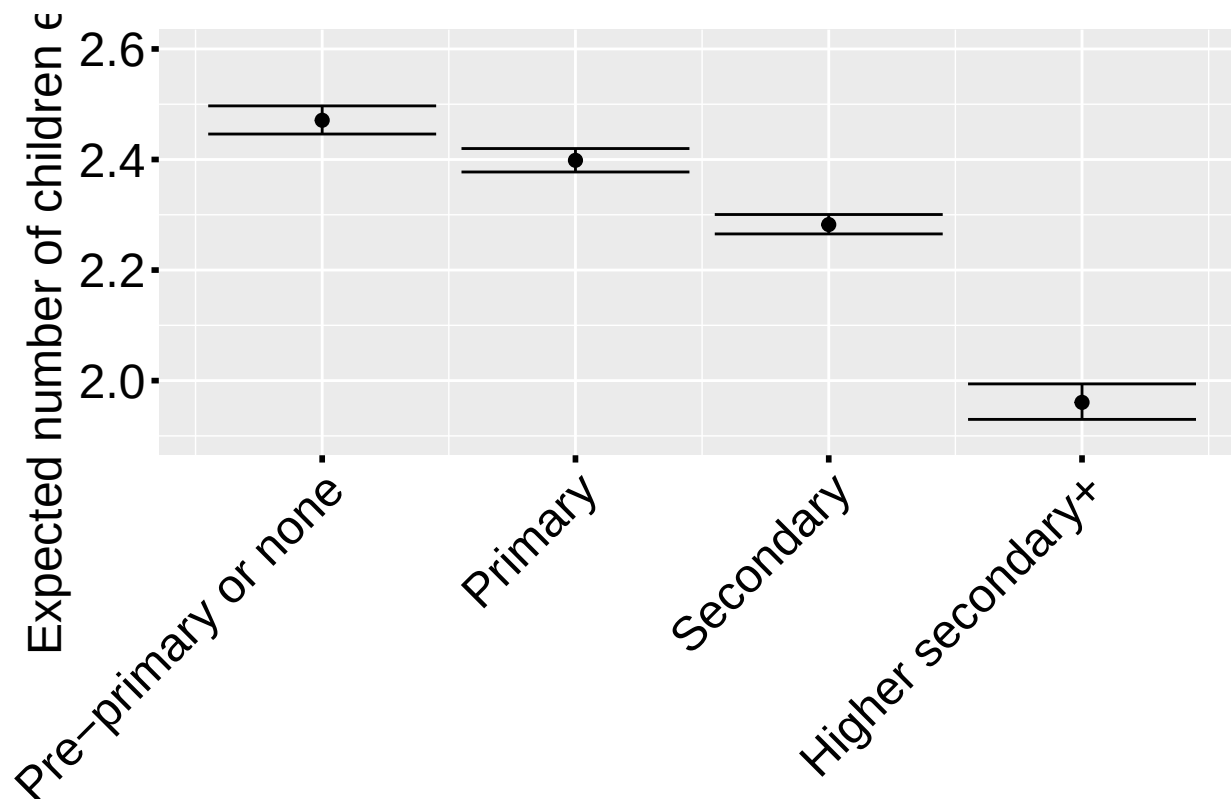
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- NULL
for (edu in 0:3){
  sample_size <- c(sample_size, sum(wm$welevel==edu))
}
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.women_edu.csv",sep=""))
```

```
BHGP.df_women_edu <- data.frame(x =1:4,
                                F = c(qE[2,])[index],
                                L = c(qE[1,])[index],
                                U = c(qE[3,])[index])
```

```
abc <- ggplot(BHGP.df_women_edu, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Women's education", cex.lab = 2.5, cex.axis = 1.5) +
  geom_point(size = 2) +
  ylim(1.9,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:4), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



Women's education

```
save(BHGP.df_women_edu, file=paste(IMAGEOUT,"BHGP.df_women_edu.RData",sep=""))

dev.print(pdf, paste(IMAGEOUT, "E_women_edu.pdf",sep=""),width=8,height=8)

## pdf
## 2
```

Expected Children Ever Born - Wealth Index

```
winames <- c("Middle", "Poorest", "Poor", "Rich", "Richest")
index <- c(2, 3, 1, 4, 5)
winames[index]

## [1] "Poorest" "Poor"      "Middle"   "Rich"     "Richest"

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 5), nrow = 5, byrow = T)
Xs[,10] <- c(0, 1, 0, 0, 0)
Xs[,11] <- c(0, 0, 1, 0, 0)
Xs[,12] <- c(0, 0, 0, 1, 0)
Xs[,13] <- c(0, 0, 0, 0, 1)

eXB.post<- exp(t(Xs*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))

sample_size <- sum(wm$windex5==3)
```

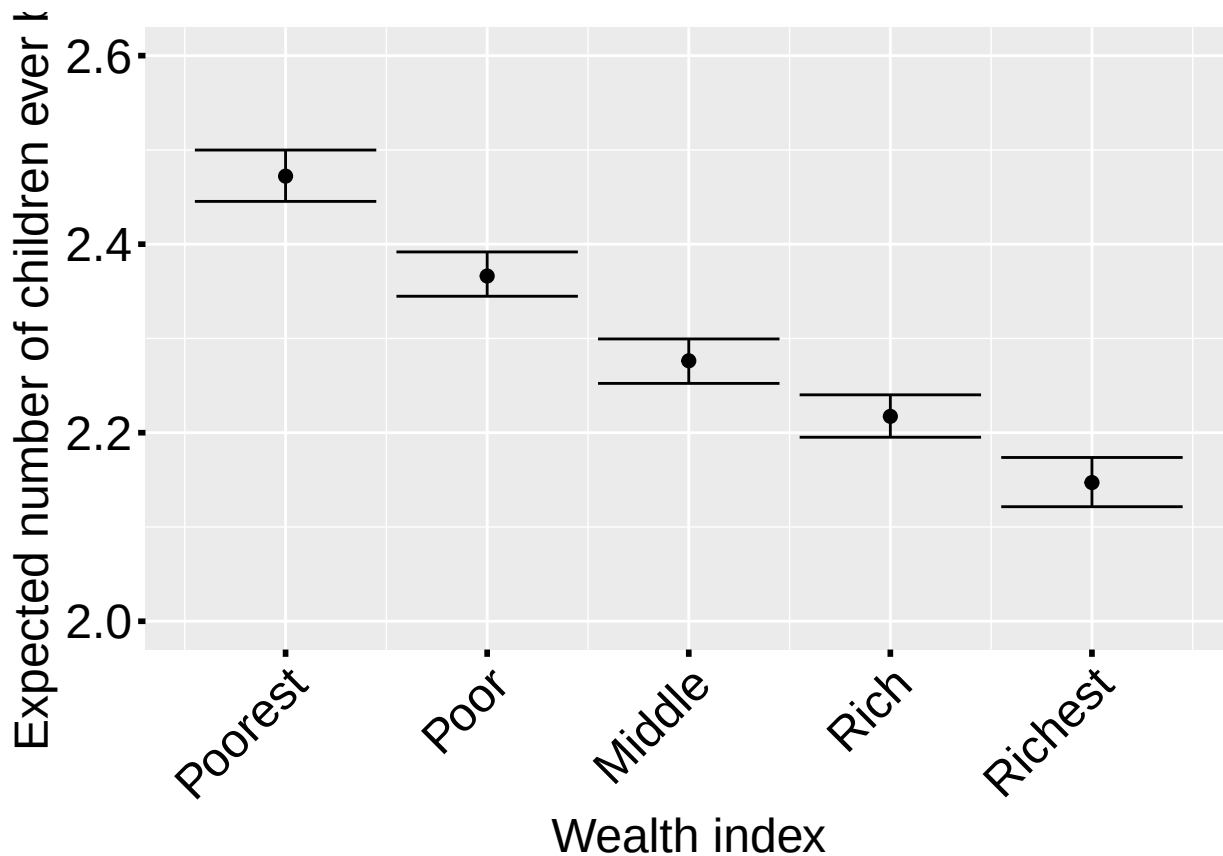
```

for (wealth in c(1,2,4,5)){
  sample_size <- c(sample_size, sum(wm$windex5==wealth))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.windex.csv",sep=""))

BHGP.df_windex5 <- data.frame(x =1:5,
                             F = c(qE[2,])[index],
                             L = c(qE[1,])[index],
                             U = c(qE[3,])[index])
abc <- ggplot(BHGP.df_windex5 , aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Wealth index", cex.lab = 2.5, cex.axis = 1.5) +
  geom_point(size = 2) +
  ylim(2,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:5), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHGP.df_windex5, file=paste(IMAGEOUT,"BHGP.df_windex5.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_windex5.pdf",sep=""),width=8,height=8)

```



```
## pdf
## 2
```

Expected Children Ever Born - other wives

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,15] <- c(1, 2) # 2 for Yes, 1 for No
```

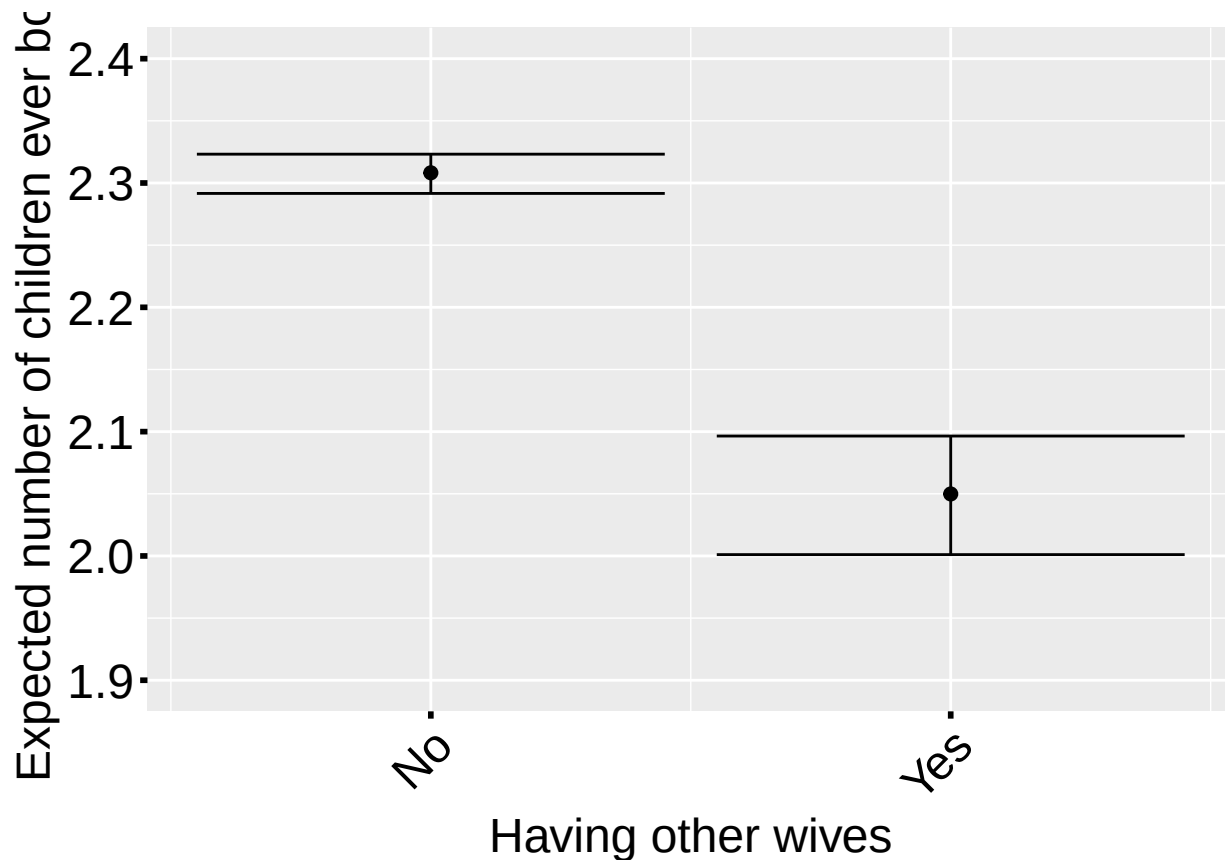
```
eXB.post<- exp(t(Xs%%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,14]==1), sum(X[,14]==2)) # No, Yes
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.other_wives.csv",sep=""))
```

```
BHGP.df_other_wives <- data.frame(x =1:2,
                                   F = c(qE[2,])[index],
                                   L = c(qE[1,])[index],
                                   U = c(qE[3,])[index])
```

```
abc <- ggplot(BHGP.df_other_wives, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Having other wives") +
  geom_point(size = 2) +
  ylim(1.9,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BHGP.df_other_wives, file=paste(IMAGEOUT,"BHGP.df_other_wives.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_other_wives.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## No
## 0.1260237
```

Expected Children Ever Born - media

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,14] <- c(1, 2) # 2 for Yes, 1 for No
```

```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

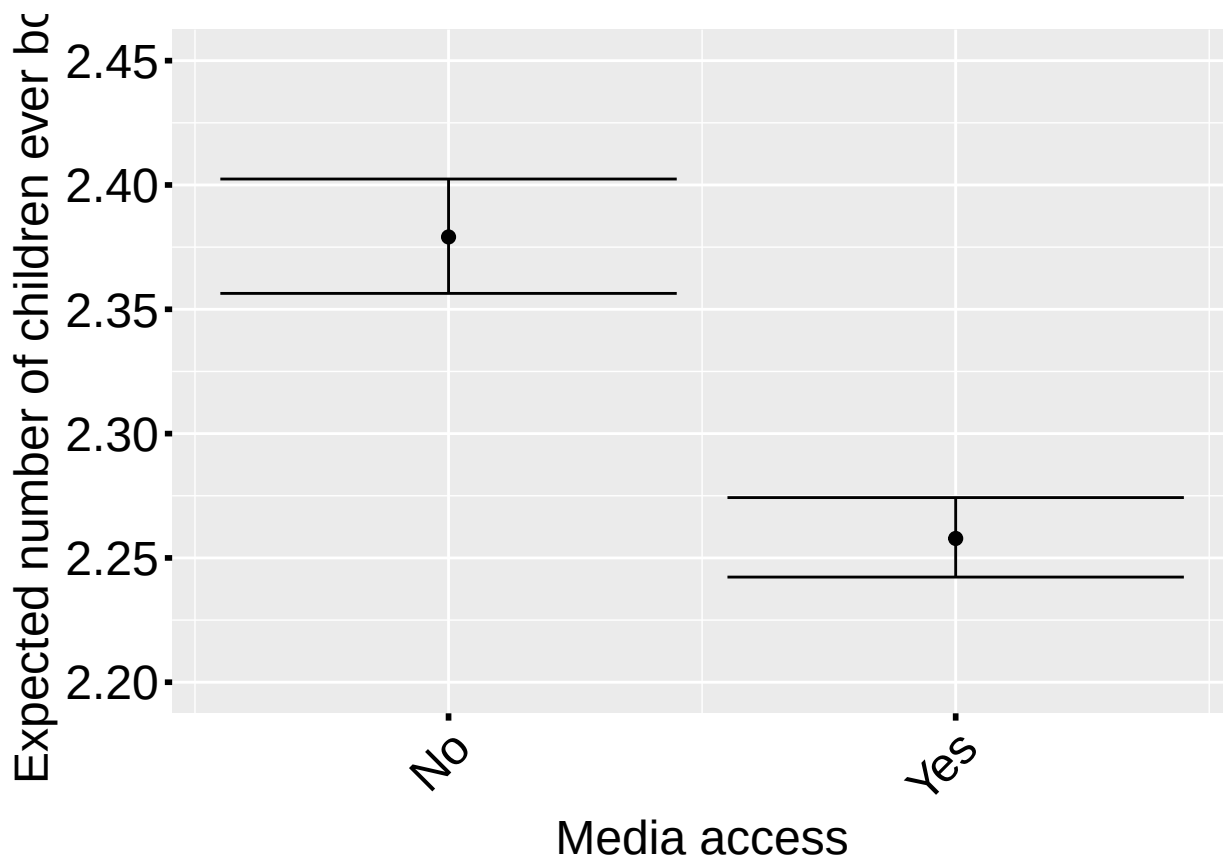
```

sample_size <- c(sum(X[,13]==1), sum(X[,13]==2)) # No, Yes
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.media.csv",sep=""))

BHGP.df_media <- data.frame(x =1:2,
                             F = c(qE[2,])[index],
                             L = c(qE[1,])[index],
                             U = c(qE[3,])[index])
abc <- ggplot(BHGP.df_media, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Media access") +
  geom_point(size = 2) +
  ylim(2.2,2.45) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHGP.df_media, file=paste(IMAGEOUT,"BHGP.df_media.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_media.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##           No  
## 0.05370963
```

Expected Children Ever Born - Household head's sex

```
anames<- c("Male", "Female")  
index <- c(1, 2)  
anames[index]
```

```
## [1] "Male"   "Female"
```

```
Xs <- apply(X, 2, mean)  
Xs[2] <- Xs[1]^2  
Xs <- as.matrix(c(1, Xs))  
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)  
Xs[,17] <- c(1, 2) # 2 for Female, 1 for Male
```

```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )  
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,16]==1), sum(X[,16]==2)) # Male, Female  
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames  
write.csv(t(qE), paste(IMAGEOUT,"qE.HH_sex.csv",sep=""))
```

```
BHGP.df_HH_sex <- data.frame(x =1:2,  
                             F = c(qE[2,])[index],  
                             L = c(qE[1,])[index],  
                             U = c(qE[3,])[index])
```

```
abc <- ggplot(BHGP.df_HH_sex, aes(x = x, y = F)) +  
  labs(y = "Expected number of children ever born", x = "Household head's sex") +  
  geom_point(size = 2) +  
  ylim(2.0,2.4) +  
  geom_errorbar(aes(ymax = U, ymin = L)) +  
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +  
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),  
        axis.ticks.x = element_line(color = "black", linewidth = 1),  
        axis.text.y = element_text(size = 18, color = "black"),  
        axis.ticks.y = element_line(color = "black", linewidth = 1))  
print(abc)
```



```

Xs[2] <- Xs[1]^2
Xs <- c(1,Xs)

#eCEB_each_woman=exp(matmul(apply(Z.list[[j]],2,mean), GAMMA.post.mat[[j]]) + matmul(apply(X.list[[j]],
eXB.post.district <- exp(GAMMA.post.mat[[j]] %*% Zs + t(Xs %*% THETA.post.mat))
qE <- apply(eXB.post.district,2,quantile,probs=c(0.025,0.5,.975))
# data_column is a 4x1 matrix, with each row represents mean, 2.5%, 50%, 97.5%
district_eCEB <- cbind(district_eCEB, qE)
yy <- c(yy, mean(y.list[[j]]))
}

district_eCEB <- t(district_eCEB)
colnames(district_eCEB) <- c("2.5%", "50%", "97.5%")
rownames(district_eCEB) <- district_names
district_eCEB[order(district_eCEB[, 2],decreasing = TRUE)[c(1:10)], ]

##           2.5%      50%      97.5%
## Sunamganj  2.996510 3.103811 3.209492
## Bandarban  2.780154 2.957894 3.147539
## Brahmanbaria 2.762770 2.859136 2.967343
## Gopalganj   2.689956 2.794683 2.902254
## Cox's Bazar 2.664290 2.761218 2.859030
## Sylhet      2.634491 2.723084 2.818196
## Kushtia     2.603527 2.714071 2.817839
## Netrokona   2.548568 2.661900 2.760700
## Khulna      2.466515 2.546906 2.630948
## Noakhali    2.453292 2.530634 2.608165

map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$eCEB <- district_eCEB[,1]
map$yy <- yy

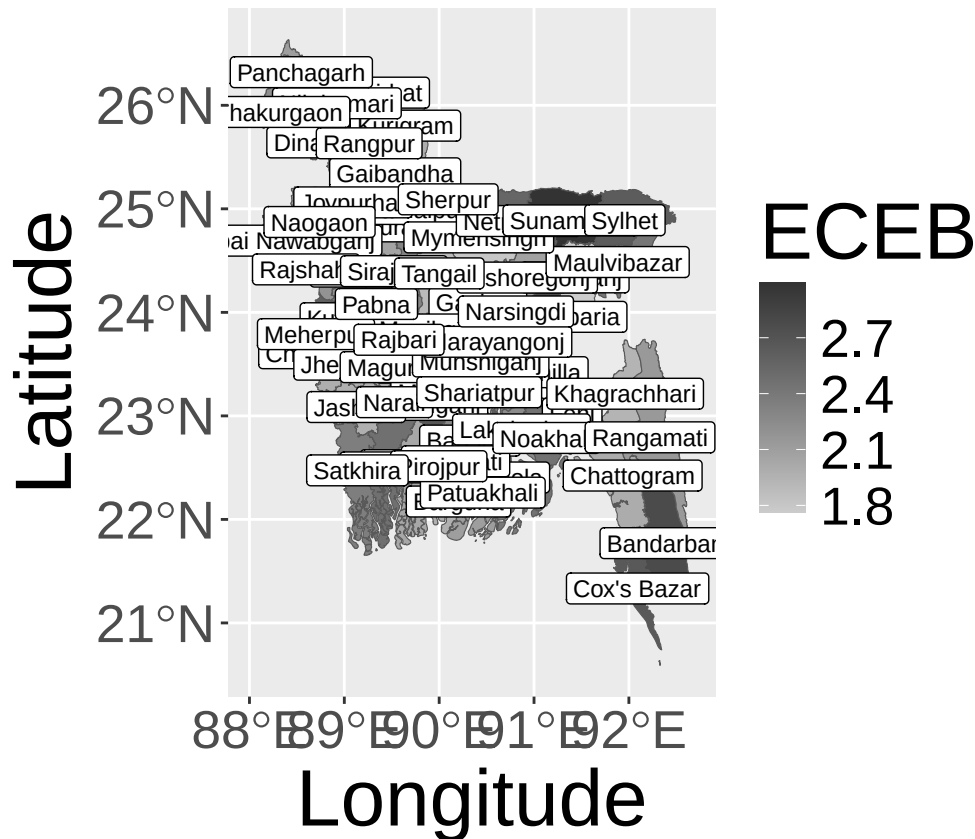
# --- Create the division boundary for Sylhet ---
Sylhet_div <- map %>%
  filter(ADM1_EN == "Sylhet") %>%
  group_by(ADM1_EN) %>%
  summarise(geometry = st_union(geometry)) # merge all district polygons

ggplot(map) +
  geom_sf(aes(fill = eCEB)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  # geom_sf(data = Sylhet_div, fill = NA, color = "red", size = 1.2, linewidth = 1.5) +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  labs(fill="ECEB", x="Longitude", y="Latitude") +
  theme(
    legend.title = element_text(size = 30),      # Legend title size
    legend.text = element_text(size = 20),       # Legend item text size
    axis.title = element_text(size = 30),
    axis.text = element_text(size = 20)
  )

```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
```

```
## correct results for longitude/latitude data
```



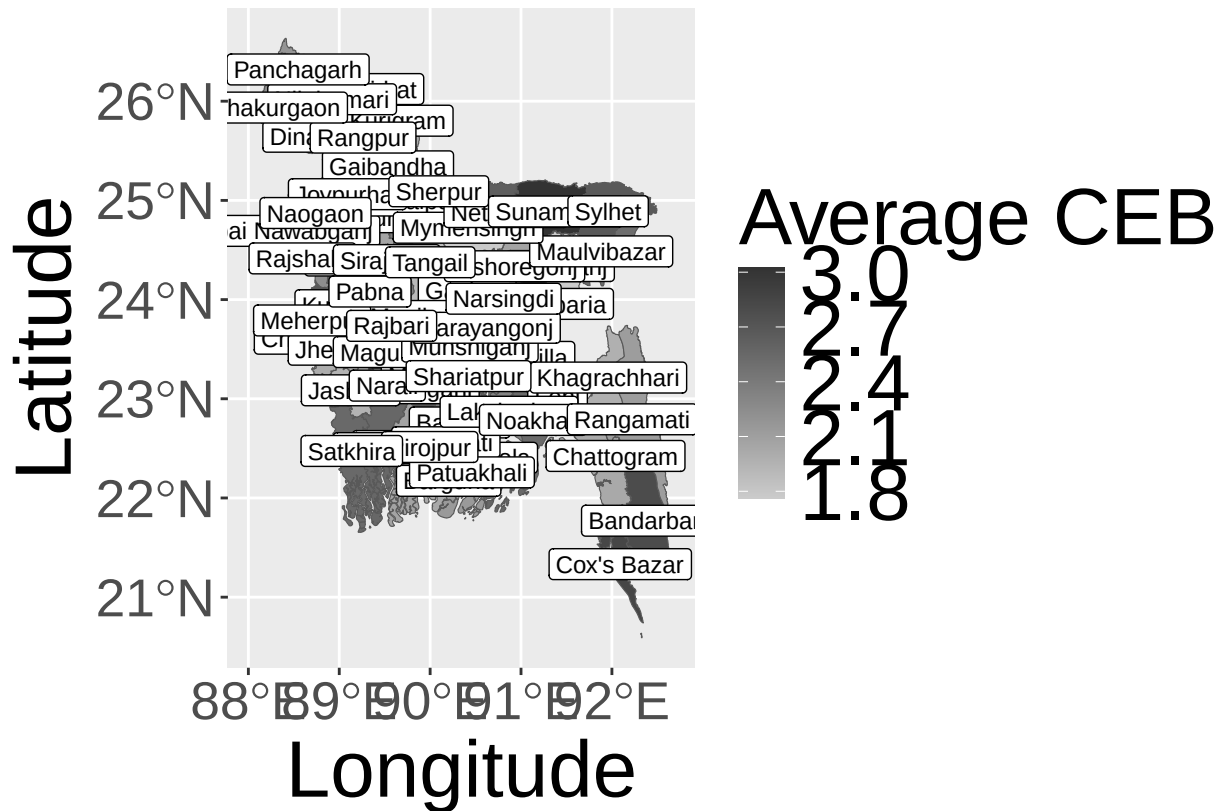
```
dev.print(pdf, paste(IMAGEOUT, "E_district_map.pdf", sep=""), width=16, height=16)
```

```
## pdf
## 2
yy <- NULL
for(j in 1:m)
{
  yy <- c(yy, mean(y.list[[j]]))
}

map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$yy <- yy

ggplot(map) +
  geom_sf(aes(fill = yy)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  labs(fill="Average CEB", x="Longitude", y="Latitude") +
  theme(
    legend.title = element_text(size = 30),      # Legend title size
    legend.text = element_text(size = 30),      # Legend item text size
    axis.title = element_text(size = 30),
    axis.text = element_text(size = 20)
  )
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
```



```
dev.print(pdf, paste(IMAGEOUT, "Avg_district_map.pdf", sep=""), width=16, height=16)
```

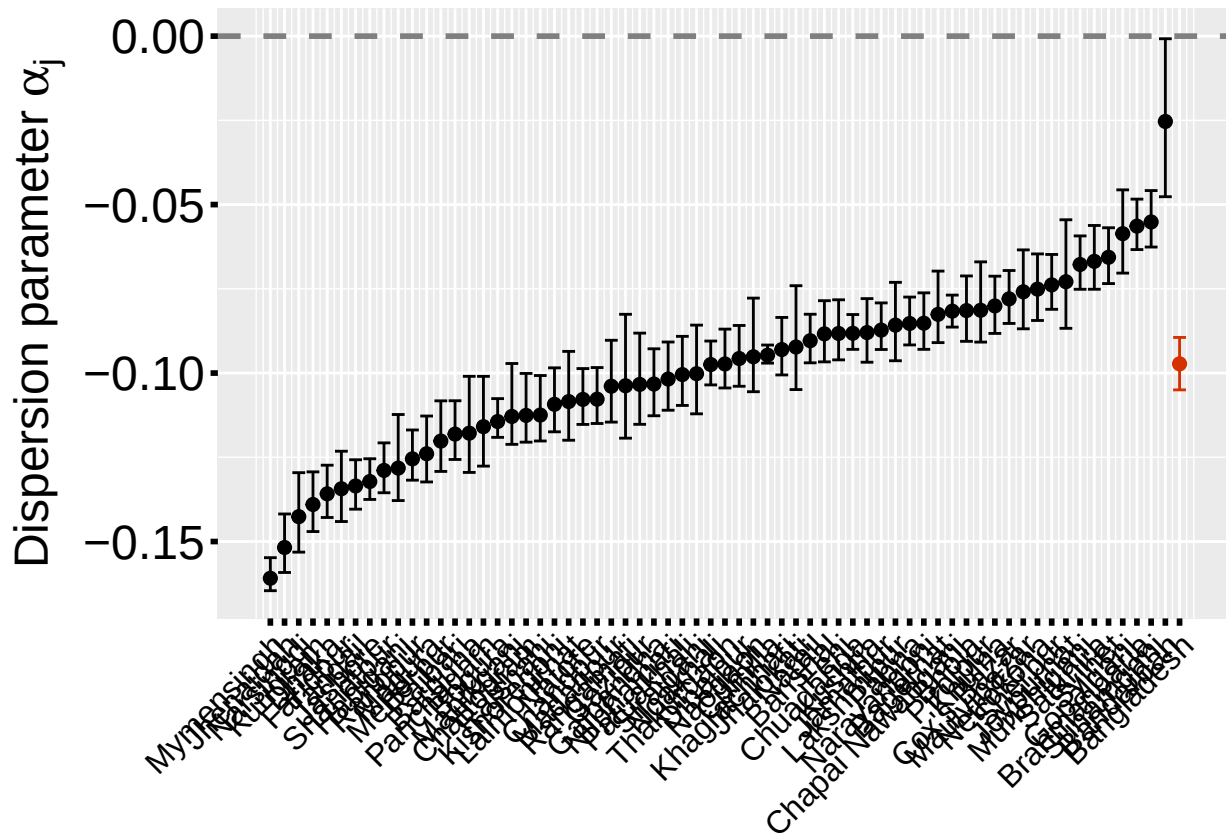
```
## pdf
## 2
```

```
q.ALPHA <- apply(ALPHA.post, 2, quantile, probs=c(.025, .5, .975))
# reorder from the most underdispersed data
index <- order(q.ALPHA[2,])
df <- data.frame(x = 1:m,
  names = district_names[index],
  L = q.ALPHA[1,][index],
  M = q.ALPHA[2,][index],
  U = q.ALPHA[3,][index])

MuAlphaCI <- quantile(MU.ALPHA.post, probs=c(.025, .5, .975))
df <- rbind(df, data.frame(x=m+1, names="Bangladesh", L=MuAlphaCI[1], M=MuAlphaCI[2], U=MuAlphaCI[3]))
df$Col <- rep("black", 65)
df$Col[65] <- "#D53100"
abc <- ggplot(df, aes(x = x, y = M)) +
  labs(y = expression(paste("Dispersion parameter ", alpha[j], sep="")), x = NULL) +
  geom_point(size = 2, color=df$Col) +
  geom_hline(yintercept = 0, color = "gray50", linetype = "dashed", size = 1) +
  geom_errorbar(aes(ymax = U, ymin = L), color=df$Col) +
  scale_x_continuous(breaks = c(1:(m+1)), labels = c(district_names[index], "Bangladesh")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
    axis.ticks.x = element_line(color = "black", linewidth = 1),
```



```
axis.text.y = element_text(size = 18, color = "black"),
axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
dev.print(pdf, paste(IMAGEOUT, "Alphaj.pdf", sep=""), width=16, height=8)
```

```
## pdf
## 2

# Plot the CI
BP_IMAGEOUT = "./pics/BP/"
BHP_IMAGEOUT = "./pics/BHP/"
BHGP_IMAGEOUT = "./pics/BHGP/"

# Plot the CI
load(paste(BP_IMAGEOUT, "BP.df_division.RData", sep=""))
load(paste(BHP_IMAGEOUT, "BHP.df_division.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_division.RData", sep=""))

# Add a new column to each data frame to identify the source
BP.df_division$Source <- "BP"
BHP.df_division$Source <- "BHP"
BHGP.df_division$Source <- "BHGP"

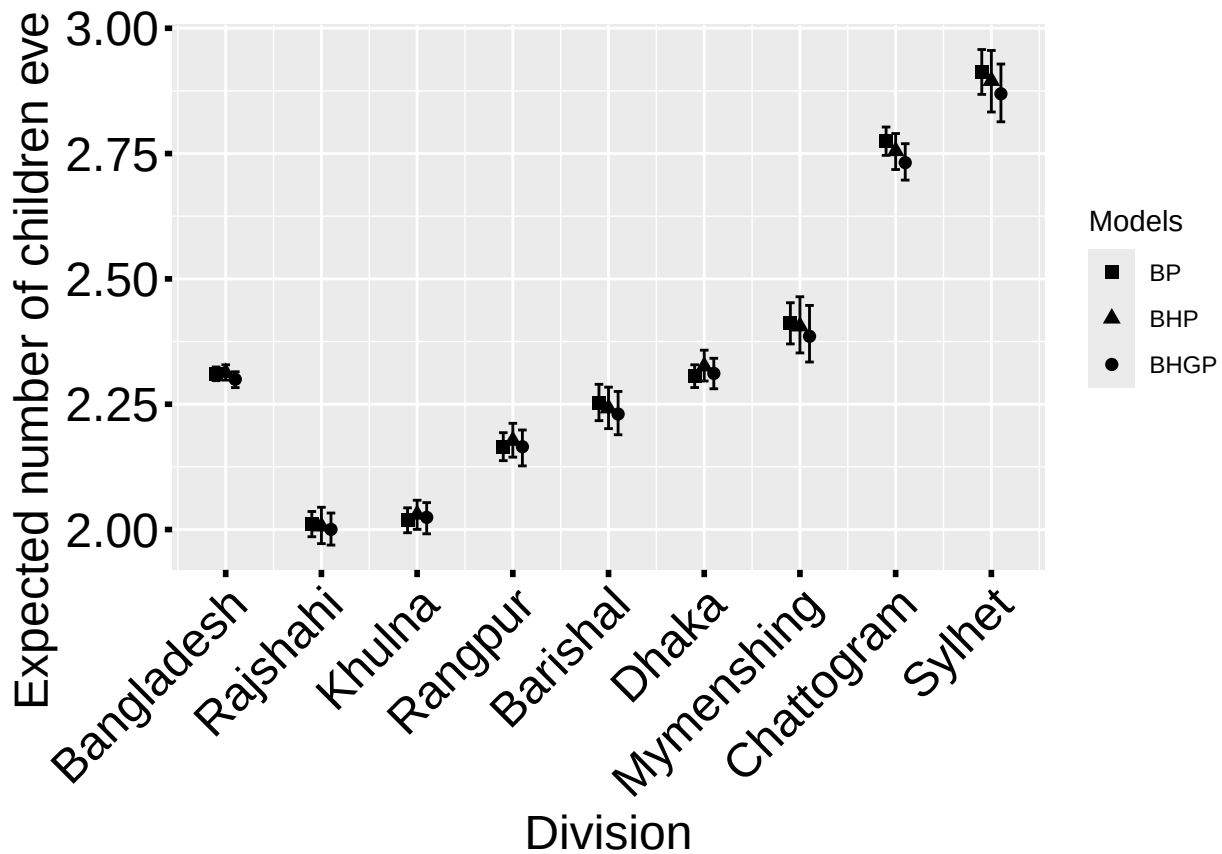
BP.df_division$x <- BP.df_division$x - 0.1
BHGP.df_division$x <- BHGP.df_division$x + 0.1
combined_df <- bind_rows(BP.df_division, BHP.df_division, BHGP.df_division)

E_division_all <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
```

```

geom_point(size = 2) +
geom_errorbar(aes(ymin = L, ymax = U)) +
scale_x_continuous(breaks = c(1:9), labels = c("Bangladesh", "Rajshahi", "Khulna", "Rangpur", "Barishal", "Dhaka", "Mymensingh", "Chattogram", "Sylhet")) +
scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
labs(x = "Division", y = "Expected number of children ever born") +
theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
      axis.ticks.x = element_line(color = "black", linewidth = 1),
      axis.text.y = element_text(size = 18, color = "black"),
      axis.ticks.y = element_line(color = "black", linewidth = 1))
print(E_division_all)

```



```

dev.print(pdf, paste(IMAGEOUT, "E_division_all.pdf", sep=""),width=8, height=8)

```

```

## pdf
## 2

load(paste(BP_IMAGEOUT, "BP.df_ethnicity.RData", sep=""))
load(paste(BHP_IMAGEOUT, "BHP.df_ethnicity.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_ethnicity.RData", sep=""))

# Add a new column to each data frame to identify the source
BP.df_ethnicity$Source <- "BP"
BHP.df_ethnicity$Source <- "BHP"
BHGP.df_ethnicity$Source <- "BHGP"

BP.df_ethnicity$x <- BP.df_ethnicity$x - 0.1

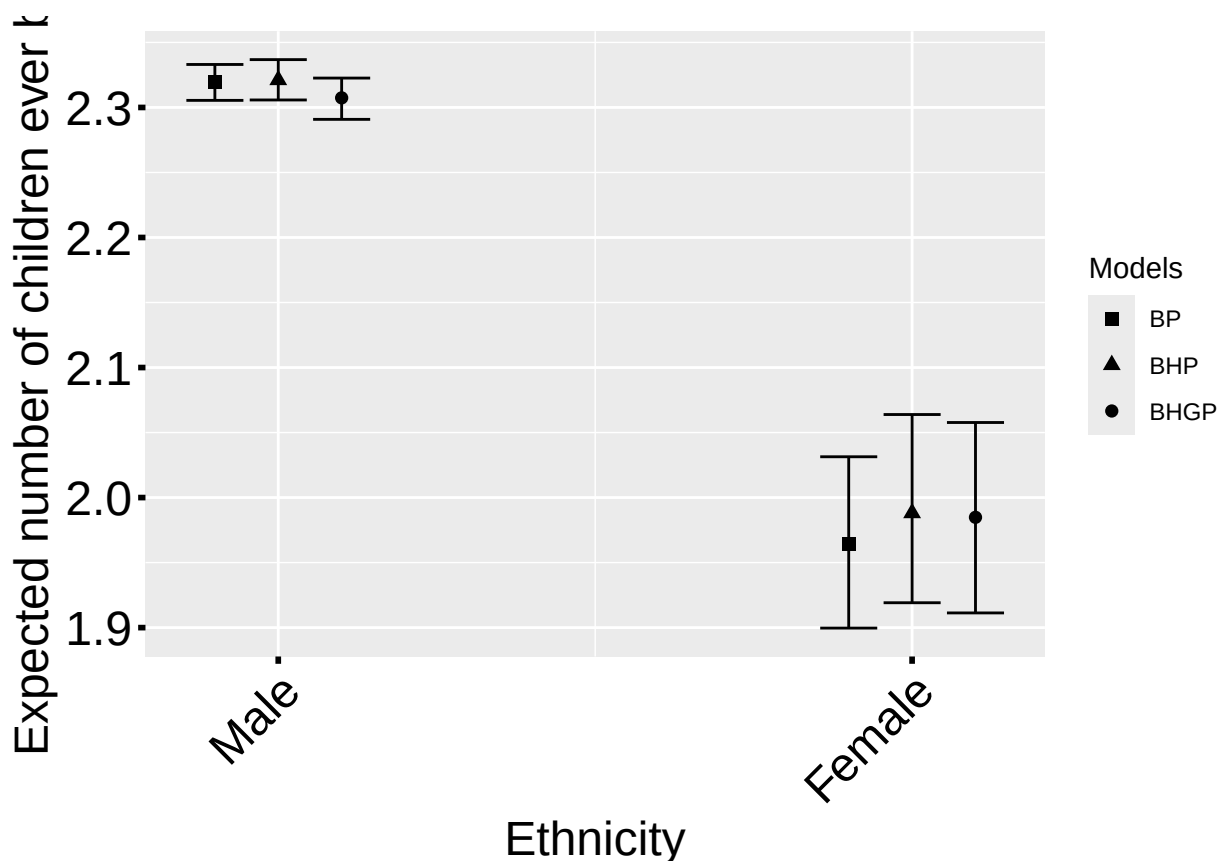
```

```

BHGP.df_ethnicity$x <- BHGP.df_ethnicity$x + 0.1
combined_df <- bind_rows(BP.df_ethnicity, BHP.df_ethnicity, BHGP.df_ethnicity)

E_Ethnicity <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U)) +
  scale_x_continuous(breaks = c(1:2), labels = c("Male", "Female")) +
  labs(x = "Ethnicity", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(E_Ethnicity)

```



```

dev.print(pdf, paste(IMAGEOUT, "E_ethnicity_all.pdf", sep=""),width=8, height=8)

```

```

## pdf
## 2

```

```

load(paste(BP_IMAGEOUT, "BP.df_area.RData", sep=""))
load(paste(BHP_IMAGEOUT, "BHP.df_area.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_area.RData", sep=""))

```

```

# Add a new column to each data frame to identify the source
BP.df_area$Source <- "BP"

```

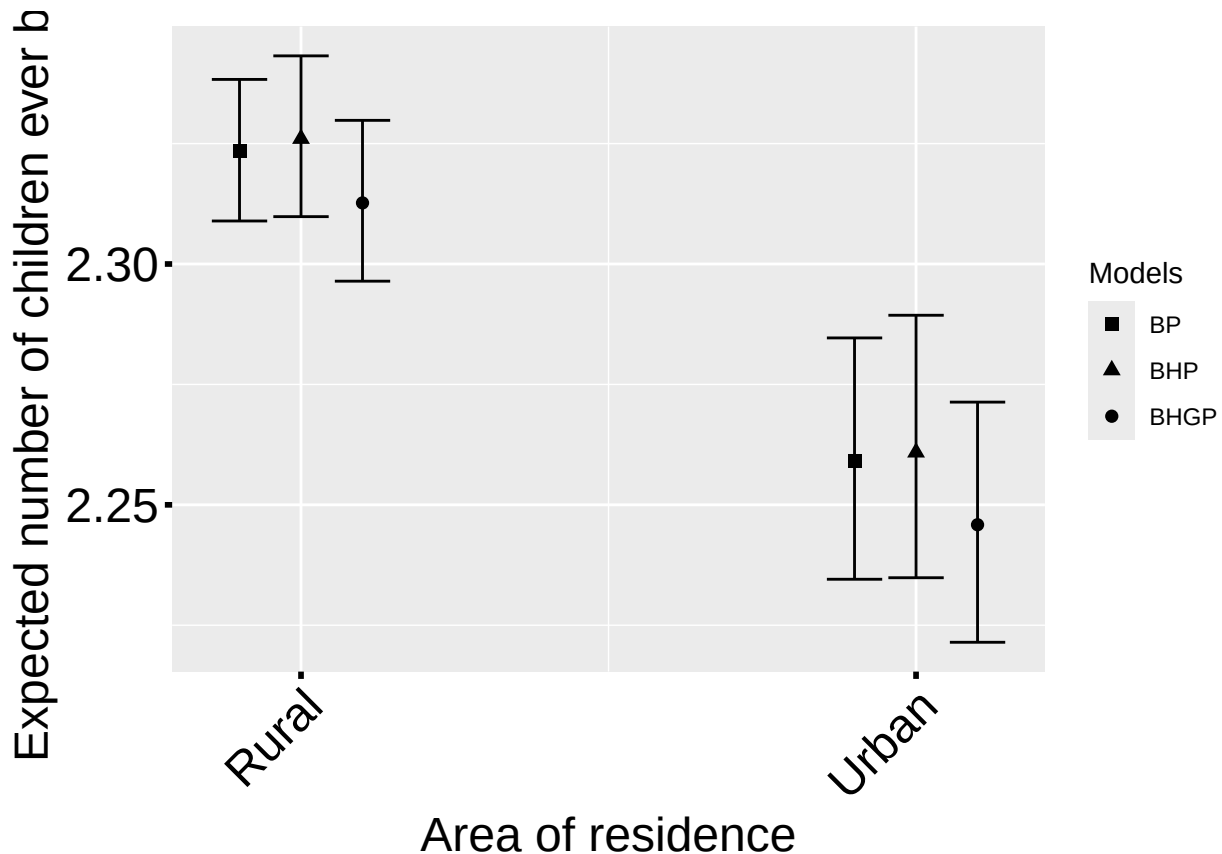
```

BHP.df_area$Source <- "BHP"
BHGP.df_area$Source <- "BHGP"

BP.df_area$x <- BP.df_area$x - 0.1
BHGP.df_area$x <- BHGP.df_area$x + 0.1
combined_df <- bind_rows(BP.df_area, BHP.df_area, BHGP.df_area)

E_area <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U)) +
  scale_x_continuous(breaks = c(1:2), labels = c("Rural", "Urban")) +
  labs(x = "Area of residence", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(E_area)

```



```

dev.print(pdf, paste(IMAGEOUT, "E_area_all.pdf", sep=""), width=8, height=8)

```

```

## pdf
## 2

```

```

# Plot the CI
load(paste(BP_IMAGEOUT, "BP.df_women_edu.RData", sep=""))

```

```

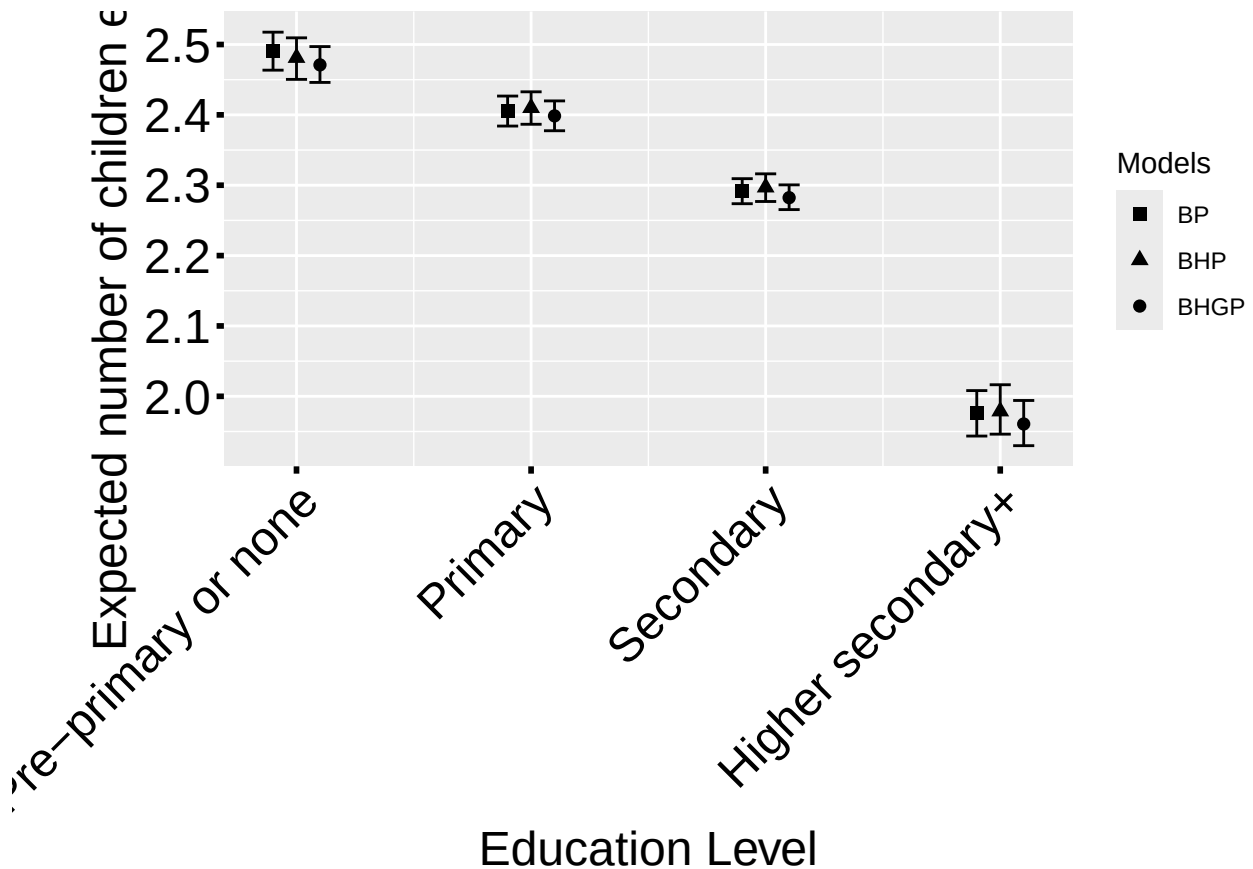
load(paste(BHP_IMAGEOUT, "BHP.df_women_edu.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_women_edu.RData", sep=""))

# Add a new column to each data frame to identify the source
BP.df_women_edu$Source <- "BP"
BHP.df_women_edu$Source <- "BHP"
BHGP.df_women_edu$Source <- "BHGP"

BP.df_women_edu$x <- BP.df_women_edu$x - 0.1
BHGP.df_women_edu$x <- BHGP.df_women_edu$x + 0.1
combined_df <- bind_rows(BP.df_women_edu, BHP.df_women_edu, BHGP.df_women_edu)

E_women_edu <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U)) +
  scale_x_continuous(breaks = 1:4, labels = c("Pre-primary or none", "Primary", "Secondary", "Higher secondary or more")) +
  labs(x = "Education Level", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1),
        plot.margin = margin(l=30))
print(E_women_edu)

```



```

dev.print(pdf, paste(IMAGEOUT, "E_women_edu_all.pdf", sep=""),width=8, height=8)

## pdf
## 2

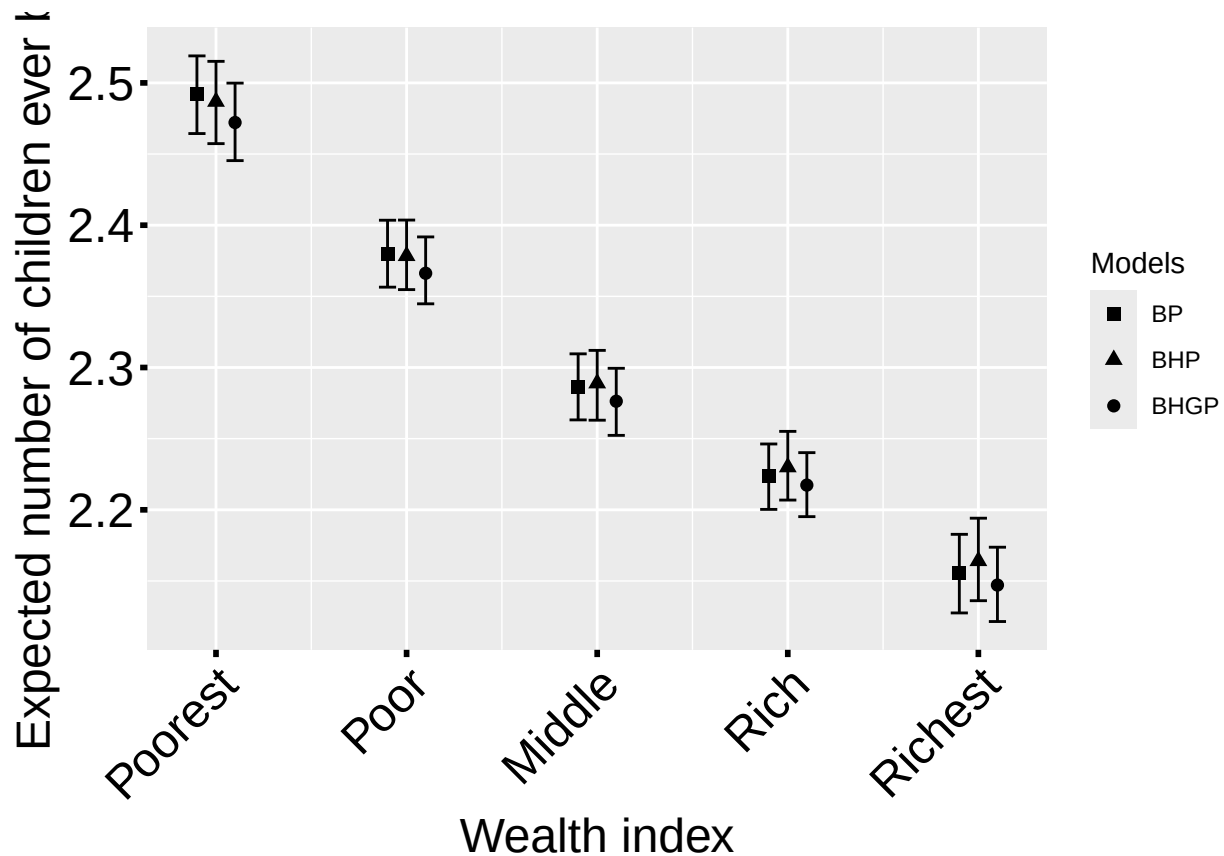
# Plot the CI
load(paste(BP_IMAGEOUT,"BP.df_windex5.RData",sep=""))
load(paste(BHP_IMAGEOUT,"BHP.df_windex5.RData",sep=""))
load(paste(BHGP_IMAGEOUT,"BHGP.df_windex5.RData",sep=""))

# Add a new column to each data frame to identify the source
BP.df_windex5$Source <- "BP"
BHP.df_windex5$Source <- "BHP"
BHGP.df_windex5$Source <- "BHGP"

BP.df_windex5$x <- BP.df_windex5$x - 0.1
BHGP.df_windex5$x <- BHGP.df_windex5$x + 0.1
combined_df <- bind_rows(BP.df_windex5, BHP.df_windex5, BHGP.df_windex5)

E_windex5<- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U )) +
  scale_x_continuous(breaks = 1:5, labels = c("Poorest", "Poor", "Middle", "Rich", "Richest")) +
  labs(x = "Wealth index", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1),
        )
print(E_windex5)

```



```
dev.print(pdf, paste(IMAGEOUT, "E_windex5_all.pdf", sep=""),width=8, height=8)
```

```
## pdf
## 2
```

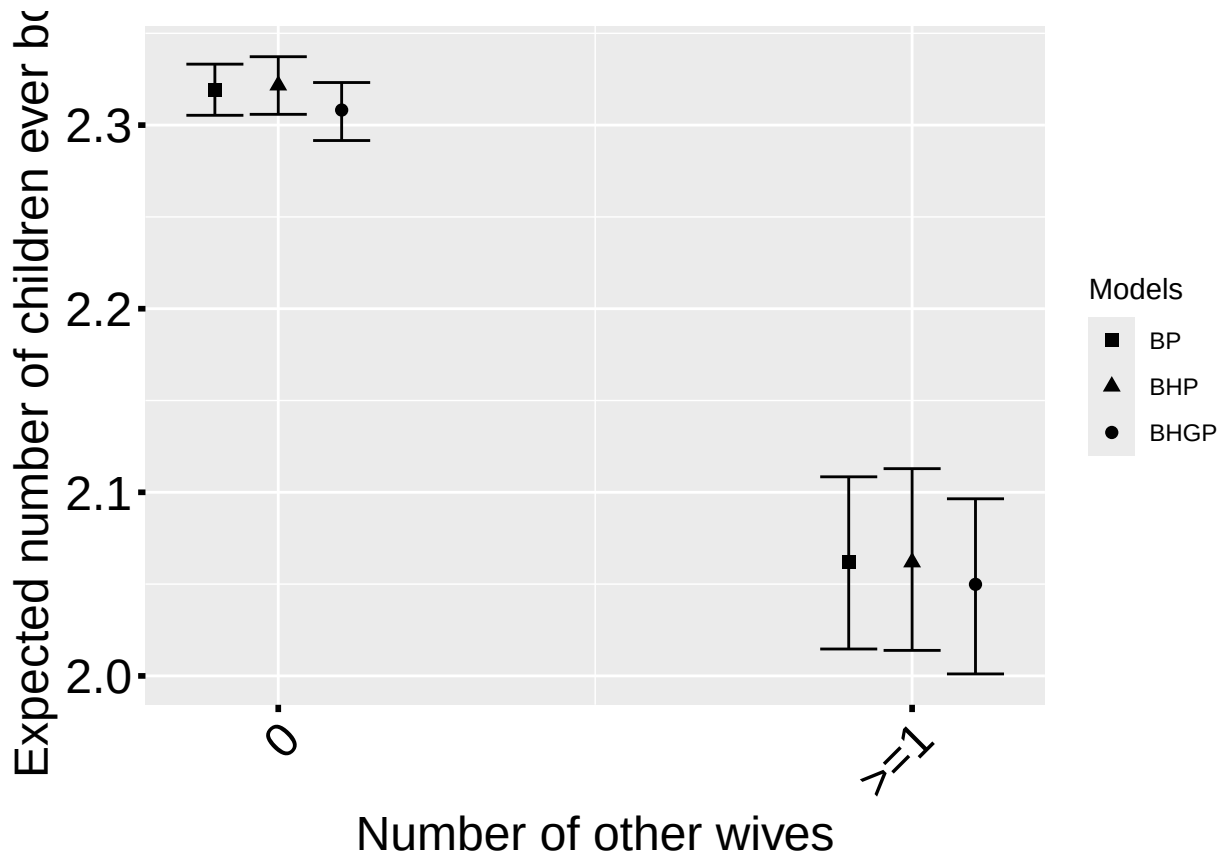
```
load(paste(BP_IMAGEOUT, "BP.df_other_wives.RData", sep=""))
load(paste(BHP_IMAGEOUT, "BHP.df_other_wives.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_other_wives.RData", sep=""))

# Add a new column to each data frame to identify the source
BP.df_other_wives$Source <- "BP"
BHP.df_other_wives$Source <- "BHP"
BHGP.df_other_wives$Source <- "BHGP"

BP.df_other_wives$x <- BP.df_other_wives$x - 0.1
BHGP.df_other_wives$x <- BHGP.df_other_wives$x + 0.1
combined_df <- bind_rows(BP.df_other_wives, BHP.df_other_wives, BHGP.df_other_wives)

E_other_wives <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U)) +
  scale_x_continuous(breaks = c(1:2), labels = c("0", ">=1")) +
  labs(x = "Number of other wives", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
```

```
axis.ticks.y = element_line(color = "black", linewidth = 1))
print(E_other_wives)
```



```
dev.print(pdf, paste(IMAGEOUT, "E_other_wives_all.pdf", sep=""),width=8, height=8)
```

```
## pdf
## 2
```

```
load(paste(BP_IMAGEOUT, "BP.df_HH_sex.RData", sep=""))
load(paste(BHP_IMAGEOUT, "BHP.df_HH_sex.RData", sep=""))
load(paste(BHGP_IMAGEOUT, "BHGP.df_HH_sex.RData", sep=""))

# Add a new column to each data frame to identify the source
BP.df_HH_sex$Source <- "BP"
BHP.df_HH_sex$Source <- "BHP"
BHGP.df_HH_sex$Source <- "BHGP"

BP.df_HH_sex$x <- BP.df_HH_sex$x - 0.1
BHGP.df_HH_sex$x <- BHGP.df_HH_sex$x + 0.1
combined_df <- bind_rows(BP.df_HH_sex, BHP.df_HH_sex, BHGP.df_HH_sex)

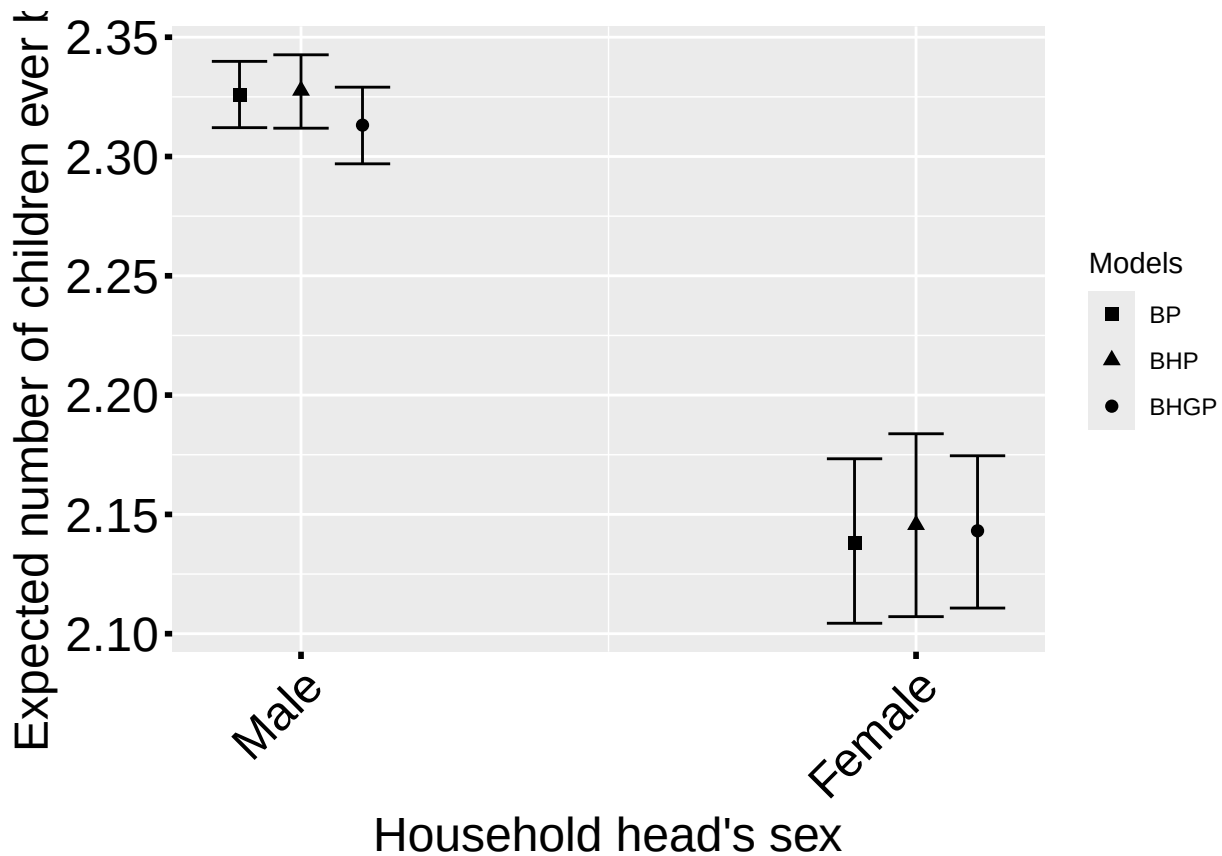
E_HH_sex <- ggplot(combined_df, aes(x = x, y = F, shape = Source, group = Source)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = L, ymax = U)) +
  scale_x_continuous(breaks = c(1:2), labels = c("Male", "Female")) +
  labs(x = "Household head's sex", y = "Expected number of children ever born") +
  scale_color_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
```



```

scale_shape_discrete(name="Models", breaks = c("BP", "BHP", "BHGP")) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(E_HH_sex)

```



```

dev.print(pdf, paste(IMAGEOUT, "E_HH_sex_all.pdf", sep=""),width=8, height=8)

```

```

## pdf
## 2
x <- 0:12
mu <- 2
alpha <- -0.09
data <- c(4663, 10933, 16565, 10670, 5029, 2029, 885, 337, 145, 52, 18, 3, 2)
xcol<-gray.colors(3, start = 0, end = .9)
plot(x+0.1, dgenpois2(x,mu,alpha),type="h", ylim=c(0,0.35), text.font = 2, lwd=6, col=xcol[3], xlab="Number of children ever born")

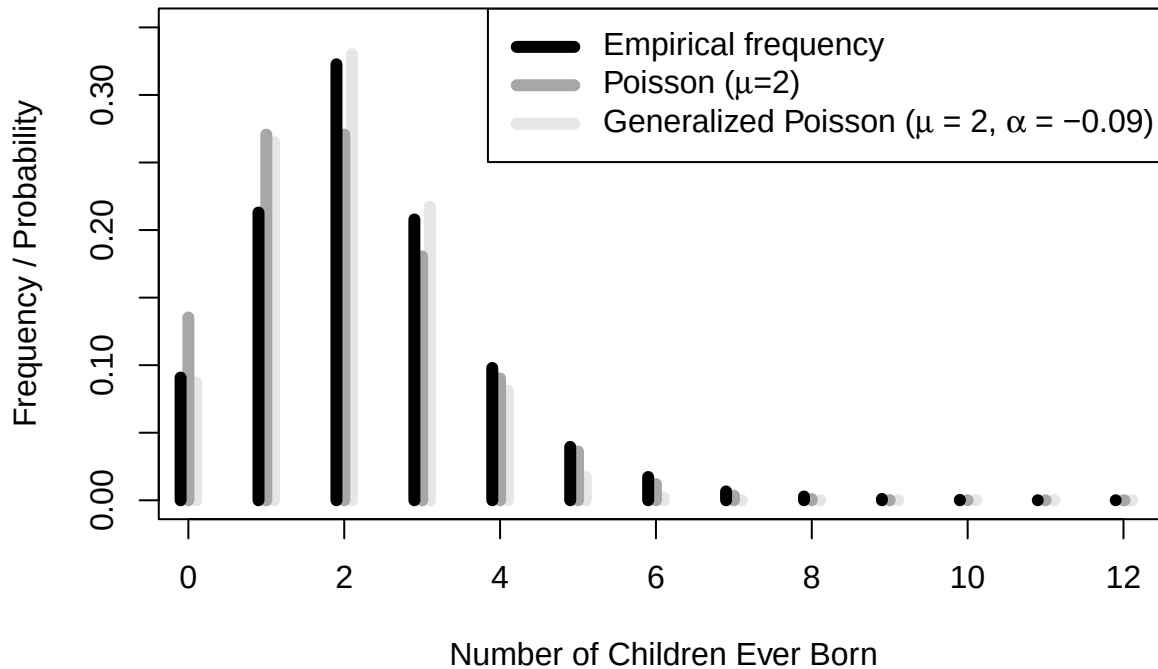
## Warning in plot.window(...): "text.font" is not a graphical parameter
## Warning in plot.xy(xy, type, ...): "text.font" is not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "text.font" is not a graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "text.font" is not a graphical parameter
## Warning in box(...): "text.font" is not a graphical parameter

```

```
## Warning in title(...): "text.font" is not a graphical parameter
```

```
points(x, dgenpois2(x,mu,0),type="h", lwd=6, col=xcol[2])
points(x-0.1, data/sum(data), type="h", lwd=6, col=xcol[1])
```

```
legend("topright", legend = c("Empirical frequency", expression(paste("Poisson (",mu,"=2)",sep="")), exp
    lwd = 6, text.font = 2 )
```



```
dev.print(pdf,"pics/CEB_histo.pdf", width=12, height=6)
```

```
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
```

```
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
## Warning in grDevices::dev.copy(device = function (file = if (onefile) "Rplots.pdf" else
## "Rplot%03d.pdf", : "text.font" is not a graphical parameter
```

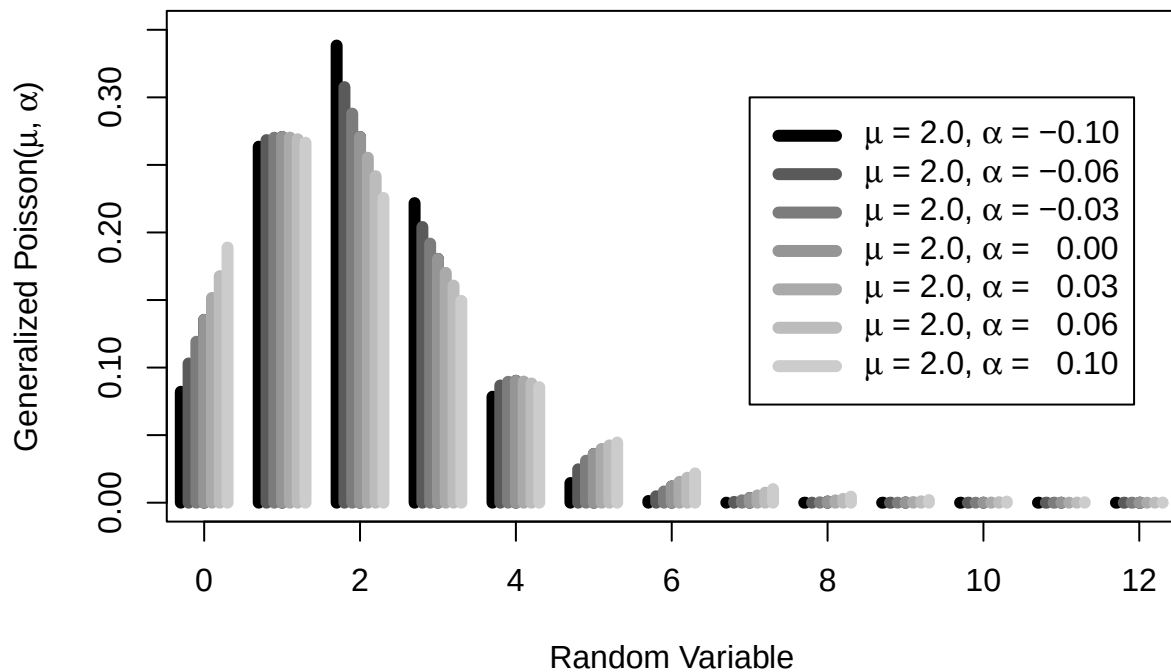
```
## pdf
## 2
```

```
x <- 0:12
mu <- c(2.0, 2.0, 2.0, 2.0, 2.0, 2.0, 2.0)
alpha <- c(-0.10,-0.06,-0.03,0,0.03,0.06,0.10)
xshift <- c(-0.3,-0.2,-0.1,0,0.1,0.2,0.3)
xcol<-gray.colors(7, start = 0, end = 0.8)
plot(x, dgenpois2(x,mu,0),type="h", col=1, ylim=c(0,0.35), lwd=6, xlab="Random Variable", ylab=expression(
for (i in 1:7){
    points(x+xshift[i], dgenpois2(x,mu[i],alpha[i]),type="h", col=xcol[i], lwd=6)
```

```

}
legend(7,0.3,
  c(
    expression(paste(mu, " = 2.0, ", alpha," = -0.10")),
    expression(paste(mu, " = 2.0, ", alpha," = -0.06")),
    expression(paste(mu, " = 2.0, ", alpha," = -0.03")),
    expression(paste(mu, " = 2.0, ", alpha," = 0.00")),
    expression(paste(mu, " = 2.0, ", alpha," = 0.03")),
    expression(paste(mu, " = 2.0, ", alpha," = 0.06")),
    expression(paste(mu, " = 2.0, ", alpha," = 0.10"))
  ), col=xcol, lwd=c(6,6,6,6,6,6,6))

```



```

dev.print(pdf,"pics/GenPois2.pdf", width=16, height=8)

```

```

## pdf
## 2

```

```

library("glmmTMB")
df.yX <- as.data.frame(cbind(yX,district))

```

```

fit_genpois<- glmmTMB(formula = y~women_age+women_age2+husband_age+HH_age+women_agem+women_edu1+women_e

```

```

fit_pois<- glmmTMB(formula = y~women_age+women_age2+husband_age+HH_age+women_agem+women_edu1+women_edu2

```

```

as.matrix(exp(summary(fit_genpois)$coeff$cond[,1]))
ranef(fit_genpois)
ranef(fit_pois)

```

Posterior CI of GAMMA + THETA

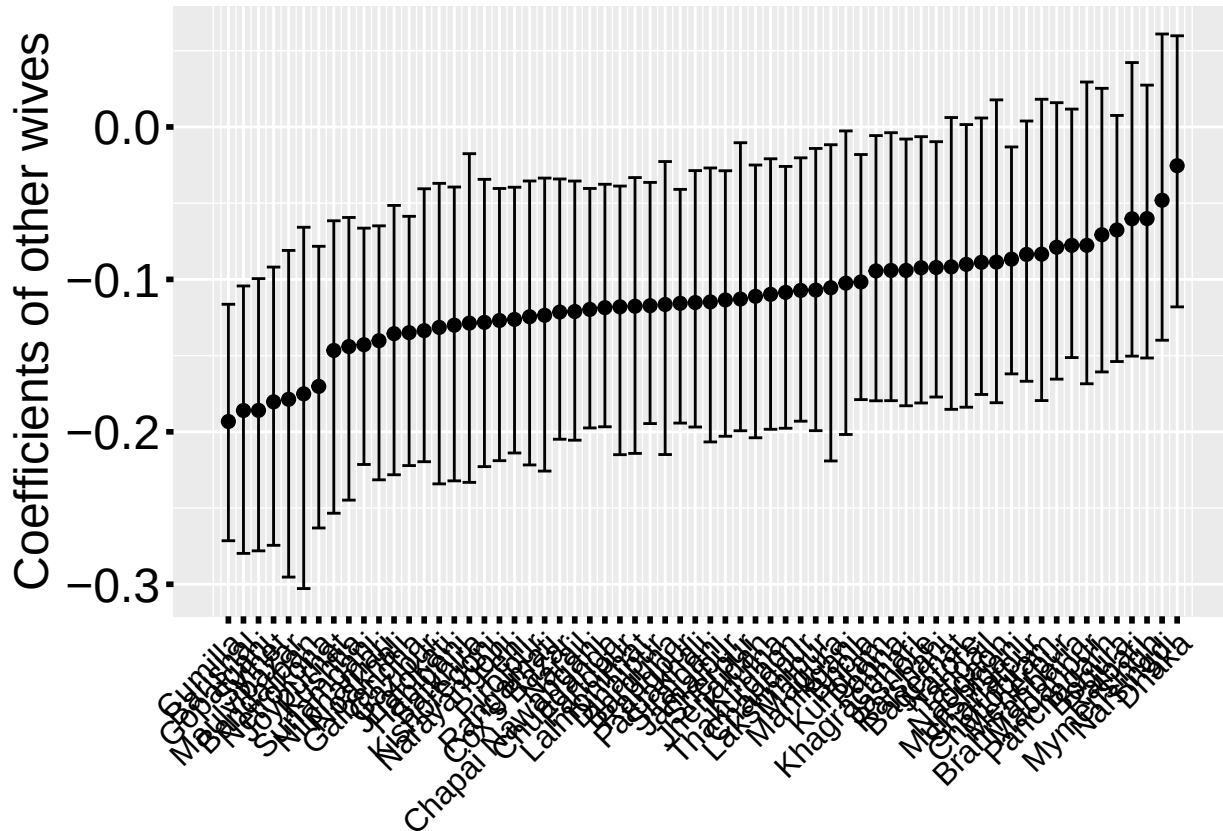
To plot the posterior CI of gamma + theta coefficients of other_wives

```
#index_other_wives is the index of other_wives in the regression coefficient vector GAMMA
index_other_wives = 15
i <- 1
GAMMA_other_wives<- NULL
for (dis in district_codes){

  gamma_ij <- NULL
  for (s in c(1:length(GAMMA.post))){
    gamma_ij <- c(gamma_ij, GAMMA.post[[s]][i, index_other_wives])
  }
  GAMMA_other_wives <- rbind(GAMMA_other_wives, gamma_ij)
  i <- i+1
}

# THETA.post.mat[2,] is the fixed effect of fixed predictor
qE<-apply(GAMMA_other_wives+rep(THETA.post.mat[index_other_wives,],m),1,quantile,probs=c(.025,.5,.975))
index <- order(qE[2,])

df <- data.frame(x = 1:m,
                 F = qE[2,][index],
                 L = qE[1,][index],
                 U = qE[3,][index])
abc <- ggplot(df, aes(x = x, y = F)) +
  labs(y = "Coefficients of other wives", x = NULL) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal =
dev.print(pdf, paste(IMAGEOUT, "BETA.other_wives.pdf",sep=""),width=16,height=8)
```

```
## pdf
## 2
```

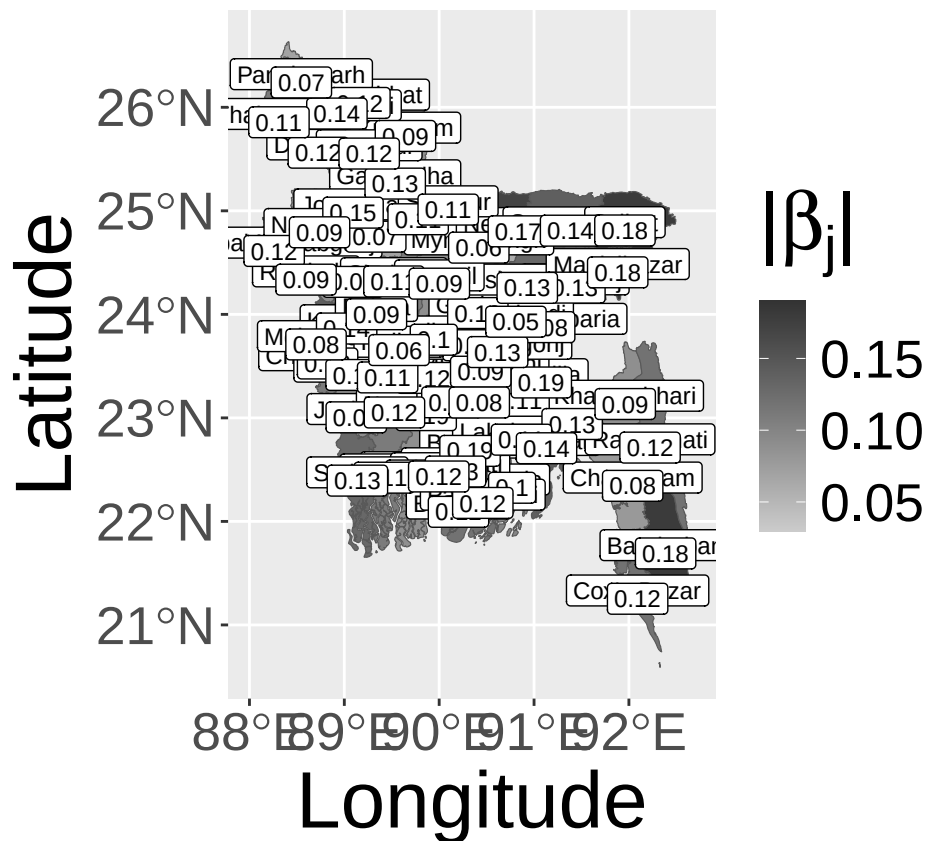
```
write.csv(df, paste(IMAGEOUT,"qE.BETA.other_wives.csv",sep=""))
```

```
district_coeff <- abs(round(qE[2,],2))
map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$district_coeff<- district_coeff
```

```
ggplot(map) +
  geom_sf(aes(fill = district_coeff)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  geom_sf_label(aes(label = district_coeff), nudge_y = -0.08, size = 3) +
  labs(fill=expression(paste("|",beta[j],"|")), x="Longitude", y="Latitude") +
  theme(
    legend.title = element_text(size = 30),      # Legend title size
    legend.text = element_text(size = 20),      # Legend item text size
    axis.title = element_text(size = 30),
    axis.text = element_text(size = 20)
  )
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
```

```
## correct results for longitude/latitude data
```



```
dev.print(pdf, paste(IMAGEOUT, "map_other_wives_coefficients.pdf", sep=""), width=16, height=16)
```

```
## pdf
## 2
```

To plot the percentage of more-than one wives per district

```
district_percentage <- NULL
for (j in 1:m){
  dis_j <- X.list[[j]][,14] # other_wives is the 14th column of X
  percentage <- round(sum(dis_j == 2)/length(dis_j) * 100,2)
  district_percentage<- c(district_percentage, percentage)
  j <- j+1
}
map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$percentage<- district_percentage

ggplot(map) +
  geom_sf(aes(fill = percentage)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  geom_sf_label(aes(label = percentage), nudge_y = -0.08, size = 3) +
  labs(fill="Percentage (%)", x="Longitude", y="Latitude") +
  theme(
```

```

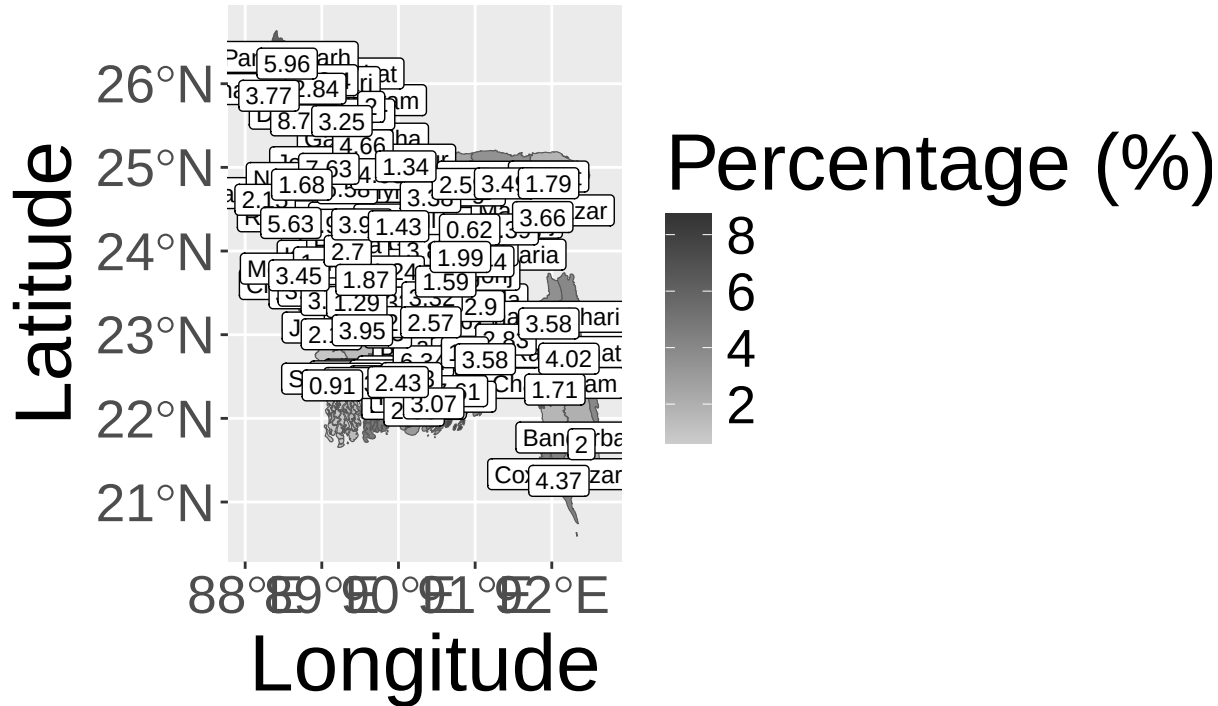
legend.title = element_text(size = 30),      # Legend title size
legend.text = element_text(size = 20),      # Legend item text size
axis.title = element_text(size = 30),
axis.text = element_text(size = 20)
)

```

```

## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data

```



```

dev.print(pdf, paste(IMAGEOUT, "map_other_wives_percentage.pdf", sep=""), width=16, height=16)

## pdf
## 2

```

To plot the posterior CI of gamma + theta coefficients of HH_sex

```

#index_HH_sex is the index of HH_sex in the regression coefficient vector GAMMA
index_HH_sex = 17
i <- 1
GAMMA_HH_sex <- NULL
for (dis in district_codes){

  gamma_ij <- NULL
  for (s in c(1:length(GAMMA.post))){
    gamma_ij <- c(gamma_ij, GAMMA.post[[s]][i, index_HH_sex])
  }
  GAMMA_HH_sex <- rbind(GAMMA_HH_sex, gamma_ij)
  i <- i+1
}

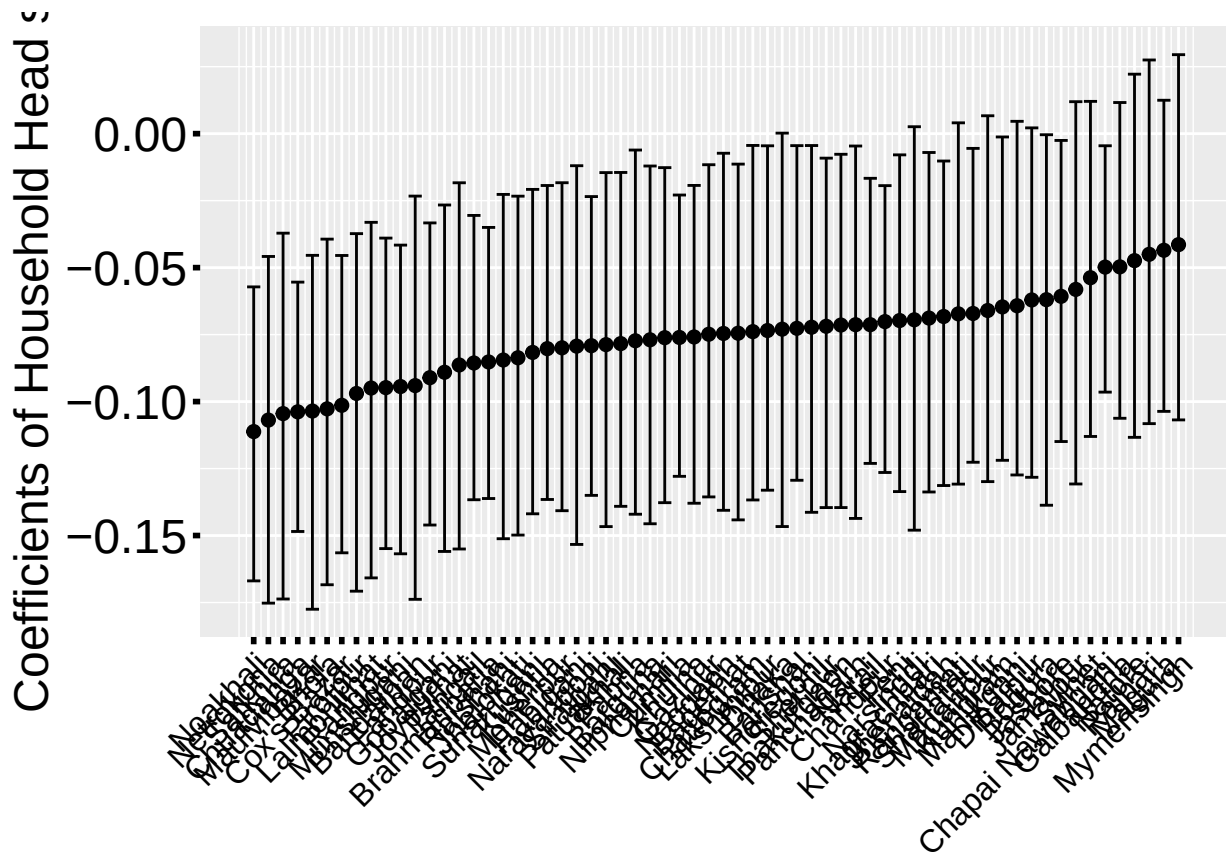
```

```

# THETA.post.mat[2,] is the fixed effect of fixed predictor
qE<-apply(GAMMA_HH_sex+rep(THETA.post.mat[index_HH_sex,],m),1,quantile,probs=c(.025,.5,.975))
index <- order(qE[2,])

df <- data.frame(x = 1:m,
                F = qE[2,][index],
                L = qE[1,][index],
                U = qE[3,][index])
abc <- ggplot(df, aes(x = x, y = F)) +
  labs(y = "Coefficients of Household Head sex", x = NULL) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal =
dev.print(pdf, paste(IMAGEOUT, "BETA.HH_sex.pdf",sep=""),width=16,height=8)

## pdf
## 2

write.csv(df, paste(IMAGEOUT,"qE.BETA.HH_sex.csv",sep=""))

```



```

district_coeff <- abs(round(qE[,2],2))
map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$district_coeff <- district_coeff

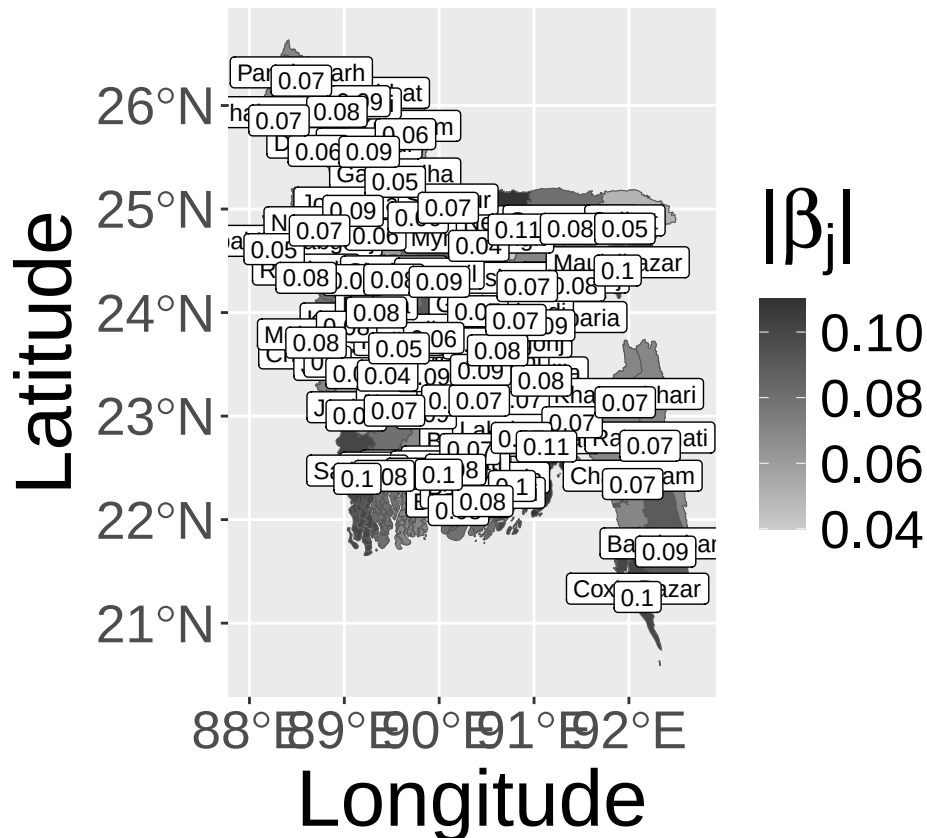
ggplot(map) +
  geom_sf(aes(fill = district_coeff)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  geom_sf_label(aes(label = district_coeff), nudge_y = -0.08, size = 3) +
  labs(fill=expression(paste("|",beta[j],"|")), x="Longitude", y="Latitude") +
  theme(
    legend.title = element_text(size = 30),      # Legend title size
    legend.text = element_text(size = 20),      # Legend item text size
    axis.title = element_text(size = 30),
    axis.text = element_text(size = 20)
  )

```

```

## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data

```



```

dev.print(pdf, paste(IMAGEOUT, "map_HH_sex_coefficients.pdf", sep=""), width=16, height=16)

```

```

## pdf
## 2

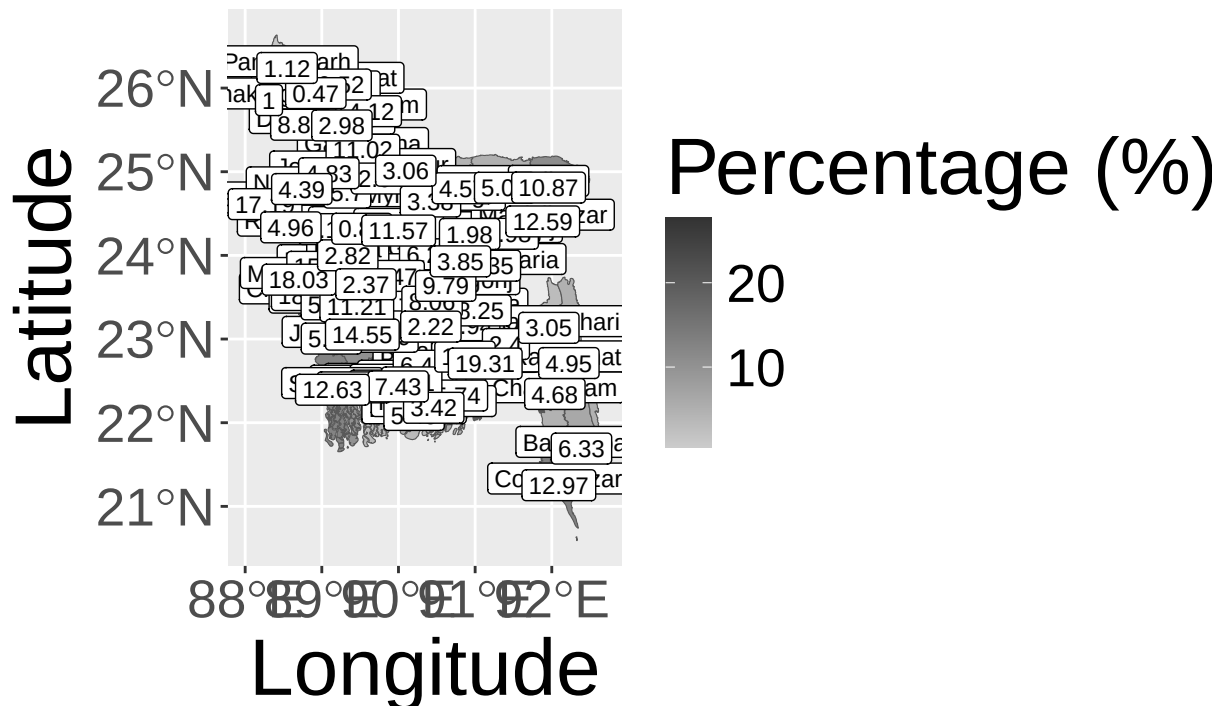
```

To plot the percentage of female-headed households in each district

```
district_percentage <- NULL
for (j in 1:m){
  dis_j <- X.list[[j]][,16] # HH_sex is the 14th column of X
  percentage <- round(sum(dis_j == 2)/length(dis_j) * 100,2)
  district_percentage<- c(district_percentage, percentage)
  j <- j+1
}
map <- read_sf("map.json")
map <- map %>% arrange(ADM2_EN)
map$percentage<- district_percentage

ggplot(map) +
  geom_sf(aes(fill = percentage)) +
  scale_fill_gradient(low="gray80", high="gray20") +
  geom_sf_label(aes(label = ADM2_EN), size = 3) +
  geom_sf_label(aes(label = percentage), nudge_y = -0.08, size = 3) +
  labs(fill="Percentage (%)", x="Longitude", y="Latitude") +
  theme(
    legend.title = element_text(size = 30),      # Legend title size
    legend.text = element_text(size = 20),      # Legend item text size
    axis.title = element_text(size = 30),
    axis.text = element_text(size = 20)
  )
)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not give
## correct results for longitude/latitude data
```



```

dev.print(pdf, paste(IMAGEOUT, "map_HH_sex_female_percentage.pdf", sep=""), width=16, height=16)

## pdf
## 2

DIC Calculation

#
thin=50
thin2=2
N_THIN=(S-B)/(thin*thin2)
DIC_array<-array(dim=N_THIN)
for(i in 1:N_THIN)
{
  DIC_i <- 0
  for(j in 1:m)
  {
    DIC_i <- DIC_i + sum(-2 * dgenpois2(y.list[[j]],
                                         meanpar=exp(matmul(Z.list[[j]], GAMMA.post.mat[[j]][i, ]) + matmul(X.list[[j]], as.matrix(
  }
  DIC_array[i] <- DIC_i
}

DIC_bar = mean(DIC_array)
D_mean<-0
for(j in 1:m)
{
  D_mean <- D_mean + sum(-2 * dgenpois2(y.list[[j]],
                                         meanpar=exp(matmul(Z.list[[j]], apply(GAMMA.post.mat[[j]], 2, mean)) + matmul(X.list[[j]],
}
DIC <- 2*DIC_bar - D_mean
# 111513.5

```